Artificial surf reefs: a review of built projects, surf quality, sediment transport and durability, with implications for the Israeli coast

Course paper, SM-52080001 Introduction to Ocean Engineering (Prof. Nitai Drimer)

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October 2026

Abstract

Artificial surf reefs are promoted as structures that create surfable waves, protect beaches and enhance habitat. This review compiles the 13 surf reefs built or reshaped since 1997 with traced plan shapes, nine three-dimensional reconstructions and traceable sources. Coastal-engineering theory (shoaling, breaker index, Iribarren number, breaking intensity, peel angle) applied to profiles cut through the surfed part of each reconstructed reef shows that the crest depth sets which waves break and the ramp how: ramp length and mean seaward slope rank with the surf outcome (Spearman $\rho = 0.93$ and 0.77, $p < 0.05$), crest depth does not ($\rho = 0.39$, $p = 0.36$), and over similar depth gaps a 1:17 ramp gives a plunging, surfable wave and a 1:3 rise a collapsing one. Surfer testimony indicates that the reef must also stay shallow just shoreward of the break for the wave to hold its shape. Reefs that work are large (25,000 m^3 or more), because volume buys ramp length; the three that do not work are 4,900 m^3 or less. No reef offers a beginner's wave. Where surveys can test it, the shoreline response is 2 to 6 % of standard salient predictions; on the Gold Coast it cannot be separated from nourishment. Six of seven geotextile or inflatable reefs suffered major damage within a few years, none of four rock reefs. In a three-criterion rating (surf quality; safety and longevity; sediment and storm protection), only Albany clearly delivers a high-quality wave, with safety problems; the other two reefs that work (Palm Beach, Burkitts) do so only in a narrow range of conditions; rock reefs rate best on durability, and the sediment benefit is unknown or confounded everywhere. At Haifa, Israel, the 0.4 m spring tide would make the wave depend on wave height, not on the tide; a rock reef with a crest 1.0 to 1.2 m below mean sea level (deep enough to leave water over the rock in winter swells and at low water levels, shallow enough to break the common seas), a 1:16 to 1:20 ramp, a crest that stays shallow shoreward of the break and about 14,000 to 20,000 m^3 per 100 m is proposed for testing.

Keywords: artificial surf reef; multi-purpose reef; surf quality; breaking intensity; peel angle; submerged breakwater; shoreline response; geotextile sand containers; Israel; Haifa

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1 Introduction

1.1 Purpose of surf reefs

A surf reef is a submerged structure of rock or sand-filled geotextile containers, placed in or just outside the surf zone so that incoming waves break over it in a controlled way and produce a rideable wave. Three purposes are attached to such structures (Mead & Borrero, 2011). The first is surf amenity: a reef can create a break where the natural one is poor, or restore one lost to coastal works. The second is coastal protection: a submerged structure dissipates wave energy and, in theory, causes sand to accumulate in its lee as a salient or tombolo; Blacka et al. (2013) review such reefs as a "softer" alternative to seawalls with negligible visual intrusion. The third is ecological enhancement: hard, stable substrate supports more species than mobile sand, and colonisation of surf reefs begins within days (Mead & Borrero, 2011). Reefs that combine the three are called multi-purpose reefs (MPRs).

The record is mixed. Of 32 submerged constructed reefs reviewed for New South Wales, 29 were intended for coastal protection, and the resulting shoreline often differed significantly from the design prediction; eight were built to improve surfing, and approximately half of these were considered at least partially successful; most structures settled or scoured, so that the crest level differed from the design (Blacka et al., 2013, p. ii).

1.2 History in brief

Artificial reefs for habitat date back to 2000 BC on the west coast of India (Kurian, 1995, as cited in Mead and Borrero, 2011). A reef combining coastal protection, ecology and surfing was first proposed in a 1969 article in Surfer Magazine, and the first methods to quantify a surfing break were developed by J. R. Walker in the 1970s and 1980s (Mead & Borrero, 2011). The first reef counted in this review, Burkitts Reef at Bargara, Australia, is a natural platform reshaped by hand in 1997 (Howells & O'Neal, 2024); the first purpose-designed structure, Cables Reef in Perth, was built in 1999 (Mead & Borrero, 2011). Between 1999 and 2010 seven reefs were built in five countries, most from sand-filled geotextile containers (Mead & Borrero, 2011). From 2011 the projects shifted to quarried rock, and an inflatable, sand-ballasted reef (the Airwave) failed on installation in 2019 (Section 3).

1.3 Motivation and research questions

This review addresses two questions: what effect a reef has on the waves and on surf quality for surfers of different levels, with insight that is computational where possible and grounded in coastal theory; and what effect it has on sand transport, including the combination of a surf reef with beach stabilisation. Published evaluations are mostly by the designers (for example Mead & Borrero, 2011) or cover single sites. The main independent synthesis (Blacka et al., 2013) concentrates on New South Wales and precedes the rock reefs of 2019 and 2025. No source found tabulates all built surf reefs with traceable geometry, cost, material and outcome, and none relates their geometry to surf quality.

The practical motivation is a planned surf reef at Haifa, on the Mediterranean coast of Israel, where conditions differ from those of the built reefs in three ways. The tide is very small, about 0.4 m at spring tide (Rosen, 2001), against 1.1 to 4.3 m at the reefs surveyed here (for the three south-west Australian reefs this is the predicted annual range, not a spring range; Section 3). Waves are mostly small, with significant heights H_s of 0.5 to 1.0 m the most common class (Bitan & Zviely, 2020; Spanier & Zviely, 2022). Sand moves strongly with the seasons; field observations by the author (Appendix G) indicate a seasonal bed-level change of the order of 1 m at Haifa (Section 7). The analysis was guided by eight working hypotheses, drawn from field experience and the literature, labelled (a) to (h): (a) reef volume, (b) sand transport, (c) soft materials, (d) the number of surfable days, (e) the failure mechanism of tubes, (f) depth gap and ramp length, (g) surfer level and (h) the Israeli setting. They served only as questions to investigate; the findings of Sections 3 to 7 are discussed in Section 8 on their own terms, and Appendix F records for each hypothesis whether the evidence supports it and whether it is consistent with published work or new.

The research questions are:

1. Which surf reefs have been built, of what size, material and cost, and how did they perform and last (Section 3)?
2. How does a reef change the waves, and what surf quality results for surfers of different levels, as set by depth gap, ramp length, breaker type, peel angle and tide (Section 4)?
3. What effect have surf reefs had on sand transport and the shoreline, and can a surf reef be combined with beach stabilisation where beaches are nourished (Section 5)?
4. Which materials survive in the breaking zone, and why do soft structures fail (Section 6)?
5. What do the answers imply for a reef at Haifa (Section 7)?

Section 2 gives the methods, Section 8 the discussion and Section 9 the conclusions. Appendix A documents the method for each reef, Appendix B shows the three-dimensional reconstructions, Appendix C holds the full data table, Appendix D covers the other structures named by Bar and Drimer (2023) and the Israeli geotube projects, Figure E1 (Appendix E) shows where the depth profiles and peel-angle lines were taken on each reef, and Appendix F lists the working hypotheses and their status.

2 Methods

2.1 Sources and the traceability standard

For each reef, the published design and monitoring papers, official reports, survey data sets, georeferenced aerial and satellite images, press articles and reef-specific web pages were collected. Peer-reviewed and agency reports were preferred; designers' papers (for example Mead & Borrero, 2011) are treated as the account of an interested party; press, surfer media and web databases are used only for facts that nothing else gives, and are named as such. Every number in the data set carries the value and unit, a full citation, the place where it was read (page, table, figure or file), the method, and a status: *sourced* (read in a cited source), *derived* (computed in this review from sourced inputs, a traced plan shape or a reconstructed three-dimensional model), *assumed* (flagged and numbered in Appendix A) or *unknown* (searched for and not found; no value given). Confidence is rated high, medium or low by a fixed rubric (Section 2.3; Appendix A.0). Where a statement rests on a second-hand source, for example Hutt et al. (2001) as reproduced by Lewis et al. (2015), the text says so. Values from secondary compilations whose cited sources could not be fetched and confirmed were not used. Costs were converted to US dollars at the World Bank annual-average exchange rate of the price year and to 2024 prices with the US consumer price index (World Bank, 2026); both steps are derived and inherit the uncertainty of the price year.

2.2 The reef sample

The sample is the 13 surf reefs built (for Burkitts, reshaped) with surfing as a primary or joint purpose, including those that failed or were removed and the unverified Xala reef (Mexico, 2025-26). It was compiled from Mead and Borrero (2011), Blacka et al. (2013), the web database of Raised Water Research and press reports, and is not claimed to be exhaustive. Borth is included because its northern mound was shaped as a surf reef; its southern oval is a breakwater. Of the eleven projects named by Bar and Drimer (2023), three are in the sample; the other eight, and the Israeli geotube projects, were not built as surf reefs and are described in Appendix D.

2.3 Plan shape, three-dimensional reconstruction and datums

Each outline was traced on the best available evidence, in a fixed order of preference: a measured survey of the structure, then a drawing of the built version, then a photograph. Satellite and aerial images (Esri World Imagery and dated Wayback releases; Esri, Maxar, Earthstar Geographics, & the GIS User Community, n.d.) were georeferenced from their tile bounds, drawings were scaled from a scale bar or a known dimension, and each outline was checked on a second image and against the dimensions in the text. Confidence is high if the structure is clearly visible on a georeferenced image or an as-built survey and the dimensions agree with the text to about 10 %, medium for a scaled design drawing, a partly visible structure or a 10-25 % disagreement, and low if the shape is inferred from text or a sketch. Outlines were obtained for nine reefs (Figure 1).

A three-dimensional model was built only where the plan, the crest level and the seabed depth were each sourced. The seabed is taken from the best measured data (soundings, lidar, multibeam read from a published figure, a coastal elevation model), otherwise from a chart (Garmin Navionics SonarChart, datum tested; Garmin Navionics, n.d.) or from design contours. The reef surface is the surveyed surface or a loft from the crest level over the traced toe outline down to the seabed; its volume is a mound envelope with voids included, not the fill volume. All heights are relative to mean sea level (MSL) through an explicit datum chain per reef (Appendix C, note 1), and only the tidal planes of the nearest standard port are used, for example those of Maritime Safety Queensland (2026) for the Gold Coast. Nine reefs were modelled (Appendix A, Table A1; Appendix B).

2.4 Depth profiles, depth gap and ramp length

The depth profile is the shore-normal section of the model through the shallowest crest cell. For Albany, Cables and Borth this automatic line did not cross the part of the reef where the wave breaks and is ridden, or was not shore-normal; for these reefs the profile is taken along a surf cut through the surfed part of the reef, perpendicular to the local shoreline (Albany: the middle of the long −2 m AHD ridge rather than the −1 m take-off plateau; Cables: through the footprint centroid, normal to the fitted shoreline; Borth: 10 m south of the centreline, a judgement because no source locates the break). The cuts are drawn in Figure E1 (Appendix E) and their reasons recorded in Appendix A; volumes and footprints do not depend on the cut. The crest depth C is the depth of the shallowest crest point below MSL (negative if the crest is above MSL); T is the seabed depth at the seaward toe of the traced outline; the depth gap is $G = T - C$. The ramp length L is the horizontal distance from the toe to the crest edge (the seaward-most cell within 0.25 m of the column maximum), and the mean slope G/L is quoted as 1:n. Six reefs (Narrowneck, Cables, Albany, Palm Beach, Borth, Boscombe) have a ramp from audited geometry; the Pratte's ramp is assumed, and Mount Maunganui and Bunbury rest on hand estimates or a stand-in surface (Appendix C, note 2). Sourced crest depths are used where they exist (Cables, Palm Beach, Albany).

2.5 Computations

Sections 4 to 7 apply standard coastal-engineering relations to the profiles; formulas, thresholds and the status of every input are given where they are used. In outline: linear shoaling and Snell refraction for deep-water heights of 0.5 to 3 m and periods of 6 to 14 s at three tide levels, with breaking at $H = \gamma d$ ($\gamma = 0.78$), the breaker type from the surf-similarity (Iribarren) number (Battjes, 1974), the breaking intensity from the seabed gradient (Mead & Black, 2001b) and the peel angle against surfer skill (Hutt et al., 2001, as tabulated by Lewis et al., 2015) (Section 4); shoreline predictors (Black & Andrews, 2001; Ranasinghe & Turner, 2006) and wave transmission, compared with the surveyed seabed and shoreline change in the lee and at control sections, and longshore transport by the CERC formula (US Army Corps of Engineers, 2002) as an order of magnitude (Section 5); single-tube sliding and uplift after Bar and Drimer (2023) with a calibrated estimate, and rock stability by the Hudson formula (Section 6); and the same calculations with the Israeli tide (Section 7). Associations with outcome are Spearman rank correlations with permutation p values.

2.6 Limitations of the method

Every value is traced to a cited source, and the design choices made in this review are listed in Appendix A. Readings from figures are by eye or by calibrated scales and carry the stated reading error. The original texts of Hutt et al. (2001), Mead and Black (2001a) and Black and Andrews (2001) were not read; their content is used as reproduced by later sources, and this is stated where it matters. No new field data were collected; the author's field observations at Haifa (Appendix G) are used only in Section 7 and are labelled there. The consequences of the small sample, the unknown wave climates and the idealised seabeds are discussed in Section 8.2.

2.7 Authorship and use of AI tools

The author selected and supplied the primary sources (papers, reports, survey data, imagery and surfer testimony collected by the author, for example viewer comments on online videos), defined the research questions and the hypotheses to be tested, and directed the analysis. AI tools assisted with source extraction, data compilation, computation and drafting. Every source added by the AI tools was validated by the author, who also reviewed and confirmed all text, data, calculations and figures, including by comparing the three-dimensional reconstructions with the published figures and imagery (a worked example is given in Appendix A, Section A.14).

3 Survey of built surf reefs

3.1 The sample at a glance

Table 1 lists the 13 reefs in order of the start of construction; every value has its source, status and method in Appendix C. Seven reefs are made of geotextile or inflatable fabric filled with sand, five of rock (Burkitts is a natural platform reshaped) and one (Xala) of quarried boulders that is not yet documented. By the classes of this review, three reefs work for surfing (Burkitts, Palm Beach, Albany), three work rarely (Narrowneck, Mount Maunganui, Boscombe), two are mixed or disputed (Cables, Kovalam), three do not work (Pratte's, Opunake, Bunbury) and two are unknown (Borth, Xala). These classes, used in the figures, follow the surf-quality rating (A) of Table 1 and Section 8.4: 'works' is A 3 to 5 with consistent reports, 'works rarely' A 2, 'does not work' A 1, and 'mixed or disputed' marks conflicting reports (Cables A 3, Kovalam A 2). Figure 1 shows the plan shapes and depth profiles of the nine reefs with a traced outline, and Figure 2 the size, crest depth and depth gap of all 13.

#	Reef	Built	Purpose	Material	Volume, reported (m³)	Volume, 3D (m³)	Crest depth (m)	Depth gap (m)	Cost as reported	Cost (USD M, 2024 prices)	USD per m³ (2024 prices)	A, surf quality	B, safety and longevity
1	Burkitts (AUS)	1997	surf	natural basalt, reshaped	300 (est.)	–	?	?	AUD 10,000 (1997)	0.015	50	3 – inconsistent	4 – intact
2	Cables (AUS)	1999	surf	granite boulders	5,500	4,683	1.55	4.96	AUD 1.8 M (1999)	2.19	398	3 – inconsistent	4 – intact
3	Narrowneck (AUS)	1999	multi	sand containers	70,000[1]	13,495	2.32	3.81	AUD 2.8 M (1999-2006)[2]	2.96	42	2 – works rarely	3 – maintained or renewed
4	Pratte's (USA)	2000	surf	sand bags	1,350	442[3]	2.68	1.35	USD 0.55 M (2000-01)	1.00	742	1 – no surfable wave	1 – failed or removed
5	Mount Maunganui (NZ)	2005	multi	sand containers	2,800	2,943	2.03	2.70	NZD 1.5 M (2005-08)	1.54	549[4]	2 – works rarely	1 – failed or removed
6	Opunake (NZ)	2006	surf	sand bags	4,900 (design)[5]	–	?	3.00†	NZD 1.7-2.1 M (2006-09)	1.55-1.92	317-392	1 – no surfable wave	1 – failed or removed
7	Boscombe (UK)	2008	multi	sand containers	12,800	13,113	0.90	4.56	GBP 3.2 M (2008-09)	7.29	569	2 – works rarely	1 – failed or removed
8	Kovalam (IND)	2009	multi	sand containers	4,000	–	?	?	INR 75 M (2009-10)	2.36	590	2 – works rarely	1 – failed or removed
9	Borth (UK)	2011	multi	quarry rock	?[6]	30,119	-0.19	4.62	(GBP 13 M, 2012)	(28.06)	–	?	4 – intact
10	Palm Beach (AUS)	2019	multi	quarry rock	25,000[7]	22,020	1.50	5.17	AUD 18.2 M (2019)[8]	15.52	621	3 – inconsistent	4 – intact

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[1] Narrowneck volume: sand pumped into the containers (Mead & Borrero, 2011, p. 2; International Coastal Management, 2025); 408 containers of about 147 m^3 give about 60,000 m^3 by 2000 (derived). The 3D value is the mound above today's bed in the field visible in 2020; see the text below the table.

[2] Narrowneck cost: incl. top-ups and maintenance (Blacka et al., 2013, p. 101); build alone AUD 2.3 M (International Coastal Management, 2025); the 2017-18 renewal of about AUD 2 M (The Inertia, 2017) is excluded.

[3] Pratte's 3D: Phase I design layout only (110 of the 200 bags).

[4] Mount Maunganui: the design was about 6,500 m^3, of which 2,800 m^3 were built (Mead & Borrero, 2011, p. 3); per design m^3 about USD 240.

[5] Opunake: about 3,400 m^3 built (Mead & Borrero, 2011, p. 3; Blacka et al., 2013, Table 4.4), i.e. USD 460-560 per built m^3; cost at least NZD 1.7 M spent by 2009, up to NZD 2.1 M (Reid & Sumner, 2021).

[6] Borth: no reef-only volume or cost; GBP 13 M (Royal Commission on the Ancient and Historical Monuments of Wales, 2020) and nearly 300,000 t cover the whole Phase 1 defence scheme. 3D: both rock mounds (about 51,000 t at 1.7 t/m^3).

[7] Palm Beach volume: "approx. 25,000 m^3" (International Coastal Management, 2025); 50,000-60,000 t are also reported, i.e. 28,000-35,000 m^3 at a bulk density of 1.7-1.8 t/m^3; council register 33,000 (units not stated).

[8] Palm Beach cost: reef phase of the shoreline project; the 2017 nourishment is excluded; no reef-only account found.

#	Reef	Built	Purpose	Material	Volume, reported (m^3)	Volume, 3D (m^3)	Crest depth (m)	Depth gap (m)	Cost as reported	Cost (USD M, 2024 prices)	USD per m^3 (2024 prices)	A, surf quality	B, safety and longevity
11	Bunbury Airwave (AUS)	2019	surf	inflatable bladder, sand ballast	70-94 (sand ballast)	93-117	0.68	2.24	approx. USD 0.4 M (2019)[9]	approx. 0.49	(4,700)	1 – no surfable wave	1 – failed or removed
12	Albany (AUS)	2025	surf	quarry rock	34,000	(34,000)	1.00	6.33	AUD 13.0 M (2024-25)[10]	8.38	246	5 – excellent	2 – hazard or repairs
13	Xala (MEX)	2025	multi	quarry boulders	?	–	?	?	?	?	?	?	?

Table 1. Master table of the 13 surf reefs. *Volume, reported*: as stated by the sources (sand fill or rock; not one common quantity, see the notes and the text below). *Volume, 3D*: envelope above the modelled seabed of this review's reconstruction, voids included (Appendix A; – = no model). Crest depth below MSL (negative: above MSL); depth gap = seabed depth at the seaward toe minus crest depth (Section 2.4); † referred to chart datum (design stage). *Cost as reported*: the whole-project total in the original currency, with the years of spending; *USD M, 2024 prices*: that total converted at the World Bank annual-average exchange rate of one price year (the construction or completion year) and raised to 2024 prices with the US consumer price index; *USD per m^3*: the 2024-price cost divided by the reported volume (both derived; Section 2.1; Appendix C). Values in brackets are not comparable with the other rows. *A, surf quality*: 5 excellent, 4 good, 3 inconsistent, 2 works rarely, 1 no surfable wave. *B, safety and longevity*: 5 no problems, 4 intact, 3 maintained or renewed, 2 hazard or repairs, 1 failed or removed. ? = unknown or no evidence. A and B are this review's ratings (Section 8.4, Table 7).

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[9] Bunbury: USD 350,000-400,000 in secondary sources (Raised Water Research, n.d.-a); the City's AUD 75,000 was a sponsorship cap (South Western Times, 2019), not the price.

[10] Albany: reported final cost (Elezovich, 2026); AUD 11.75 M announced in 2024 (Government of Western Australia, 2024). The 3D surface was fitted to the stated volume.

Reading the volume columns. The reported and the modelled volumes agree within 15 % at Cables (5,500 m³ stated; 4,671 to 5,219 m³ surveyed; 4,683 m³ modelled), Mount Maunganui and Boscombe; Albany's model was fitted to the stated volume and is not an independent check. Four reefs differ. (1) Narrowneck's 70,000 m³ and the 13,495 m³ of the 3D model measure different things. The literature value is the sand pumped into the containers. The model measures only the mound that stands above today's seabed within the container field visible on the aerial of 8 August 2020 (6,239 m²). Three effects separate the two: the containers have settled 1 to 2 m and are partly buried in the sand bed (Jackson et al., 2012, p. 6), so much of the fill lies below the modelled seabed; containers spread beyond the visible field, since the City of Gold Coast reef polygon covers 34,565 m² (City of Gold Coast, n.d.), over which 60,000 m³ is a mean thickness of about 1.7 m, whereas the visible field alone would need a mean height of almost 10 m in 4 to 6 m of water; and a stack of flattened bags encloses voids, which the envelope counts and the fill does not, a smaller effect in the opposite direction. In plan, the visible Narrowneck mound is therefore of the order of Palm Beach and Albany (Figure 1), but the sand placed is three to five times larger (derived). (2) The Pratte's model covers only the Phase I design layout. (3) At Palm Beach the stated 25,000 m³ and 60,000 t imply 2.4 t/m³, a solid-rock density rather than that of placed rock with voids (1.7 to 1.8 t/m³), so at least one of the two is wrong; the model (22,020 m³) lies below every reported basis. (4) Borth has no reef-only quantity, and Bunbury's bladder volume is not a fill volume.

Reading the cost columns. Most reefs cost USD 250 to 750 per m³ at 2024 prices, and the outliers have identifiable causes. Narrowneck is the cheapest built reef (USD 42 per m³) because the volume is sand dredged nearby and pumped into containers dropped from a split-hull barge, with no quarried rock; the price year is 1999-2000, and a consumer price index need not follow the cost of marine construction; and the 2017-18 renewal is excluded. Per m³ of the exposed mound it would be about USD 220. Burkitts (USD 50 per m³) is two days of excavator hire on a natural boulder platform, not a construction project. The small sand-bag reefs are the most expensive per m³ (Pratte's 742, Kovalam 590, Boscombe 569, Mount Maunganui 549): design, consent, mobilisation and monitoring are largely fixed costs spread over small volumes; Pratte's was placed in two campaigns, Mount Maunganui's budget was for a design more than twice the built volume, and Boscombe's cost rose from an original GBP 0.6 M to GBP 3.2 M during the project (Rendle, 2016, pp. 65-67). The two modern rock reefs differ by a factor of 2.5: Albany USD 246 per m³ (AUD 382) against Palm Beach USD 621 (AUD 728; about AUD 700 by International Coastal Management (2025)). The sources do not explain the gap; Albany's rock came from a quarry opened near the site, and the Palm Beach volume is uncertain (USD 470 per m³ if the register's 33,000 m³ holds). Borth (the whole defence scheme) and Bunbury (a manufactured bladder of about 100 m³) are not comparable and are bracketed. Total cost does not follow outcome: the best wave (Albany) cost about half as much as Palm Beach.

3.2 Australia

Cables Reef, Perth (1999). After a surfers' campaign from 1975, the Western Australian Department of Sport and Recreation financed a reef on the site of a natural break (Raised Water Research, 2019a). It was designed for swells of 0.5-3.5 m, a peel angle of about 45 degrees and a 1:20 ramp (Lyons, 1992; Pattiaratchi, 1999). Granite boulders of 1.5 t (body) and 3.0 t (perimeter) were placed on limestone in 1999 (Raised Water Research, 2019a) for AUD 1.8 M (Blacka et al., 2013). The reef is undamaged: surveyed rock volumes of 4,671 m³ (1999) and 5,219 m³ (2002) differ by less than the survey error. Its surf record is disputed: the designers report 142 surfable days a year against 5-7 before, counting days with surfers present (Mead &

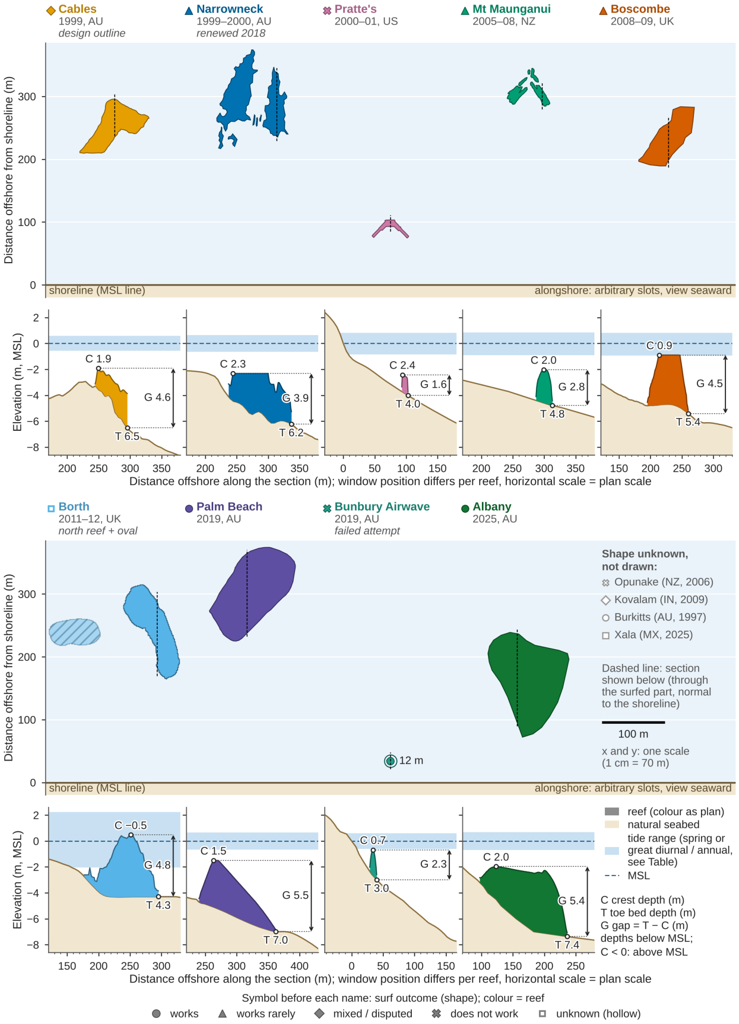
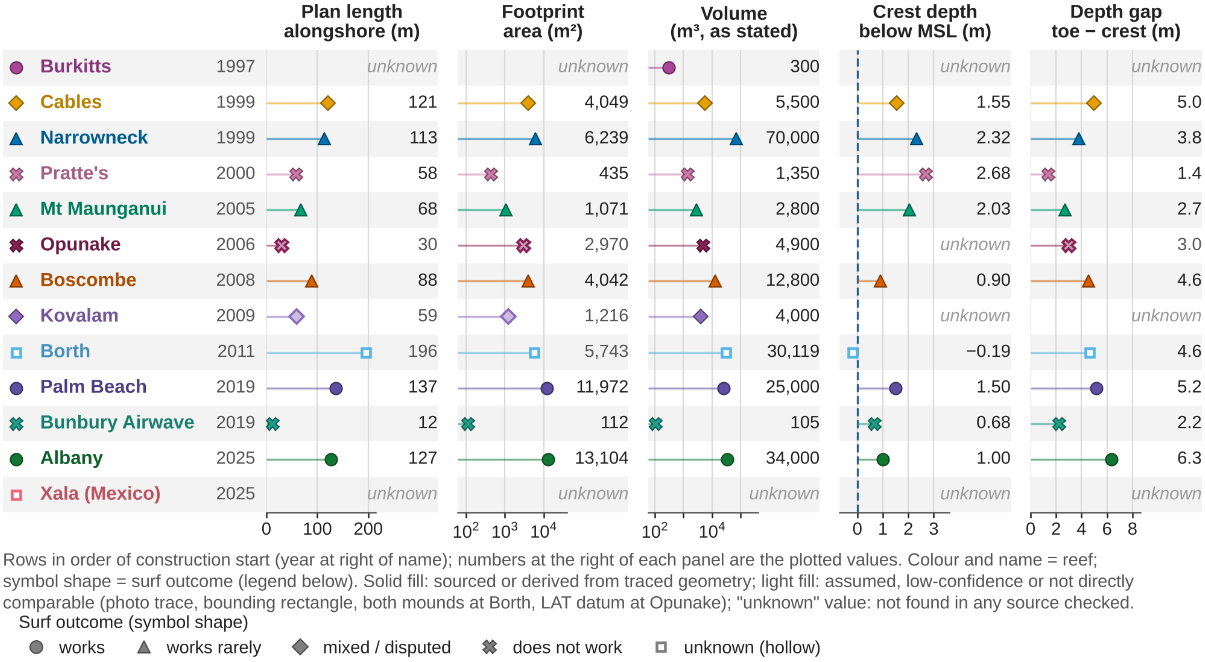


Figure 1. Plan shapes (upper part of each row) and cross-shore depth profiles (lower part) of the nine surf reefs whose outline is known, in order of construction; Opunake, Kovalam, Burkitts and Xala (Mexico) are not drawn because no outline could be traced. Colour identifies the reef (the same in all figures) and the symbol before each name its surf outcome (legend at the bottom: works, works rarely, mixed or disputed, does not work, unknown). All plans share one metric scale in x and y (100 m bar at lower right), are viewed looking seaward (unmirrored), and each outline is set at its distance from the MSL shoreline (thick line, $y = 0$) with arbitrary alongshore slots; the hatched shape is Borth's southern oval, a shore-parallel breakwater rather than a surf reef, and Bunbury is the 12 m Airwave bladder, torn on installation. Each profile is the section of the 3D reconstruction marked by the dashed line: a cross-shore cut through the part of the reef where the wave breaks and is ridden, perpendicular to the local shoreline (Albany: middle of the long −2 m AHD ridge along which the left-hander is ridden, not the −1 m take-off plateau; Cables: through the middle of the reef, turned 17.5° from the design frame to the shore normal; Borth: middle of the main ridge of the northern reef), and for the other six reefs the section whose crest depth best matches the master table (line positions: Figure E1 (Appendix E)). It is drawn at the plan's horizontal scale with a vertical exaggeration of about 21:1; elevations are relative to MSL, and the blue band is the tide range (spring range MLWS-MHWS; great diurnal range MLLW-MHHW at Pratte's; predicted annual range, about LAT-HAT, at Cables 1.14 m, Bunbury 1.20 m and Albany 1.38 m, centred on MSL, from the 2026 predictions of the Western Australia Department of Transport, & Bureau of Meteorology, 2026; tide-range table in Appendix C). C, T and G are the crest depth, the seabed depth at the seaward toe and the depth gap (T − C) read from the plotted section. They agree with the master table within 0.35 m except where the cut deliberately avoids the shallowest point: Albany (C 1.97 m on the ridge against the 1.0 m take-off crest; G 5.4 against 6.3 m) and Cables (C 1.92 against 1.55 m). Sources and methods: Appendix A.



Rows in order of construction start (year at right of name); numbers at the right of each panel are the plotted values. Colour and name = reef; symbol shape = surf outcome (legend below). Solid fill: sourced or derived from traced geometry; light fill: assumed, low-confidence or not directly comparable (photo trace, bounding rectangle, both mounds at Borth, LAT datum at Opunake); "unknown" value: not found in any source checked.

Figure 2. Size overview of the 13 surf-reef projects (in order of construction): plan length along the shore and footprint area of the traced outline, volume as stated by the sources (rock or sand fill, not one common quantity), crest depth below MSL (negative: crest above MSL, Borth) and depth gap between the seabed at the seaward toe and the crest. Colour and name identify the reef; symbol shape gives the surf outcome (legend below the panels; unknown = small hollow square). Light-filled symbols mark assumed, low-confidence or not directly comparable values, and 'unknown' means that no source checked gives the value; the four reefs without a traced shape (Opunake, Kovalam, Burkitts, Xala) therefore have no length or footprint. Values are those of the master table of this review (sources and status of every value in Appendix C; reading of the outlines and 3D models in Appendix A; main sources include Blacka et al., 2013; Mead and Borrero, 2011; Rendle, 2016; Corbett et al., 2023); lengths, areas, crest depths and gaps are derived from traced outlines and 3D models, and volumes are sourced except Borth and Bunbury (derived). The crest depths are the shallowest points of the master table (Albany 1.00 m at the take-off, Cables 1.55 m surveyed), not the surf cuts of Figure 1 and Table 2. Borth's plan area and volume come from different outlines: the area is the rock edge visible on the photograph of 17 September 2024, the volume (30,119 m^3) that of the 2022 laser-survey edge (8,675 m^2).

Borrero, 2011), whereas surfers report good waves only in big swells (Blacka et al., 2013, p. 93), and a 2009 critique puts the crest about 1.5 m below the lowest tide, in the swell shadow of Rottnest Island (Pitt, 2009, as cited in Raised Water Research, 2019a).

Narrowneck Reef, Gold Coast (1999-2000, renewed 2017-18). The primary aim was to protect the eroding Narrowneck isthmus, with surfing second (Blacka et al., 2013; Jackson et al., 2007). About 400 containers holding about 70,000 m^3 of sand were laid to a design crest of -1.0 m AHD tested on a 1:50 physical model (Jackson et al., 2007; Mead & Borrero, 2011), for AUD 2.3 M, or AUD 2.8 M with top-ups and maintenance (Blacka et al., 2013, p. 101); the 2017-18 renewal cost about AUD 2 M more (Appendix C). Non-uniform placement "detracted from the surfing amenity" (Mead & Borrero, 2011, p. 3); the containers settled by 1-2 m, the 2018 crest is at -2.2 m AHD, and 84 containers were renewed in 2017-18 (Blacka et al., 2013, pp. 97-98; Corbett et al., 2023). The reef "is rarely surfed" because world-class breaks work in the same conditions (EcoShape, n.d.). Its protective effect is disputed (Section 5.4).

Palm Beach Reef, Gold Coast (2019). The City of Gold Coast built the reef to protect the beach and to give a right-hand wave (Hunt et al., 2022), on a concept design by DHI (Mortensen et al., 2015). About 60,000 t of quarried boulders (stated as 25,000 m^3; Table 1, note) were placed to a crest at -1.5 m MSL for AUD 18.2 M (Appendix C). The reef is stable, and a year of shore-camera tracking found the waves "desirable" when easterly or southerly swells had H_s above 0.8 m and T_p above 7 s, with rides longer than in the natural surf zone (Hunt et al., 2022; Prenzler et al., 2023).

Bunbury Airwave (2019). The Airwave, patented by T. Bottegal (Waveco), is a Hypalon bladder 12 m across ballasted with 140-150 t of beach sand, intended to turn a closeout into a peeling wave. The City of Bunbury invested AUD 75,000 (ABC News, 2019); the cost was about USD 350,000-400,000 (Raised Water Research, n.d.-a; sources disagree). The bladder tore along a glued seam during installation in December 2019 and was removed within about three days (ABC News, 2019); according to the founder, the seam opened a couple of days after installation and the wave "worked amazingly" until then (T. Bottegal, Airwave founder, pers. comm., 8 Oct 2026). The two accounts differ on timing and are both given.

Southern Ocean Surf Reef, Albany (2025). Bluecoast Consulting Engineers designed the reef from the feasibility study of Royal HaskoningDHV (2015) as a long peeling wave for surfing; the reported final cost of AUD 13 M (Elezovich, 2026) was shared by the Australian Government (AUD 5 M), the Government of Western Australia (AUD 4.75 M; AUD 4.5 M in the City's figures), the City of Albany (AUD 1.995 M) and Albany Boardriders, against AUD 11.75 M of funds committed before construction (City of Albany, n.d.; Government of Western Australia, 2025). Rock of three grades (62,800 t, design volume 34,000 m^3) was placed to a crest of -1.0 m AHD (City of Albany, n.d.). It opened in July 2025 with faces of 0.8-1.3 m and rides of 100-120 m (Raised Water Research, n.d.-d; whether the wave is the "Long Right" of the 2015 option or the left described at opening was not reconciled), but a designer calls it "intermediate-to-advanced on most days" (Mercer, 2025). A storm in May 2025 moved some core rock, and about 18 high rocks were repositioned in February 2026 after injuries (Sheil, 2026).

Burkitts Reef, Bargara (1997). G. Redgard, a sugarcane cutter, moved about 300 m^3 of boulders on a natural basalt platform with a hired excavator, at about AUD 10,000 of his own money and with no engineering design (Howells & O'Neal, 2024; Raised Water Research, n.d.-b). Surfable days rose from "a handful" to about 30 a year, a researcher's estimate (A. Pitt, as reported by Raised Water Research, n.d.-b), and the site has needed no maintenance in nearly three decades (Howells & O'Neal, 2024). No crest level, depth or plan is known.

3.3 New Zealand

Mount Maunganui Reef (2005-2008). Designed by K. Black and S. Mead as a symmetrical "delta" with a wave-focusing and a wave-breaking zone (Mead & Black, 1999; Mead et al., 2007), it was financed by a community trust at about NZD 1.5 M (SunLive, n.d.). Of 6,500 m^3 designed, 2,800 m^3 were built as 24 containers (Mead & Borrero, 2011, p. 3); containers ruptured during filling and the apex deflated to about -2.0 to -2.4 m CD by 2013 (Dahm & Gibberd, 2013). In October 2006 the right-hander broke "in a hollow, tubing manner", but after the crest slumped by about 1 m it "breaks during only a very confined set of conditions" (Mead & Borrero, 2011, p. 3). The reef was removed in 2014 (SunLive, n.d.).

Opunake Reef (2006-2008, never completed). Financed by the South Taranaki District Council and a local trust (Reid & Sumner, 2021), it was designed for a right-hand wave of 1.8-2.5 m face, to raise surfable days from about one in eight to one in three (Moores, 2006). Only 27 bags were in place in April 2008, costs approached twice the estimate (Mead and Borrero, 2011, p. 4; at least NZD 1.7 M, Appendix C), and the structure broke apart; the designers later called bag failure the "fatal flaw" of the technology (Reid & Sumner, 2021).

3.4 United Kingdom

Boscombe Surf Reef (2008-2009). Bournemouth Borough Council commissioned ASR Ltd, as part of a regeneration scheme, to improve a site with “mediocre” surfing; the reef was designed primarily for surfing (Mead et al., 2010). It cost GBP 3.2 M on a fixed-price contract (Mead & Borrero, 2011; Raised Water Research, n.d.-e) and aimed to double the good surfing days from about 77 to 154 a year (Raised Water Research, n.d.-e). Sand containers (12,800 m^3) were placed with a crest designed at +0.5 m CD to break small waves at low tide; whether it was built higher is unresolved (Rendle, 2016, p. 70; Mead et al., 2010). The independent evaluation found the wave “steep and frequently plunging” and restricted to more advanced surfers (Davidson, 2010, as cited in Rendle, 2016, pp. 70-71; Section 4.3). The containers failed from March-April 2011 and the reef was closed on 31 March 2011 (Rendle & Davidson, 2012).

Borth reef (2011-2012). The reef is part of the Phase 1 coastal defence of Borth, Wales, designed by Royal Haskoning with ASR as sub-consultant, physically modelled by HR Wallingford and funded by a Welsh Government grant and EU convergence funds (Borth Community, n.d.). The whole scheme cost GBP 13 M; about 51,000 t of rock lie in two mounds (30,119 m^3, derived), of which the northern is shaped as a surf reef (Appendix A.9). The crest is 0.19 m above MSL in a 4.3 m spring range, so it dries at low water. The defences withstood the storms of December 2011 and show no damage in imagery of 2012-2024 (Hansford, 2011; Daily Post, 2013; this review), but no surf record exists, and Blacka et al. (2013) rate the surfing objective “unknown”.

3.5 United States, India and Mexico

Pratte’s Reef, El Segundo (2000-2001). The first structure in US waters built for surfing, it mitigated the loss of surf caused by the El Segundo groin (Borrero & Nelsen, 2003) and was financed by Chevron as legal mitigation, with the California Coastal Conservancy and the Surfrider Foundation (Raised Water Research, n.d.-f). It was a V of 110 sand-filled polypropylene bags in Phase I and 90 in Phase II (1,350 m^3), costing USD 550,000 (Appendix C). Fewer than 25 people surfed it in two years, in about 10 sessions, because of its “small size”: 30 m cross-shore against about 150 m at Narrowneck (Borrero & Nelsen, 2003, pp. 13-14). Anecdotal witness accounts (evidence grade D: viewer comments on the YouTube video *The $500k Surf Reef Failure*, Dan Harmon, 2026, retrieved 19 Oct 2026) agree: a worker at the adjacent power plant “never saw” a surfable wave, others describe it as breaking rarely, “really tide sensitive and wave height sensitive” and “barely surfable”, and attribute this partly to shadowing of the southern swells by Catalina Island. The bags disintegrated within about two years, the crest dropped 1.5 m, and the reef was removed in 2008 and 2010 (Blacka et al., 2013; Fontaine, 2008); one commenter on the same video, by contrast, states that the bags he saw removed “were still intact” (Dan Harmon, 2026), so the failure is better documented for the wave than for the fabric.

Kovalam Reef, India (2009-2010). ASR built India’s first multi-purpose reef from 5-6 fanned strips of sand containers (about 4,000 m^3; INR 75 M) for a left-hander of about 100 m, financed by the post-tsunami rehabilitation programme of the Government of India (Mead & Borrero, 2011; Raised Water Research, n.d.-c). The first season was praised as a wave that “could very well be the best wave an artificial reef has ever created” (Collins, 2010), but later “most of the waves just roll over the reef” (Raised Water Research, n.d.-c); the containers were torn, deflated and sunk by 2013 (E. Rendle, pers. comm., 2013, in Blacka et al., 2013, p. 70). The designers’ claim of a beach 10-30 m wider (Mead & Borrero, 2011) has not been independently assessed.

Xala Reef, Jalisco, Mexico (2025-26; unverified). Announced as part of a private resort development and built from quarried boulders (“3,000 truckloads”; Romeyn, 2025; Indelicato, 2026), the reef has no published crest level, volume, cost, surf report or monitoring. It carries no weight in the analysis.

3.6 What the survey shows

Four patterns emerge, to be tested in Sections 4 to 7. First, size separates the outcomes: the three reefs that do not work are 4,900 m^3 or less, whereas the two constructed reefs that work are 25,000 m^3 or more (Burkitts is a reshaped natural platform, not a built volume). Second, six of the seven fabric-filled or inflatable reefs suffered major damage or loss within a few years, whereas none of the four rock reefs failed (Appendix C, Table C4); material is confounded with date and budget. Third, reported surfable days are not comparable: they are a count of days with surfers present (Cables), a design baseline (Boscombe), a share of the year of unstated basis (Albany) or an estimate (Burkitts). Fourth, no reef was built or reported as a beginner wave. Design wave heights and periods are unknown for most reefs, and most performance claims come from the designers (Mead & Borrero, 2011).

3.7 Structures that are not surf reefs

Bar and Drimer (2023, p. 2) summarise the built record in one paragraph and borrow the statement that most projects “have failed” from the Raised Water Research website, without a list or a definition of failure. Of the eleven projects they name, three are among the 13 reefs above. Of the other eight, four are geotextile-tube shore-protection breakwaters (Korea, Mexico, the UAE, Indonesia), one is a nature-based breakwater (New York), two are unbuilt design studies (Portugal, Spain), and one is the Ashkelon North cliff-foot geotube scheme in Israel, where several tubes tore in the first winter (State Comptroller of Israel, 2020). None was built for surfing; they are documented in Appendix D.

4 Reef effect on waves and surf quality

A surf reef must make a swell break at a chosen place, with a chosen breaker shape, and with a break that peels along the reef at a speed surfers can follow. Section 4.1 summarises the theory; Sections 4.2 to 4.6 apply it to the nine reconstructed reefs (Section 2; Appendix A). Outcomes are scored 3 (works), 2 (works rarely or mixed) and 1 (does not work) from the survey (Appendix C).

4.1 Theory: shoaling, refraction, breaking, breaker type and peel

Shoaling. In linear (Airy) theory the wave number k follows from $\omega^2 = gk\tanh(kh)$, and conservation of energy flux, without refraction, friction or reflection, gives the shoaling coefficient

$$K_s = \frac{H}{H_0} = \sqrt{\frac{c_{g0}}{c_g}},$$

with H_0 the deep-water height and c_g the group velocity at depth h (Dean & Dalrymple, 1984/1991). Over a reef the depth falls from toe to crest within tens of metres, so K_s rises quickly at the end of the wave’s path.

Refraction and focusing. By Snell’s law, $\sin\theta/c = \text{const}$, oblique crests turn towards the depth contours, and convex contours make rays converge on a shoal (Dean & Dalrymple, 1984/1991). Mead and Black (2001a) classified the seabed of 34 world-class breaks into ramp, platform, focus, wedge, ledge, ridge and pinnacle components, each of which pre-conditions the swell or sets the break point or breaker type (as summarised by Blacka et al., 2013, Section 2.6); a surf reef is designed to reproduce them.

Breaking criterion. A wave breaks when $H_b = \gamma h_b$. The flat-bed limit $\gamma = 0.78$ (McCowan, 1894, as given by Weggel, 1972) rises on slopes; Weggel (1972) gives γ_b as a function of bed slope and $H_b/(gT^2)$, and Battjes (1974) reports values of about 0.7 to 1.2. The criterion sets *which* waves break on a crest.

Breaker type. The surf-similarity (Iribarren) parameter at the break point,

$$\xi_b = \frac{\tan\beta}{\sqrt{H_b/L_0}}, \qquad L_0 = \frac{gT^2}{2\pi},$$

separates spilling ($\xi_b < 0.4$), plunging ($0.4 < \xi_b < 2.0$) and surging or collapsing breakers ($\xi_b > 2.0$) (Battjes, 1974, after Galvin, 1968). It sets *how* a wave breaks. For a 1 m breaker, $\xi_b = 0.4$ corresponds to a slope of 1:31 at $T = 10$ s and 1:22 at 7 s, and $\xi_b = 2.0$ to 1:6.2 and 1:4.4 (derived in this review).

Breaking intensity. Mead and Black (2001b) related the hollowness of plunging waves, measured by the vortex ratio Y (length to width of the tube), to the seabed gradient along the wave orthogonal, $Y = 0.065\,X + 0.821$ ($R^2 = 0.71$), where X is the gradient "given as the horizontal distance to one vertical unit", i.e. a slope of 1:X (Mead, 2000, Eq. 6.5, p. 115). Intensity is "extreme" for $Y = 1.6$ to 1.9, "very high" for 1.91 to 2.2, "high" for 2.21 to 2.5, "medium/high" for 2.51 to 2.8 and "medium" for 2.81 to 3.1; waves with $Y < 1.6$ are likely to collapse and be unsurfable (Mead, 2000, Table 6.2, p. 121). The fit covers slopes of about 1:9 to 1:40 and scatters widely (Robertson et al., 2014, found $R^2 = 0.27$ on their field data), so it is used as an indicator only.

Peel angle and surfer skill. The peel angle α is the angle between the unbroken crest and the path of the break, which moves along the crest at the peel speed $V_p = c_b/\sin\alpha$, so small angles are fast (Lewis et al., 2015, p. 3). Hutt et al. (2001) gave the smallest peel angle that each surfer rating can ride: 90, 70, 60, 55, 50, 40, 29 and 27 degrees for ratings 1 (beginner) to 8 (professional), with no limit for ratings 9 and 10 (as reproduced by Lewis et al., 2015, Table 3, and Barlow, 2013, Table 1.1, which agree). Walker (1975, p. 5) described the 90-degree wave as suited to "a beginning surfer who has not developed skill levels sufficient to negotiate a turn" and suggested 30 degrees "as a minimum for a plunging breaker at 6-foot height": a hydrodynamic minimum, not a skill class. On a reef arm, Snell's law refracts the angle ϕ_0 between the incident crest and the arm at the toe (depth h_0) to $\sin\alpha_b = \sin\phi_0\, c(h_b)/c(h_0)$ at the break (method of this review).

4.2 Where waves break: crest depth, depth gap and ramp length

Waves of $H_0 = 0.5$ to 3 m and $T = 6$ to 14 s were shoaled along the depth profile of each reef (Section 2.4; for Albany, Cables and Borth a surf cut through the part of the reef where the wave is ridden; line positions in Figure E1 (Appendix E)) at low, mean and high water (normal incidence, $\gamma = 0.78$; sensitivity with $\gamma = 0.55$ and the Weggel index). At MSL a wave needs H_0 of 0.3 to 1.3 m to break on the reef, and above 1.6 to 5.2 m it breaks seaward of it (Table 2; Borth, whose crest dries, excepted). Over all cases, 34 to 84 % break on the ramp or crest.

Hypothesis (f; Appendix F) holds that the depth gap alone does not decide the wave: a long, gradual ramp from 5 m to 2 m can give a good wave, because most waves do not break at 2 m, whereas a sudden rise to 1 to 1.5 m gives a too-hollow wave. Figure 3 tests this on two reefs with comparable depth gaps (5.49 m at Palm Beach, 4.51 m at Boscombe) under the same 1 m, 10 s wave. Both waves break at the same height and depth ($H_b = 1.39$ m, $h_b = 1.78$ m): the crest depth fixes where the wave breaks. But over Palm Beach's 89 m ramp (1:17) the breaker plunges with $\xi_b = 0.56$ and "very high" intensity ($Y = 2.05$), while over Boscombe's 13 m ramp (1:3.1) it collapses ($\xi_b = 3.00$, $Y = 1.05$, extrapolated).

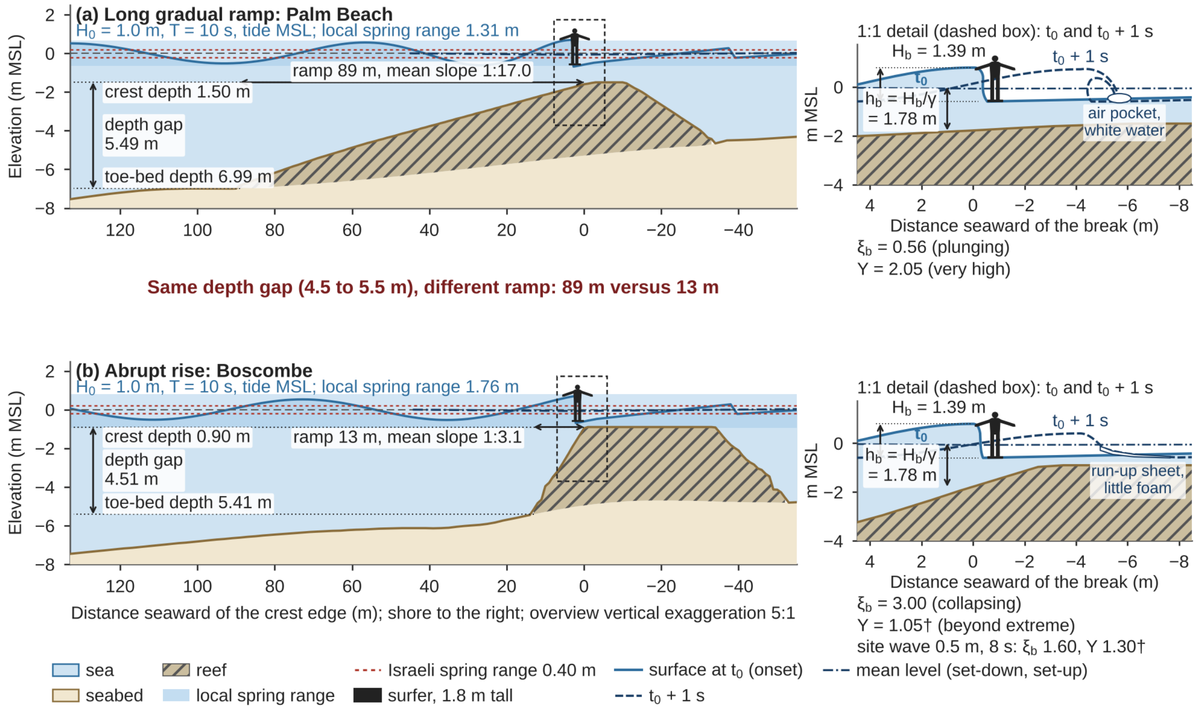

(a) Long gradual ramp: Palm Beach
H_0 = 1.0 m, T = 10 s, tide MSL; local spring range 1.31 m
ramp 89 m, mean slope 1:17.0
crest depth 1.50 m
depth gap
5.49 m
toe-bed depth 6.99 m
Elevation (m MSL)
2
0
−2
−4
−6
−8
120
100
80
60
40
20
0
−20
−40
1:1 detail (dashed box): t_0 and t_0 + 1 s
H_b = 1.39 m
t_0
t_0 + 1 s
$h_b = H_b/\gamma$
= 1.78 m
air pocket,
white water
m MSL
4
2
0
−2
−4
−6
−8
Distance seaward of the break (m)
ξ_b = 0.56 (plunging)
Y = 2.05 (very high)
Same depth gap (4.5 to 5.5 m), different ramp: 89 m versus 13 m
(b) Abrupt rise: Boscombe
H_0 = 1.0 m, T = 10 s, tide MSL; local spring range 1.76 m
crest depth 0.90 m
ramp 13 m, mean slope 1:3.1
depth gap
4.51 m
toe-bed depth 5.41 m
Distance seaward of the crest edge (m); shore to the right; overview vertical exaggeration 5:1
1:1 detail (dashed box): t_0 and t_0 + 1 s
H_b = 1.39 m
t_0
t_0 + 1 s
$h_b = H_b/\gamma$
= 1.78 m
run-up sheet,
little foam
ξ_b = 3.00 (collapsing)
Y = 1.05† (beyond extreme)
site wave 0.5 m, 8 s: ξ_b 1.60, Y 1.30†
sea
seabed
reef
local spring range
Israeli spring range 0.40 m
surfer, 1.8 m tall
surface at t_0 (onset)
t_0 + 1 s
mean level (set-down, set-up)

Figure 3. Shoaling and breaking over a long gradual ramp (a, Palm Beach, Gold Coast) and over an abrupt rise (b, Boscombe), both along the depth profile of the reef's 3D model, drawn at one horizontal scale and with the same incident wave ($H_0 = 1.0$ m, $T = 10$ s, normal incidence, mean sea level). The overview panels are vertically exaggerated 5:1 (linear shoaling, $H = H_0 K_s$, Dean and Dalrymple, 1984/1991; the wave breaks where $H = \gamma d$ with $\gamma = 0.78$, McCowan, 1894, as given by Weggel, 1972), and the dashed boxes are enlarged 1:1 to show the breaker at two instants, onset (t_0, filled) and about 1 s later ($t_0 + 1$ s, dark dashed; the crest has moved $c_b \times 1 \text{ s} = \sqrt{g h_b} \times 1 \text{ s} \approx 4.2$ m shoreward), with the breaking depth $h_b = H_b/\gamma = 1.78$ m ($H_b = 1.39$ m in both panels) and a 1.8 m standing surfer. In (a) the crest sharpens and the front turns near-vertical at t_0, and 1 s later a thin lip (about 0.15 H_b) has been thrown onto the trough ahead, enclosing an air pocket (plunging; Galvin, 1968; Peregrine, 1983); in (b) the steep front does not overturn, and 1 s later it collapses from its base, which surges ahead over the crest as a thin run-up sheet with little foam (collapsing or surging; Galvin, 1968; Battjes, 1974). Landward of the break the surface drops to a trough about 0.38 H_b below the local mean level and then rises over the back of the previous, already broken wave to its bore front one wavelength ($L_b = T\sqrt{g h_b} \approx 42$ m) ahead; the mean water level (dash-dot) dips by about 0.05 H_b at the break (set-down) and rises toward the shore (set-up, about 0.25 H_b at the shoreline, an order-of-magnitude value) after Longuet-Higgins and Stewart (1964) and Bowen et al. (1968); the trough shape follows Dean and Dalrymple (1984/1991). Lip, foam and bore shapes are schematic; heights and positions of the break are model values. A comparable depth gap from the crest to the seabed at the seaward toe (5.49 m and 4.51 m) spread over an 89 m ramp (mean slope 1:17) gives a plunging breaker with surf similarity $\xi_b = 0.56$ (Battjes, 1974) and a "very high" breaking intensity, $Y = 2.05$ (Mead & Black, 2001b), whereas the 13 m ramp (1:3.1) gives a collapsing front with $\xi_b = 3.00$ and $Y = 1.05$†; under Boscombe's own typical sea (0.5 m, 8 s) $\xi_b = 1.60$ and $Y = 1.30$†. † Outside the 1:9 to 1:40 slope range of the Mead and Black (2001b) fit, i.e. extrapolated. Blue band: local spring tidal range (MLWS to MHWS); dashed red lines: the Israeli spring range of 0.40 m (Rosen, 2001).

Across the reefs (Figure 4; cuts in Figure E1 (Appendix E)), the depth gap ranks with the outcome (Spearman $\rho = 0.93$, permutation $p = 0.005$, $n = 8$), as do the ramp length ($\rho = 0.93$, $p = 0.005$) and the mean seaward slope ($\rho = 0.77$, $p = 0.039$), but not the crest depth ($\rho = 0.39$, $p = 0.36$). The same 2 to 4 m gap gives a surfable wave over a long ramp and a collapsing wave over an abrupt rise. Albany, which works, plots beside Palm Beach, the other reef that works: on its surf cut along the long -2 m ridge the ramp is 99 m at a mean 1:19.8 ($Y = 2.11$ for the mean face, “very high”). On the former automatic cut through its -1 m take-off plateau (22 m, 1:4.0; hollow symbol in Figure 4) it sat among the collapsing reefs; that plateau is the short focusing peak where the surfer takes off, not the surfed face, which shows how much the result depends on cutting the profile where the wave is actually ridden. Three caveats apply to Albany: its 3D surface is fitted to a design rule (low confidence; Appendix A.12); the ridge is nearly flat (-2.0 to -2.4 m AHD) for about 80 m, so the mean slope hides a steep outer step (up to 1:3.5) near the toe; and the swell probably reaches the reef obliquely after refraction in King George Sound, which a shore-normal one-dimensional cut ignores. The ramp is a recognised seabed component (Mead and Black, 2001a, as summarised by Blacka et al., 2013), but “there are no models available off the shelf which directly assess surfability” (Blacka et al., 2013, p. 56), and no source found quantifies the contrast between a gradual ramp and a sudden rise; the quantification is new to this review.

4.3 Breaker type, hollowness and the Boscombe case

Over all cases that break on the reef (Table 2), Palm Beach gives a median ξ_b of 0.49 and Y of 2.01 with no case below $Y = 1.6$, whereas Bunbury (1:2.7) and Boscombe (1:3) give $Y < 1.6$ in every case and Mount Maunganui (1:9) in 91 %. The median ξ_b ranks with the outcome at $\rho = -0.77$ ($p = 0.039$): the reefs that work break more gently. Period matters as well as slope: the surging share rises from 0.21 at 6 s to 0.41 at 10 to 12 s, so the same reef gives gentler waves in short-period seas such as Israel’s. Hollowness alone is not failure: Mount Maunganui first “broke as designed in a hollow, tubing manner” (Mead & Borrero, 2011, p. 3), and Albany, which plunges on its ridge (median ξ_b 0.56, median Y 1.75, “extreme”, with no surging case), works but is “intermediate-to-advanced on most days” (Mercer, 2025).

Boscombe tests hypothesis (d; Appendix F), that its structure rose too steeply and made a wave mostly for bodyboarders. The computed 1:3 flank collapses in every case, Boscombe’s typical sea (0.5 m, 8 s) gives $\xi_b = 1.60$ and $Y = 1.30$, and even the 2011 survey of the damaged reef (mean ramp 1:12) gives $Y < 1.6$ in 85 % of cases. The independent audit found the wave “steep and frequently plunging”, so that “novice surfers and intermediates would not be able to surf the Boscombe Reef” (Davidson, 2010, as cited in Rendle, 2016, pp. 70-71); it required Hutt level 7 or 8 against a design level of 5 or 6 (Rendle, 2016, p. 205; the designers give 3 to 6, Mead et al., 2010). Bodyboarders were more satisfied than stand-up surfers, who called it “a bodyboarders wave” (Rendle, 2016, pp. 178, 204–205), and the designers note that outside the design swell the waves were “frequently more suitable to body-boarding due to the intense plunging nature of the wave breaking” (Mead et al., 2010, p. 5); yet advanced stand-up surfers rode it (Davidson, 2010, as cited in Rendle, 2016, p. 70). The evidence supports “too steep” and “mostly bodyboarders”. The crest was set shallow so that the common 0.5 m waves would break at low tide (Rendle, 2016, p. 71), exactly the condition in which a short ramp collapses the wave, and, exposed at low tide and too deep at high tide, it left a mid-tide window of “1 to 2 hours” (Rendle, 2016, p. 71).

The depth shoreward of the break matters as well as the ramp seaward of it. A breaking wave loses energy until its height falls to a stable fraction of the local depth, about $H = 0.4h$, and then stops breaking; where the water deepens again after the break, as over the trough behind a bar, the wave drops below that ratio, stops breaking and re-forms as an unbroken wave (Dally et al., 1985). Dally et al. (1985) model this as a decay of the energy flux towards that of the stable wave at a rate inversely proportional to the depth (decay coefficient $K = 0.15$,

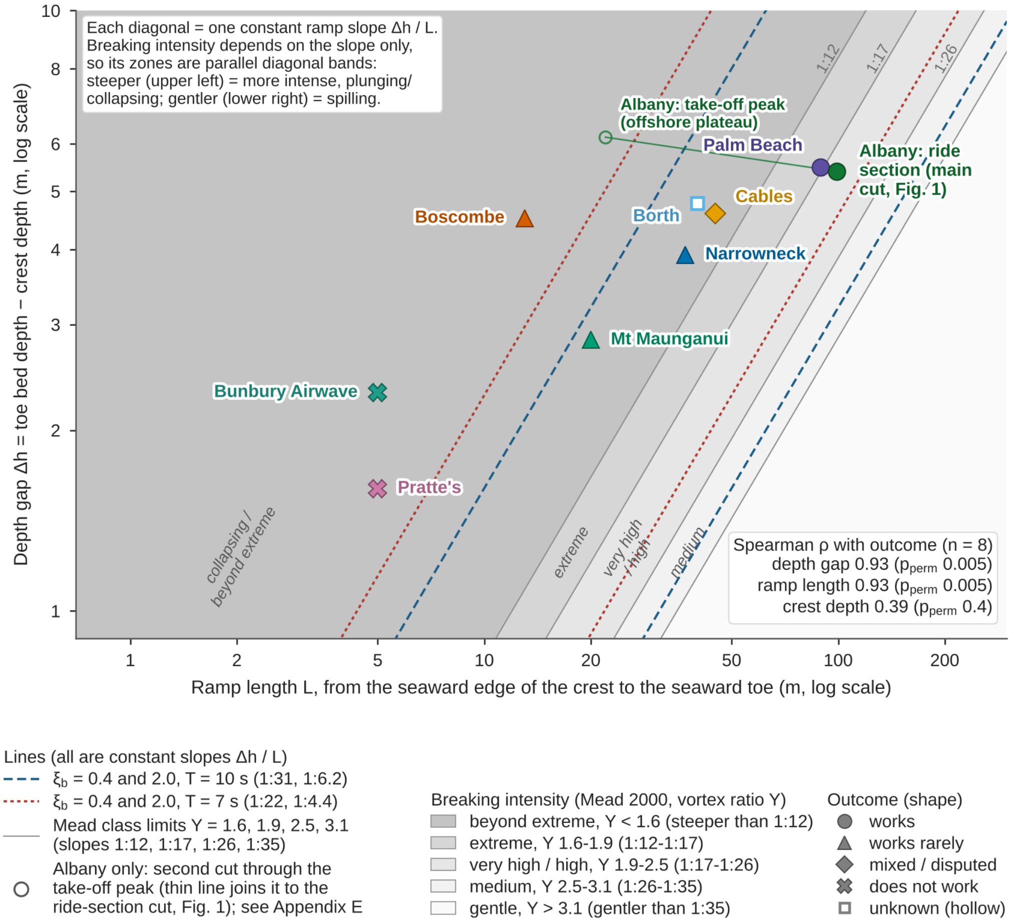


Figure 4. Depth gap Δh (toe bed depth minus crest depth) against ramp length L (from the seaward edge of the crest to the seaward toe) for the nine modelled reefs, read on the surf cut of each reef (Figure 1; line positions in Figure E1 (Appendix E)), on logarithmic axes. On these axes every line of constant mean gradient $\Delta h/L = 1/X$ is a straight diagonal of the same direction, so the breaker-type zones, which depend on the gradient only, appear as parallel diagonal bands: steeper (upper left) means a more intense, plunging to collapsing break, gentler (lower right) a spilling break. The grey bands are the breaking-intensity classes of Mead and Black (2001b), from the vortex ratio $Y = 0.065X + 0.821$ (Mead, 2000, Eq. 6.5; class limits $Y = 1.6$, 1.9, 2.5 and 3.1 at $X = 12.0$, 16.6, 25.8 and 35.1, i.e. slopes 1:12 to 1:35); $Y < 1.6$ is "beyond extreme", likely to collapse and unsurfable (Mead, 2000, p. 121). The dashed and dotted lines are the Iribarren limits $\xi_b = 0.4$ (spilling/plunging) and 2.0 (plunging/surging) (Battjes, 1974; Lewis et al., 2015) for a breaking height of 1 m with $T = 10$ s and with $T = 7$ s, representative of Israeli seas. Colour and label identify the reef; symbol shape gives the outcome (circle works, triangle works rarely, diamond mixed or disputed, cross does not work, small hollow square unknown). Each reef is plotted once, for its surf cut, the profile line of Figure 1; where the lines and arms of every reef run is shown in Figure E1 (Appendix E). Only Albany carries a second, hollow symbol, joined to the main one by a thin line, because its two cuts have a different physical meaning: the main cut (filled circle: 99 m, 5.40 m, very high intensity) runs along the long −2 m ridge where the left-hander is ridden (ride section), and the second cut (hollow: 22 m, 6.15 m, in the collapsing band) crosses the short, shallow −1 m offshore plateau where the surfer takes off, which is steep because it is the focusing peak of the reef and not the surfed face. The gradient $\Delta h/L$ includes the natural bed slope between crest and toe, so it is steeper than the reef-face slope at some reefs (Cables 1:9.8 here against 1:23.7 on the face); geometry is idealised for six of the nine reefs. Spearman correlation with the ordinal outcome ($n = 8$, Borth unknown): depth gap $\rho = 0.93$ (permutation $p = 0.005$), ramp length 0.93 (0.005), crest depth on the cut 0.39 (0.36).

stable ratio $\Gamma = 0.4$), solved for a flat shelf and a plane slope and run on a profile with two bars and troughs, where it reproduces the shoaling, breaking and re-forming seen in nature. Applied from the break point shoreward (breaking heights $H_b = 1$, 1.5 and 2 m at low, mean and high water; Appendix A), the model gives the length over which the reef keeps the wave breaking before the water behind the break deepens or the reef ends (Table 3). The result depends on the line, so it is read on five lines per reef (Figure E2, Appendix E): the Figure 1 cut; three shore-normal lines at 25, 50 and 75 % of the reef's alongshore extent, counted from its left end as seen looking seaward; and the longest crest line, the longest path along the shallow ridge (reef within 1 m of its shallowest point) from its most seaward point, where the wave first finds the reef. The two kinds of line measure different things. On a shore-normal line the broken wave travels shoreward over the reef, and the length says how long the reef keeps the broken wave breaking. On a peeling reef the wave also breaks progressively along the arm: each point of the crest line breaks when the still-unbroken part of the wave reaches it, so along the crest line the relevant length is that of the stretch where the arm keeps finding breaking depth ($h \leq H_b/0.78$), the peel length that a surfer rides, not a decay distance. On the shore-normal lines the lengths are short and similar within each reef: the median over the three lines is 35 m at Albany, 23 to 25 m at Palm Beach, Narrowneck and Boscombe, 14 and 17.5 m at Cables and Mount Maunganui, and 5 and 7 m at Pratte's and Bunbury, the two reefs that do not work. At Pratte's even a 2 m wave that breaks on the crest has only 4 to 6 m of reef left inside the break and goes on breaking, if at all, on the natural bed. Albany's Figure 1 cut, through the middle of its ridge, gives 90 m, because the ridge stays nearly flat at −2.0 to −2.4 m AHD for about 80 m inside the steep outer face; its 25 % line gives 29 m. Over the eight reefs with a known outcome the shore-normal median ranks with the outcome at $\rho = 0.77$ (permutation $p = 0.037$), the same value as on the Figure 1 cut and a weaker link than that of the ramp length; this shore-normal metric is the one used in Sections 8 and 9. Along the crest lines the picture changes where the ridge runs obliquely to the shore. Palm Beach gives only 19 to 28 m on any shore-normal line, because its crest is a mound with a trough about 4.5 m deep behind it, but its 92 m crest line keeps the wave peeling for a median of 84 m: the 100 m or so of shallow reef seen in Figure 1 lies along the ride, not across it. Albany's 155 m crest line, from the take-off plateau along the −2 m ridge, peels over its whole length for 1.5 and 2 m waves at low water but over only 8 to 13 m for 1 m waves at mean and high water, when the ridge is too deep for them to break (median 45 m). Boscombe (98 m) and Narrowneck (77 m), which work rarely, also have long shallow ridges, whereas Mount Maunganui (16 m), Pratte's (11.5 m) and Bunbury (9 m) stay short. The crest-line peel length therefore ranks with the outcome at only $\rho = 0.62$ ($p = 0.15$; 0.69, $p = 0.08$, with ridge bands of 0.5 or 1.5 m): a long shallow arm is needed for a long ride but does not by itself make a good wave, since Boscombe's arm sits on a collapsing 1:3 flank (above). The cruder test that holds the height at its breaking value (breaking stops where $h > H_b/0.4$) separates nothing for 2 m waves, since the water behind every reef is shallower than 5 m; the decay model is what distinguishes the reefs. Surfers who rode Pratte's describe exactly this: the wave "came from deep water, hit the shallow reef quickly, then instantly went into deep water again. The wave would suck up fast, but before it could actually barrel, it would mush out and disappear", and "the reef was shaped like a V instead of a triangle" (viewer comments, grade D; Dan Harmon, 2026). Two narrow arms of a V leave deep water between and behind them, whereas a filled triangular plan that stays shallow toward the shore keeps the bed shallow under the breaking wave along the ride. The computation is one-dimensional and idealised: normal incidence, no refraction, set-up or friction, a stable ratio fixed at 0.4 although it varies with the bed slope, idealised model profiles for Pratte's and Bunbury (Section 2.4), and no account of the depth lost to the sides of a V or a narrow arm, which a single line cannot see; along the crest line the wave height is taken as uniform along the arm, and the crest-line rule is a geometric judgement (Appendix E).

4.4 Peel angle and surfer skill

The designed arm angles of 45 to 81 degrees refract to 16 to 62 degrees at the break (Figure 5; arm and cut positions in Figure E1 (Appendix E)). Small waves break in shallow water, where refraction is strongest: at $H_0 = 0.5$ m every reef reaches α_b below 27 to 29 degrees (Hutt ratings 8 to 9), and only large waves keep α_b above 50 degrees. Peel speeds are 6.9 to 11.4 m/s. Only one case reaches the 60 to 90 degrees of ratings 1 to 3: Borth at $H_0 = 3$ m (61.5 degrees, rating 3), a large wave at the upper end of Borth's window that breaks at 3.2 m, above the 2.5 m height limit of that rating, on a reef with no surf record; the incidence angle ϕ_0 is assumed for five of the eight reefs (Appendix A). The observations agree: no reef was built or reported as a beginner wave (Albany "intermediate-to-advanced", Mercer, 2025; Boscombe "an intermediate to expert break", Rendle, 2016, p. 273). The literature defines the beginner's wave by a large peel angle rather than by a spilling breaker; both are missing here, since spilling breakers are 0 to 29 % of on-reef cases, and a spilling ramp of about 1:30 needs 35,000 to 49,000 m^3 per 100 m in Israeli conditions (Section 7).

4.5 Reef size and volume

Hypothesis (a; Appendix F) holds that reefs need a large volume and that small reefs such as Pratte's do not work. Volume ranks with outcome (Figure 6): $\rho = 0.72$ ($p = 0.021$, $n = 10$) for the stated volume of the built reefs, 0.93 ($p = 0.005$, $n = 8$) for the model volume and 0.87 ($p = 0.003$, $n = 10$) when Narrowneck is counted by its exposed mound and Opunake by its built volume (Table 1, notes). All three reefs that do not work are 4,900 m^3 or smaller and both built reefs that work are 25,000 m^3 or larger, because volume buys ramp length (ramp length $\rho = 0.93$ against relief 0.62, $p = 0.15$): a 1:16 to 1:20 ramp on a 4 to 5 m bed needs about 14,000 to 25,000 m^3 per 100 m, depending on the crest depth (Section 7). Burkitts, a natural reef reshaped by moving about 300 m^3 of rock, works, so volume matters when the reef must create the whole relief; and size is collinear with material and budget, so the association is descriptive. The direction is in the literature: "the single largest limitation to building a successful artificial reef in terms of surfing is the issue of scale" (Shand, 2011, as cited in Mead and Borrero, 2011, p. 2), and Pratte's failure was attributed to its "small size" (Borrero & Nelsen, 2003, pp. 13-14). Volume is not sufficient: Narrowneck, the largest reef by its sand fill (70,000 m^3; the exposed mound is about 13,500 m^3), works only rarely (Mead & Borrero, 2011, p. 2).

4.6 Surfable window and tide

Reef waves work within narrow windows of wave height and water level. Pratte's broke "1-5 times per year" (Borrero and Nelsen, 2003, as cited in Rendle, 2016, p. 39), which local witnesses confirm anecdotally: it "never produced a surfable wave" in one account and was "really tide sensitive and wave height sensitive" in another (viewer comments, grade D; Dan Harmon, 2026, retrieved 19 Oct 2026); and Palm Beach works for easterly or southerly swells above H_s 0.8 m with peak periods above 7 s (Prenzler et al., 2023); the designers attribute the narrow windows to small volume (Mead & Borrero, 2011, pp. 2, 6). This supports the first part of hypothesis (d), that reef waves are active on few days.

The computation steps the water level through the tidal range and counts a level as surfable when the wave breaks on the reef with $\xi_b < 2$ and $Y \geq 1.6$, weighted by a synthetic spring-neap

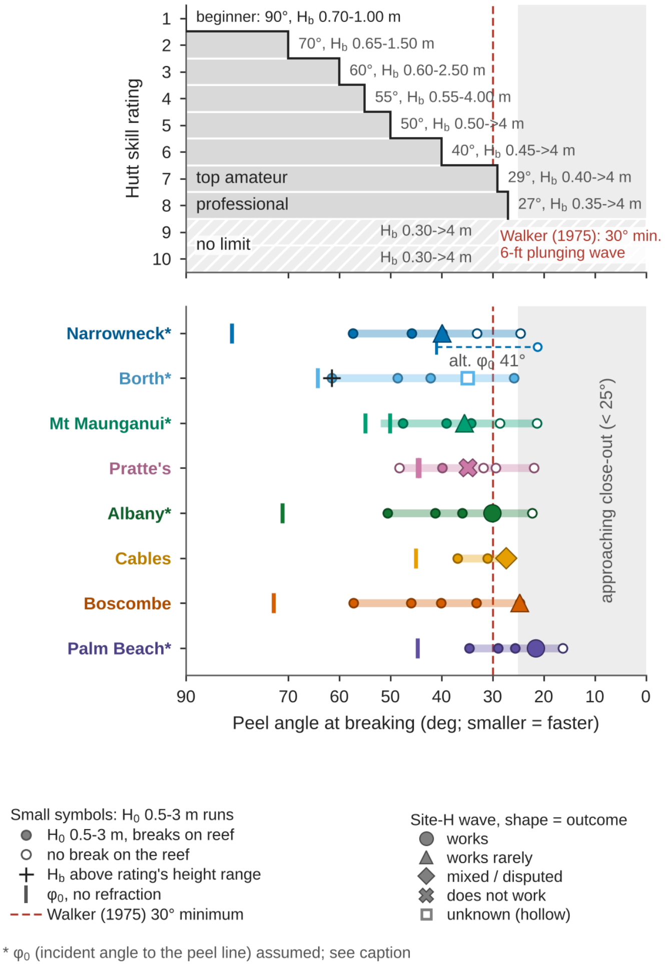


Figure 5. Peel angle at breaking and the surfer skill that can use it. Upper panel: the surfer-skill table of Hutt et al. (2001, Table 3, as reproduced by Lewis et al., 2015, Table 3, and Barlow, 2013, Table 1.1): a surfer of rating r can ride waves peeling at an angle of at least 90, 70, 60, 55, 50, 40, 29 and 27 degrees (ratings 1 to 8; ratings 9 and 10 have no limit), within the breaking-wave height range of the rating printed beside each step (units not printed in the reproduction, taken as metres). Lower panel: for each reef the bar spans the peel angle at the site period for incident heights $H_0 = 0.5$ to 3 m (dots; open dots: the wave does not break on the reef), obtained from the angle ϕ_0 between the incident crest and the reef arm and refracted to the break depth by Snell's law without diffraction (vertical ticks: ϕ_0, unrefracted; arm positions in Figure E1 (Appendix E)); the large symbol is the site-wave value, its shape the outcome (circle works, triangle works rarely, diamond mixed or disputed, cross does not work, hollow square unknown) and its colour and the label colour the reef; + marks a case whose breaking height lies above the Hutt height range of its rating (Borth at $H_0 = 3$ m reaches 62 degrees but breaks at 3.2 m, above the 2.5 m limit of rating 3). The angle ϕ_0 is assumed for five of the eight reefs (*; dashed: the alternative of 41 degrees for Narrowneck); the grey zone below 25 degrees approaches close-out (Mead and Black, 2001b, as cited by Lewis et al., 2015); the red dashed line is Walker's (1975, p. 5) 30-degree minimum peel angle for a 6-ft plunging wave, a limit that rises with wave height, close to the rating-7 limit (29 degrees) of Hutt et al. (2001).

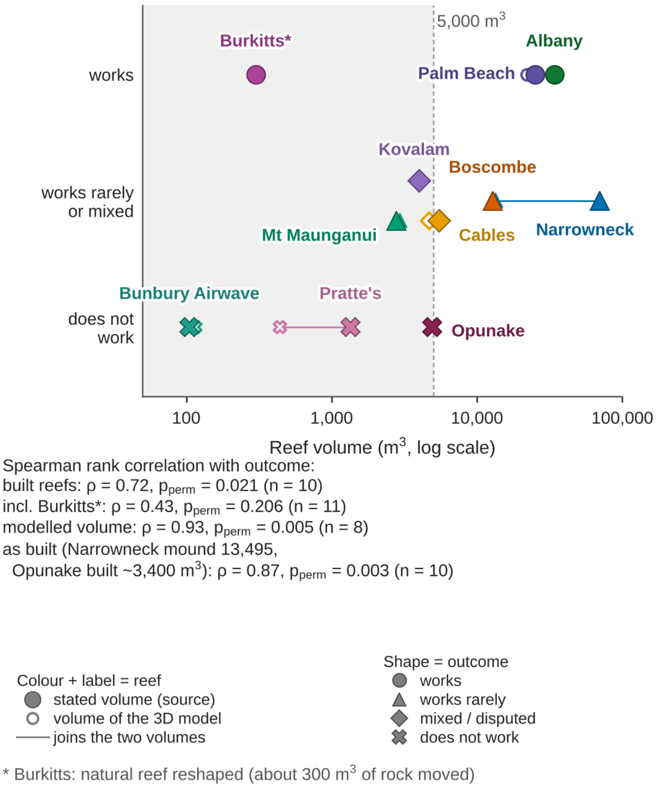


Figure 6. Reef volume against outcome for the eleven projects with a known outcome (filled symbols: volume stated in the sources; open symbols joined by a line: volume of the reconstructed 3D model, which differs most at Narrowneck, 70,000 against 13,496 m³). Colour and label identify the reef, symbol shape the outcome (circle works, triangle works rarely, diamond mixed or disputed, cross does not work), repeating the vertical axis. The three reefs that do not work are 4,900 m³ or smaller (Bunbury 105, Pratte's 1,350, Opunake 4,900 m³) and both built reefs that work (Palm Beach 25,000 and Albany 34,000 m³) are 25,000 m³ or larger; Burkitts (*) works at 300 m³ because it is a natural reef reshaped by moving about 300 m³ of rock, so volume matters when the reef has to create the whole relief. The Spearman rank correlation with the ordinal outcome (works 3, works rarely or mixed 2, does not work 1) is $\rho = 0.72$ (permutation $p = 0.021$, $n = 10$) for the stated volume of the built reefs, 0.43 ($p = 0.21$, $n = 11$) when Burkitts is included and 0.93 ($p = 0.005$, $n = 8$) for the model volume, and 0.87 (p = 0.003, n = 10) when the exposed mound of Narrowneck (13,495 m³) and the built volume of Opunake (about 3,400 m³) replace the sand fill and the design volume (Table 1, notes); the sample is small, the outcome is a judgement and volume is collinear with material (the large reefs are recent rock reefs) and budget, so the association is descriptive only (Blacka et al., 2013; Mead and Borrero, 2011; Corbett et al., 2023; sources in Appendix C).

tide (Appendix A). Given a suitable swell, Palm Beach, Albany and Cables are surfable 0.55 to 0.70 of the time (0.70, 0.62 and 0.55), Borth 0.19, Mount Maunganui and Narrowneck 0.15 to 0.16, Pratte's 0.04, Bunbury 0.01 and Boscombe never (Table 2). Under the Israeli tide (spring range 0.40 m; Rosen, 2001) the average does not rise (mean change +0.01; 66 % of cases change by less than 0.05), but the outcome becomes a function of wave height alone: for a given swell the reef works all day or not at all (Figure 7). The Palm Beach geometry under a 1 m wave rises from 0.69 to 1.00, while Mount Maunganui, which breaks only at low water, falls from 0.41 to 0. A synthetic Haifa-type reef keeps 0.87 to 0.98 of its no-tide surfable time at a 0.2 to 0.4 m range, against 0.64 to 0.68 at 4.3 m. The tidal control is documented: before its renewal Narrowneck broke regularly above an RMS height of about 0.8 m at low tide and 1.2 m at high tide (Corbett et al., 2023, p. 668), and the Albany design relied on a "low tidal range" (City of Albany, 2023, p. 2). Without a wave climate, the fractions are shares of the time with a suitable swell, not days per year.

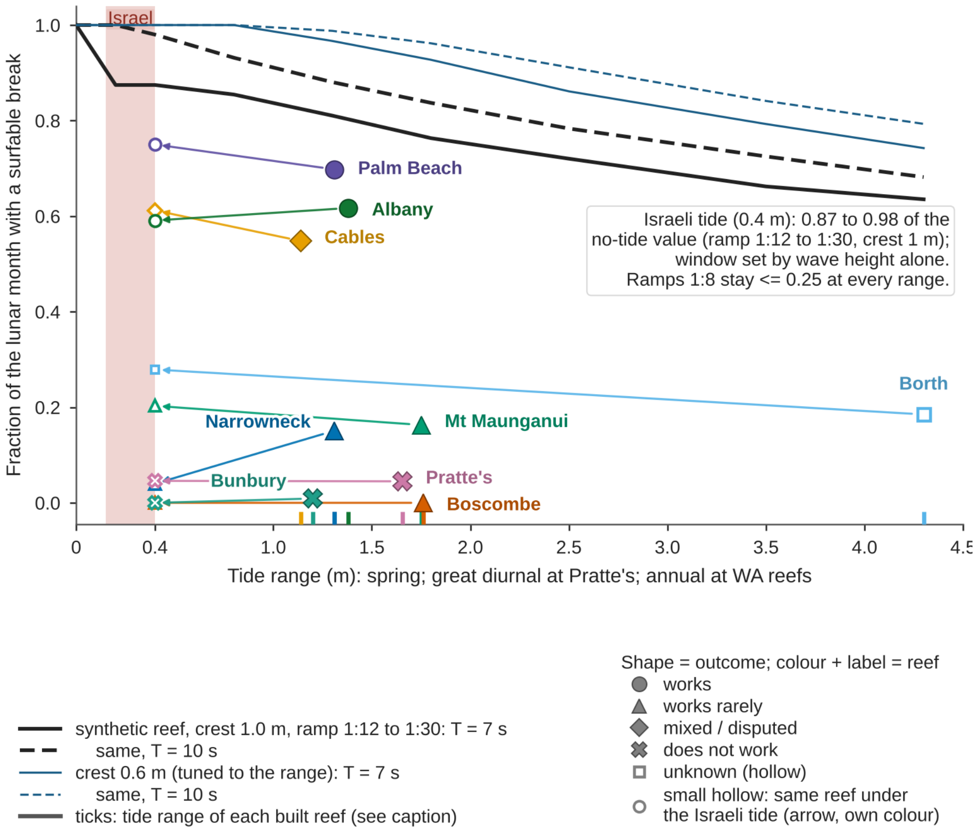


Figure 7. Fraction of the lunar month during which a reef gives a surfable break, given that the swell is present, as a function of the tide range (spring range; great diurnal range at Pratte's; predicted annual range of 2026 at the Western Australian reefs Cables, Bunbury and Albany). Lines: a synthetic Haifa-type reef (1:50 bed, crest 10 m wide; ramps from 1:12 to 1:30 give identical fractions), mean over incident heights $H_0 = 0.6$ to 1.5 m, for a crest 1.0 m below MSL (black) and for the crest depth that is best at each range (0.6 m, blue), at $T = 7$ s (solid) and 10 s (dashed); a break counts as surfable when it occurs on the ramp or crest with $\xi_b < 2$ and $Y \geq 1.6$ (Battjes, 1974; Mead & Black, 2001b), on a synthetic semidiurnal tide modulated between neap and spring. Symbols: the nine modelled reefs, each at its own tide range (ticks on the lower axis) and surfable fraction (mean over $H_0 = 0.5$ to 2 m, best default section; colour and label: reef; shape: outcome, circle works, triangle works rarely, diamond mixed or disputed, cross does not work, hollow square unknown), joined by an arrow to the same geometry and swell under the Israeli tide (spring 0.40 m, neap 0.15 m; Rosen, 2001; shaded band). The Israeli tide leaves 0.87 ($T = 7$ s) to 0.98 ($T = 10$ s) of the no-tide value for the synthetic reef but does not raise the mean surfable time of the built reefs (mean change +0.01); it makes the break a function of wave height alone. The fractions are shares of the time with suitable swell, not days per year, because no wave climate is included.

4.7 Synthesis

The crest depth decides which waves break and the length and gradient of the ramp how they break (ramp length $\rho = 0.93$, mean slope 0.77, median ξ_b −0.77; crest depth 0.39, not significant); every reef refracts its waves to peel angles for advanced surfers; and a small tide makes the surfable time depend on wave height alone (Table 2). The model is one-dimensional with linear shoaling, ignores refraction and focusing across the reef (except for the peel angle) and rests on idealised geometry for six of the nine reefs, so its classes are indicators, not predictions of ride quality.

Reef (outcome)	Crest depth (m)	Depth gap (m) / ramp (m), slope	H_0 window (m)	ξ_b	Y ($Y < 1.6$)	α_b (deg) / rating	Surfable, local / Israeli	Volume (m³)
Palm Beach (works)	1.50	5.49 / 89, 1:17	0.81-5.16	0.49	2.01 (0.00)	16-35 / 7-9	0.70 / 0.75	25,000 / 21,948
Albany (works)	1.97	5.40 / 99, 1:19.8 (take-off line 6.15 / 22, 1:4.0)	1.00-4.92	0.56	1.75 (0.29)	22-51 / 5-9	0.62 / 0.59	34,000 / 33,999
Cables (mixed)	1.92	4.59 / 45, 1:24	1.01-3.49	0.61	1.69 (0.39)	27-37 / 7-8	0.55 / 0.61	5,500 / 4,683
Narrowneck (rarely)	1.92	2.08 / 16, 1:15 (second line 3.92 / 37)	1.10-4.15	0.97	1.36 (0.72)	25-57 / 4-9	0.15 / 0.04	70,000 / 13,496
Mt Maunganui (rarely)	2.08	2.83 / 20, 1:9.3	1.16-3.04	1.62	1.14 (0.91)	21-52 / 5-9	0.16 / 0.20	2,800 / 2,943
Boscombe (rarely)	0.90	4.51 / 13, 1:3.1	0.48-4.09	2.00	1.06 (1.00)	25-57 / 4-9	0 / 0	12,800 / 13,112
Pratte's (does not work)	1.86	1.60 / 5, 1:4.1 (centreline)	1.31-2.28	1.13	1.37 (0.63)	22-48 / 6-9	0.04 / 0.05	1,350 / 442
Bunbury (does not work)	0.68	2.31 / 5, 1:2.7	0.28-1.62	2.97	1.05 (1.00)	dome, not computed	0.01 / 0	105 / 116
Borth (unknown)	crest 0.47 above MSL	4.77 / 40, 1:9.5	0.05-3.01	1.11	1.35 (0.84)	26-62 / 3-9	0.19 / 0.28	30,119 / 19,954

Table 2. Computed wave indicators per reef (derived in this review; Appendix A). Depth profile at MSL unless noted: the surf cut for Albany, Cables and Borth, the crest line for the other reefs (Figure E1 (Appendix E)). H_0 window: smallest H_0 that breaks on the reef and largest that does not break seaward of it (site period, MSL). ξ_b and Y: medians over all cases that break on the reef (H_0 0.5-3 m, T 6-14 s, three tide levels), in brackets the share with $Y < 1.6$. α_b: refracted peel angle for H_0 0.5-3 m and the Hutt rating range. Surfable: fraction of time with a suitable swell under the local and the Israeli tide (at Cables, Bunbury and Albany the local tide is the 2026 predicted annual range, a wider band than the spring range). Volume: stated / model (m³).

Notes: Pratte's crest line gives a 1 m ramp (a tip artefact); its centreline is shown. Narrowneck's stated volume is "some 70,000 m^3" (Mead & Borrero, 2011, p. 2); the model covers the visible container field only. Albany, Cables and Borth values are read on the surf cut (Section 2.4), so their crest depths differ from Table 1 (Albany 1.00 m at the take-off, Cables 1.55 m surveyed). Borth values come from the analysis model (surf cut through the main ridge of the northern reef, crest 0.47 m above MSL; model volume of the 7,435 m^2 analysis footprint), whereas Table 1 gives the reef crest (0.19 m above MSL) and the volume of the 2022 laser-survey edge (30,119 m^3, 8,675 m^2); the outlines differ (Appendix C, note 3). Y on slopes steeper than 1:9 is extrapolated from the Mead and Black (2001b) fit (share of cases outside the fit range: 0 at Palm Beach, 0.32 at Cables, 0.40 at Albany, 0.58 to 1.00 at the other reefs).

Reef (outcome)	Figure 1 cut (m)	25 % line (m)	50 % line (m)	75 % line (m)	Shore-normal, three lines (m)	Longest crest line: length / peel median (max) (m)
Palm Beach (works)	27 (6)	19 (6)	26.5 (6)	28 (1)	23 (13)	92 / 83.5 (92)
Albany (works)	90 (4)	29 (6)	98 (3)	– (0)	35 (9)	155 / 45 (155)
Cables (mixed)	21 (5)	10 (6)	23 (5)	9 (6)	14 (17)	87 / 46 (87)
Narrowneck (rarely)	57 (3)	38 (5)	– (0)	6 (1)	25 (6)	80 / 77 (80)
Mt Maunganui (rarely)	18.5 (4)	20.5 (4)	7 (2)	17.5 (4)	17.5 (10)	48 / 16 (23)
Boscombe (rarely)	34.5 (8)	18 (8)	37.5 (8)	33 (8)	24 (24)	106 / 97.5 (106)
Pratte's (does not work)	9 (3)	4 (3)	9 (3)	4 (3)	5 (9)	42 / 11.5 (42)
Bunbury (does not work)	7.5 (8)	7 (7)	7.5 (8)	7 (7)	7 (22)	9 / 9 (9)
Borth (unknown)	17 (7)	21 (6)	17 (7)	25 (7)	18 (20)	109 / 55 (109)

Table 3. Sustained breaking after the break point on five lines per reef (derived in this review; Section 4.3, Appendix A, Figure E2). A wave of breaking height $H_b = 1$, 1.5 or 2 m breaks at the first point from offshore where $h \leq H_b/0.78$, at low, mean and high water (9 cases per line). Shore-normal lines (Figure 1 cut, as in Table 2: Narrowneck second crest line, Pratte's centreline; and the lines at 25, 50 and 75 % of the alongshore extent of the reef from its left end, looking seaward): distance from the break shoreward until breaking stops in the decay model of Dally et al. (1985) ($K = 0.15$, $\Gamma = 0.4$) or the reef ends, whichever comes first. Longest crest line: length along the line over which the arm keeps $0.05 < h \leq H_b/0.78$ from the first breaking point (peel length). Values are medians over the cases that break on the reef, with their number in brackets; cases that break seaward of the reef or not at all are left out. One-dimensional and idealised (Section 4.3).

Notes: Spearman ρ of the medians against the outcome as ranked for Figure 4 (works 3, works rarely or mixed 2, does not work 1; Borth excluded; $n = 8$, permutation p): Figure 1 cut 0.77 ($p = 0.037$); shore-normal, three lines 0.77 ($p = 0.037$; the metric used in the text); 25 % line 0.69 ($p = 0.073$); 50 % and 75 % lines 0.66 and 0.68 ($n = 7$, $p = 0.16$ and 0.10); crest-line peel length 0.62 ($p = 0.15$; ridge band 0.5 or 1.5 m instead of 1 m: 0.69, $p = 0.08$); crest-line length itself 0.85 ($p = 0.015$), which largely restates reef size (Section 4.5). "–": no case breaks on the reef on that line (Narrowneck 50 %: the line crosses only a deep fringe of the container field, 4.8 m below low water; Albany 75 %: the reef on that line lies more than 3 m below MSL, and the waves break on the natural bed before they reach it). Pratte's and Bunbury: the breaking length on the shore-normal lines exceeds the reef left inside the break, and the rest is on the natural bed. Borth: its crest line dries at low and mean water. Method and results: Appendix A (item 4) and Figure E2.

5 Sediment transport and coastal stabilisation

Most surf reefs were promoted as multi-purpose structures that would also stabilise the beach. This section compares the standard shoreline predictors with the surveyed response and asks how a surf reef can be combined with coastal stabilisation.

5.1 Theory: transmission, salients and erosion in the lee

A submerged reef dissipates part of the incident energy by breaking and transmits the rest. For low-crested structures d'Angremond et al. (1996) fitted

$$K_t = -0.4\,\frac{R_c}{H_i} + c\left(\frac{B}{H_i}\right)^{-0.31}\left(1 - e^{-0.5\,\xi}\right), \qquad 0.075 \le K_t \le 0.8,$$

with freeboard R_c (negative when submerged), crest width B, surf similarity ξ and $c = 0.64$ (permeable) or 0.80 (impermeable); Briganti et al. (2003) give a variant for wide crests (both as reproduced by Blacka et al., 2013, Eqs. 25-29). The transmitted energy is K_t^2.

The shoreline responds to gradients of the wave-driven flow in the lee. In the models of Ranasinghe et al. (2006), a reef close to shore drives diverging currents and erosion, and a reef farther offshore drives two converging cells and accretion (as summarised by Rendle, 2016, p. 43); erosion is likely when the reef's distance from the shoreline is less than the natural surf-zone width ($S_a/SZW < 1$) and accretion when $S_a/SZW > 1.5$, peaking near 2 (as reproduced by Blacka et al., 2013, pp. 22-23). Reviewing built structures, Ranasinghe and Turner (2006, abstract) found that "contrary to expectations, a majority of the submerged structures constructed to date have resulted in shoreline erosion in their lee". The empirical method of Black and Andrews (2001) predicts little change for an alongshore length to offshore distance ratio B/S below about 0.1 and a salient amplitude from $X_{off}/B = 0.5\,(B/S)^{-1.27}$ (Eq. 30, as reproduced by Blacka et al., 2013, p. 21); Blacka et al. (2013, p. 22), citing an earlier evaluation by Evans and Ranasinghe, note that the method over-predicts at built reefs.

Figure 8a shows the mechanism behind this distance criterion. ten Voorde et al. (2009, p. 77) quote Ranasinghe and Turner (2006): the principal mode of shoreline response to submerged structures "can vary between erosive and accretive, depending on the offshore distance to the structure", and they reproduce numerically modelled currents for a distance from the apex of the structure to the coast of 100 m (erosion) and 250 m (accretion), for a peak period of 10 s and a significant wave height of 1.5 m (ten Voorde et al., 2009, pp. 77-78, Fig. 14). On this basis they adopt $S_a/SZW > 1.5$ as a preliminary design rule, with S_a the distance from the apex to the undisturbed shoreline and SZW the natural surf-zone width, and add that S_a "should clearly not be too large", because the effect of the structure on the shore declines with distance (ten Voorde et al., 2009, p. 77). The figure supports the theory above only qualitatively: it shows modelled flow, not measured sediment transport or shoreline change. Our reefs span both regimes if the distance from the waterline to the crest (Appendix C) is taken as a rough proxy for S_a: Bunbury (38 m), Kovalam (90 m) and Pratte's (103 m) lie close to the 100 m case, whereas Albany (189 m), Narrowneck (212 m), Boscombe (239 m), Palm Beach (243 m), Cables (268 m), Opunake (300 m), Mount Maunganui (308 m) and Borth (331 m) lie near or beyond the 250 m case. The wave conditions of Figure 8a (1.5 m, 10 s) are, however, a single mild case, and the surf zone widens in storms (Section 5.2), so the reefs that sit in the accretion regime for typical waves may sit in the erosion regime for storm waves.

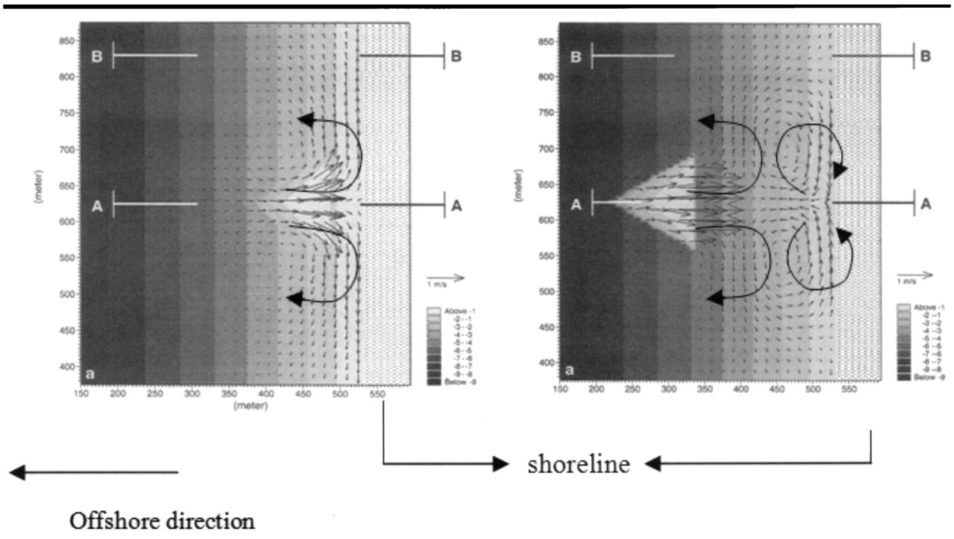


Figure 14. Erosion for distance apex structure-coast of 100 m (left) and accretion for distance apex structure-coast of 250 m (right; Ranasinghe and Turner, 2006).

Figure 8a. Numerically modelled wave-driven currents (arrows, scale 1 m/s) and depth bands (-1 m to below -9 m) around a submerged multifunctional reef for a distance from the apex of the structure to the coast of 100 m (left; erosion) and 250 m (right; accretion), for a significant wave height of 1.5 m and a peak period of 10 s. At 100 m the return currents carry flow away from the shore; at 250 m they turn towards it, with a sheltered zone behind the apex. The figure shows flow, not measured sediment. Reproduced from ten Voorde et al. (2009, Fig. 14, p. 78), after Ranasinghe and Turner (2006); the figure belongs to the original publication and is reproduced here for private research purposes only.

5.2 Predictions for the built reefs

Table 4 applies these predictors to the reconstructed reefs. At mid tide the submerged reefs transmit 41 to 80 % of the site-wave height, i.e. 17 to 64 % of its energy; only Borth, whose crest dries, blocks most waves. Both predictors expect a salient at every reef (B/S 0.23 to 0.78; amplitude 11 to 114 m; S_a/SZW 1.5 to 5.5 under typical waves), but storms widen the surf zone until $S_a/SZW = 1$ at H_0 of 1.5 to 4.5 m, putting every reef in the erosion-likely regime.

Reef	K_t low / mid / high	B/S	Predicted salient (m)	S_a/SZW (H / $2H$)	Observed lee response
Boscombe	0.05 / 0.48 / 0.47	0.39	84 (Eq. 31: 11-35)	5.5 / 3.7	+3.4 m (95 % CI +1.3 to +5.3), before/after with controls
Palm Beach	0.29 / 0.52 / 0.75	0.43	114	4.8 / 2.0	2 m or less (chart, low confidence)
Cables	0.38 / 0.52 / 0.67	0.47	98 (Eq. 31: 14-41)	3.6 / 2.0	lee high pre-dates the reef; beach bulge shrinking
Borth	0.05 / 0.05 / 0.66	0.39	88	1.8 / 1.4	+30 m bulge (+12 to +48), single survey, no baseline
Pratte’s	0.80 (upper limit)	0.63	40	2.8 / 1.3	+13 m in 5 months, no control; “no discernible impact”
Narrowneck	0.42 / 0.64 / 0.68	0.39	104	3.3 / 1.3	not testable (continuous nourishment)
Mt Maunganui	0.44 / 0.66 / 0.80	0.23	79	2.1 / 1.1	“subtle and within the natural variation”
Albany	0.39 / 0.41 / 0.40	0.78	77	2.4 / 1.6	no post-construction data (opened 2025)

Bunbury	0.46 / 0.64 / 0.80	0.34	11	1.5 / 0.7	removed within days

Table 4. Predicted and observed shoreline response (predictions derived in this review; observations from Section 5.3). K_t for the site wave at low, mid and high tide (Briganti et al., 2003; d'Angremond et al., 1996); B/S and the salient amplitude of Black and Andrews (2001, Eq. 30), in brackets the lower reading of their Eq. 31 (with an assumed total salient length of 1 to 3 B); S_a/SZW for the site wave and twice the site wave (Ranasinghe et al., 2006).

5.3 Observed response: the seabed and beach surveys

Only five of the 13 reefs have measured seabed or beach data that can test a reef effect (Appendix A). For each, this review compared the lee with control stretches 1.5 to 3.5 reef lengths alongshore and, where possible, before with after.

Boscombe has the best-controlled record, 316 beach-profile surveys from 2004 to 2026 (Channel Coastal Observatory, 2026a). Relative to four control lines, the 0 m ODN contour behind the reef advanced by +3.4 m (95 % confidence interval +1.3 to +5.3 m) after construction (Figure 8): 0.04 of the 84 m predicted by Eq. 30, while beach management moved the whole frontage by +72 m^3/m in 2006-07 and +40 m^3/m in 2020-21. This agrees with Rendle (2016, p. 102): "no significant shoreline response". The designers' 60 m salient after six months (Mead & Borrero, 2011, p. 5) coincided with 11,100 m^3 of surplus fill sand left on the beach and dispersed after the first winter (Rendle, 2016, pp. 125-128).

Cables, the only reef with offshore soundings before and after construction (Western Australia Department of Transport, 2020), has a lee seabed 0.60 m higher than the controls, but the offset already existed in 1995 (Figure 9): the lee bed changed by +0.03 m (standard error 0.04 m) from 1995 to 2002, and the beach bulge shrank from +26 m (2001) to +10 m (2012), whereas a reef-driven salient would grow.

Borth gives the one positive signal: in the 2022 lidar survey (Welsh Government, 2022) the 0 m contour bulges +30 m behind the reef (+12 to +48 m depending on the control; Figure 10), but without a pre-construction survey, and within a scheme of groynes, breakwaters and shingle recharge, the bulge cannot be attributed to the reef. At Palm Beach chart contours are offset by 2 m or less (low confidence).

The literature agrees. At Mount Maunganui effects were "subtle and within the natural variation" with "no noticeable salient" (Weppe et al., 2007, as cited in Blacka et al., 2013, p. 103); at Pratte's monitoring found "no discernible impact on either the surrounding bathymetry or the beach" (Blacka et al., 2013, p. 117); and about half of the protection structures reviewed by Blacka et al. (2013, p. 73) had no significant accretionary impact. The one positive report, a wider beach at Kovalam, rests on the designers' photographs over less than three years (Mead & Borrero, 2011, p. 5; Blacka et al., 2013, p. 70). The observed lee response is thus 0.02 to 0.06 of the Black and Andrews prediction, because transmission is high, the reefs are short (12 to 132 m alongshore) and diffract the waves, and beach management produces signals an order of magnitude larger.

5.4 The Gold Coast and the nourishment confound

Hypothesis (b; Appendix F) holds that the Gold Coast reefs are claimed to protect the coast although the shoreline shows little change, and that the claim cannot be attributed to the reef because sand is nourished continuously. The record supports it. Narrowneck was built as "a coastal control point to stabilize the nourished beaches to the south" after 1.3 million m^3 of

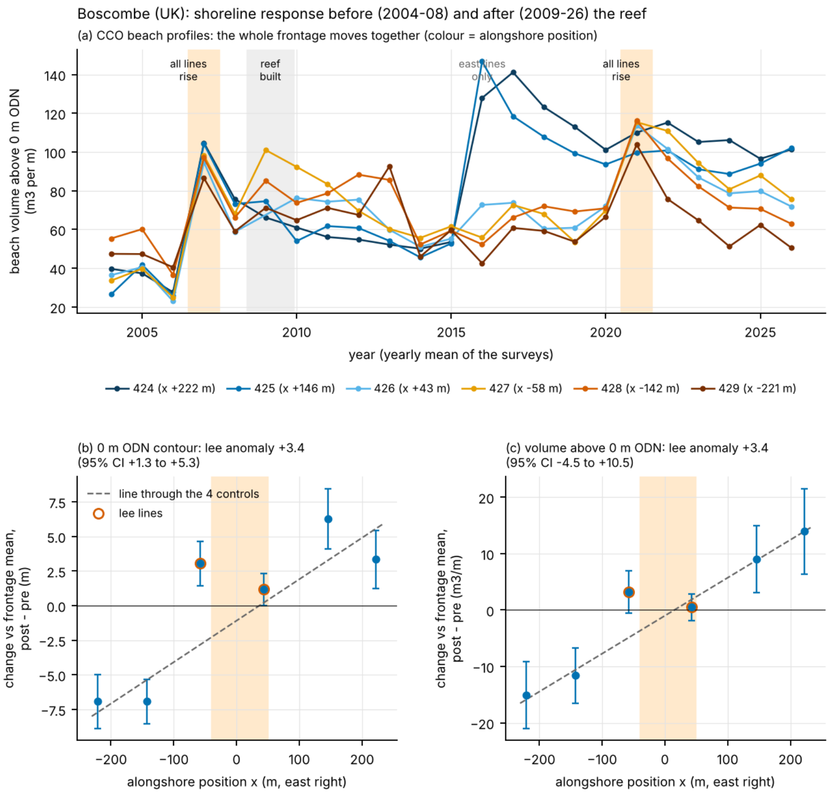


Figure 8. Shoreline response at Boscombe before (2004-2008) and after (2009-2026) the reef, from the beach-profile surveys of the Channel Coastal Observatory (2026a), analysed in this review. (a) Beach volume above 0 m ODN on six profile lines (yearly means; colour: alongshore position x): the whole frontage rises together in 2006-07 and 2020-21, a beach-management signal. (b) Change of the 0 m ODN contour (post minus pre, relative to the frontage mean) against alongshore position: the two lee lines (rings) lie +3.4 m (95 % confidence interval +1.3 to +5.3 m, bootstrap over survey dates) above the line through the four control lines. (c) The same for the beach volume: +3.4 m^3/m (−4.5 to +10.5). Black and Andrews (2001, Eq. 30) predict a salient of 84 m.

nourishment in 1999-2000, with at least 60,000 m^3 a year planned thereafter (Jackson et al., 2012, pp. 1-2). The designers report a beach 30 to 50 m wider (Mead & Borrero, 2011, p. 4), whereas Blacka et al. (2013, pp. 98-100) found “only a minor planform perturbation” of about 10 m and called the published widening “a likely misreading” of an arbitrary reference line, and Jackson et al. (2012) found “a minor groyne effect” and no long-term salient. Palm Beach was part of an AU$18 million nourishment project that placed about 470,000 m^3 on the beach (Royal HaskoningDHV, n.d.); the City’s statement that the reef “retained about 550,000 cubic metres” in two years (ABC News, 2022) is of the same order as the volume placed, and there is no nourishment-only control (Prenzler et al., 2023).

5.5 Burial, settlement and scour

What reefs change reliably is the bed near the structure: “vertical displacement of the structures of the order of 0.5 to 2 m as well as localised bed scouring” at most of them (Blacka et al., 2013, p. 72). At Pratte’s the bed scoured to about 4 m, the crest dropped 1.5 m in a year and many bags were buried at removal (Blacka et al., 2013, pp. 116-117); at Narrowneck a storm bar buried the seaward containers (Jackson et al., 2012, pp. 4-5; Corbett et al., 2023, p. 663); at Boscombe the −3.5 m contour moved 50 m shoreward in the lee (Rendle, 2016, p. 102). The rock reef at Cables, on limestone bedrock, shows no toe scour or burial (this review). Settlement and

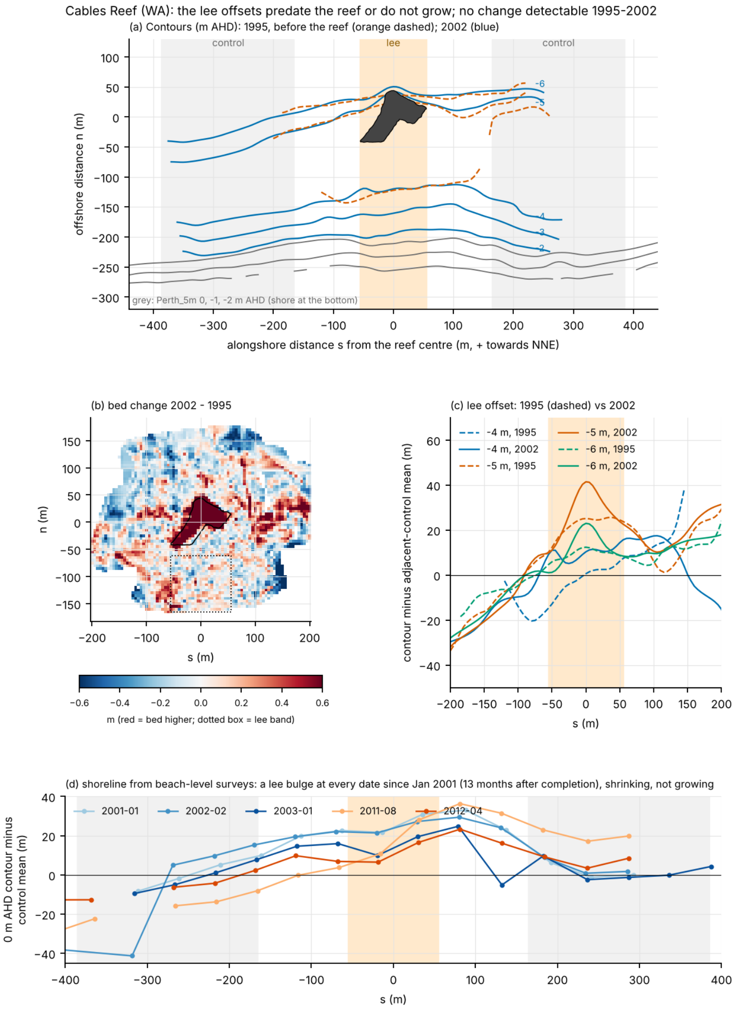


Figure 9. Seabed and shoreline around Cables Reef from the single-beam soundings of the Western Australia Department of Transport (2020) and beach-level surveys, analysed in this review. (a) Contours (m AHD) in December 1995, before the reef (dashed), and February 2002 (solid), with the reef footprint, the lee band (orange) and the control windows (grey). (b) Bed change 2002 minus 1995 (red: bed higher). (c) Offset of the −4, −5 and −6 m contours in the lee relative to the controls in 1995 (dashed) and 2002 (solid): the offset pre-dates the reef. (d) Shoreline (0 m AHD) relative to the control mean, 2001-2012: a lee bulge that shrinks from +26 m to +10 m rather than growing.

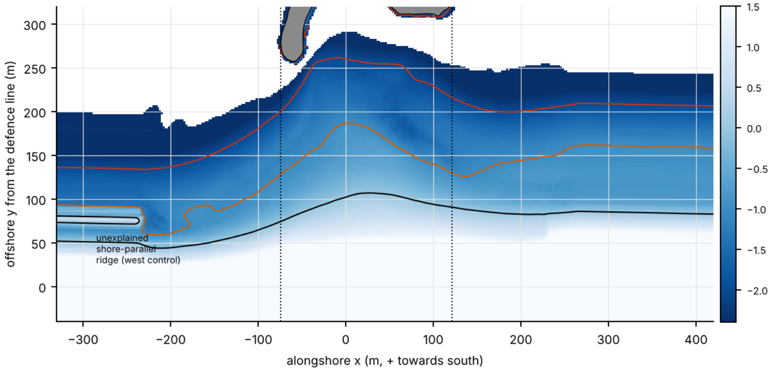
burial remove relief, which Section 4 found decisive for surfing; Section 7.4 quantifies the effect for Haifa.

5.6 Combining a surf reef with coastal stabilisation

The two functions pull the design apart. A surfing crest lies about 1 to 2 m below MSL (Table 2) and therefore transmits 17 to 64 % of the energy, and a short reef changes the planform little; placing it at about twice the surf-zone width favours accretion under typical waves (Ranasinghe et al., 2006), but not in storms. The Gold Coast schemes therefore combine nourishment with a reef acting as a control point (Jackson et al., 2012; Prenzler et al., 2023); the engineers' claim that over 30 years the reef "definitely stacks up against beach nourishment" (ABC News, 2022) has not been tested against a nourishment-only baseline. A surf reef can hold nourished sand and add amenity within a stabilisation scheme, but it should not be relied on as stand-alone protection, and a protective effect should be claimed only from before/after surveys with controls and a nourishment record, which no reef examined has published in full.

Borth (Wales): a lee bulge in the 2022 survey; no pre-construction survey

(a) LiDAR DTM 19 Mar 2022 (m MSL); contours 0 black, -1 orange, -2 red; grey = reef



(b) lee offset (trend through both controls): +30 / +40 / +57 m for 0/-1/-2 m
range over control choices: +12..+48; +1..+77; +22..+93

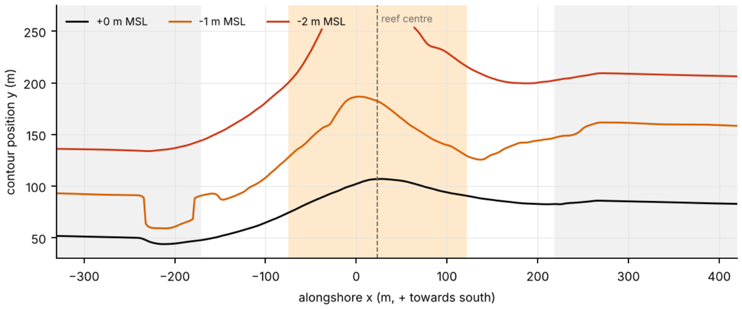


Figure 10. Seabed behind the Borth reef from the lidar terrain model of 19 March 2022 (Welsh Government, 2022), analysed in this review. (a) Bed level (m MSL) with the 0, −1 and −2 m contours and the two reef mounds (grey). (b) Contour positions alongshore, with the lee window (orange) and the control windows (grey): relative to the trend through both controls, the lee contours lie +30, +40 and +57 m farther offshore (+12 to +48, +1 to +77 and +22 to +93 m over control choices). A single survey without a baseline cannot separate the reef from the groynes, breakwaters and shingle recharge of the same scheme.

6 Materials and durability

Surf reefs have been built from geotextile sand containers (GSCs, also called geotubes or bags), a sand-filled rubber bladder (the Airwave), quarried rock and concrete. This section compares their record, the wave loads in the breaking zone, and their ecology and cost.

6.1 Failure record

Hypothesis (c; Appendix F) holds that soft structures do not survive long in the breaking zone. Table 5 summarises the fate of the reefs by material: six of the seven geotextile or inflatable reefs suffered major damage or loss within a few years, and the seventh, Narrowneck, survives only through continuous maintenance; none of the four rock reefs failed (Fisher exact test, $p = 0.015$; derived in this review). The comparison is confounded, because the rock reefs are newer and better funded. The designer of several of these reefs now calls the sandbags “an evolutionary deadend” (Reid & Sumner, 2021) and writes that “the fatal flaw of the bags was their propensity for failure” (Mead, as quoted in Geosynthetics Magazine, 2023).

Reef (year)	Material	Fate	Named cause
Narrowneck (2000)	about 400 GSCs, 20 m long, 3-4.5 m diameter (Blacka et al., 2013, p. 97)	containers compromised and seaward units buried by 2011; 84 containers added 2017-18 (Corbett et al., 2023; Jackson et al., 2012)	anchors, propellers, spears and knives (Jackson et al., 2012, p. 7)
Pratte's (2000)	woven polypropylene and polyester bags, 3.0 x 2.1 x 1.2 m	crest down 1.5 m in a year; fabric "could be torn by hand"; removed 2008-2010 (Blacka et al., 2013, pp. 116-117)	settlement, scour, degradation of the fabric
Mount Maunganui (2008)	large GSCs, up to 7.2 x 3.2 m in section	2,800 of 6,500 m^3 in place (Mead & Borrero, 2011, p. 3); largest containers removed 2014 (Rendle, 2016, p. 41)	ruptured containers and unsealed ports during construction
Opunake (2008, incomplete)	27 GSCs	never completed (Taranaki Regional Council, 2009)	funding and construction
Boscombe (2009)	about 55 GSCs, the largest 65.5 m long	fill reworked ("pillowing") in the first winter; one container lost in 2011; closed 2011 (Rendle, 2016, pp. 94-98)	contested: weakness, propeller strike or vandalism
Kovalam (2010)	GSCs, about 4,000 m^3	bags washed ashore after the first monsoon (Rendle, 2016, p. 42)	debris strikes (designer's view)
Bunbury Airwave (2019)	Hypalon bladder filled with sand (and air)	seam tore during installation; removed within days (ABC News, 2019)	seam glued because the Hypalon could not be welded (Bunbury Mail, 2022)
Cables (1999)	rock, 1.5 t and 3 t, on limestone bedrock	in place; crest change -0.16 m in 2.2 years, within survey error (Appendix C)	none reported
Borth (2012)	graded rock, units of several tonnes	in place	none reported
Palm Beach (2019)	rock, about 25,000 m^3	in place (Prenzler et al., 2023)	none reported
Albany (2025)	rock in three graded layers	core rock moved in a storm before armouring (Appendix C); 18 rocks repositioned for safety (Sheil, 2026)	storm during construction; surfer injuries

Table 5. Material and fate of the surf reefs (sources as cited).

The GSC failures follow a common sequence: point damage from anchors, propellers, knives, debris or construction faults, then loss of fill and change of shape, because "as soon as even a single sand-filled container broke and lost sand, the whole structure changed shape due to the fluid nature of sand" (Mead, as quoted in Geosynthetics Magazine, 2023). At Boscombe the damage to one large container led to the loss of two, 12.5 % of the reef's volume, and the reef was judged “unlikely” to survive its proposed 25-year life “without multiple repairs” (Rendle, 2016, pp. 98, 121). The manufacturers’ design life of about 20 years for outer-layer containers (Geofabrics, 2013, as cited in Blacka et al., 2013, p. 35) contrasts with one to eight years before major loss in the field.

The Airwave tore along a glued seam near completion in December 2019 (ABC News, 2019); the Hypalon rubber could not be welded (Bunbury Mail, 2022). Its founder states that the final design was filled with sand instead of air, which did not respond well to the waves, that the reef made perfect waves until a seam opened a couple of days after installation, and that the next version will be concrete (T. Bottegal, Airwave founder, pers. comm., 8 Oct 2026). The timing differs from the press record and the wave quality is unverified; both accounts agree that the seam, not the rubber, failed.

6.2 Wave loading in the breaking zone

Bar and Drimer (2023) computed the loads on a submerged, impermeable geotube with a numerical wave flume and defined a sliding index $SI = F_y/(W' - F_z)$ (horizontal force F_y, uplift F_z, submerged weight W') and a variant SI_c that adds the uplift of the dynamic pressure on the contact area. They found that “the most critical state, regarding the sliding stability of the Geotube, occurs, typically just before breaking” (p. 1), when an up-wave horizontal load coincides with uplift (p. 8), and advised assuming that the dynamic pressure penetrates under the tube (p. 10). They name “too-small size Geotubes” and “sand coverage” as the common causes of failure (p. 2). The tube is a rigid body in the model, which gives forces but not fabric stress or tearing. A later paper by the same authors on tube tearing, reported in correspondence, could not be located; the analysis rests on Bar and Drimer (2023).

6.3 Stability calculations

This review estimated the loads by integrating the linear incident-wave pressure over a rigid tube section, calibrated against values read from Bar and Drimer (2023), and bounded the uplift ratio $U = \max F_z/W'$ between a sealed base (upper bound) and a base reached by the wave pressure (U_c, lower bound) (Appendix A). Under depth-limited breakers ($H_b = 0.78\,h$), tubes 1.0 to 1.65 m high reach $U = 1$ at base depths of 2.3 to 3.9 m and tubes 2.2 to 3.2 m high at 5.3 to 8.0 m (Figure 11a): the sealed-base uplift scales as $U \approx H/(2c\,\Delta\,D\cosh kh)$ (D tube height, Δ relative submerged density, $c \approx 0.87$), so small tubes lift first. If the pressure reaches the base, $U_c \leq 0.22$ and the calibrated SI_c of 0.37 to 0.56 makes sliding marginal against a friction coefficient of about 0.5 (0.48 measured bag on bag by Corvaro et al., 2022). The single-tube criterion of van Steeg et al. (2011), $H_s/(\Delta\sqrt{BD}\,(f\cos a + \sin a)) \leq 0.65$, gives critical heights of 0.34 to 1.65 m for the built reefs’ tubes, which depth-limited seas ($H_s \approx 0.55\,h$) reach at base depths of 0.6 to 3.0 m, inside every surf-reef crest. By the Hudson formula with $K_D = 2$ (US Army Coastal Engineering Research Center, 1984, as tabulated by Verhagen, 2012), rock of 1 to 2 t resists the same waves; the Albany armour (D_{50} 0.94 m) is stable to H_s 2.4 to 4.3 m on the faces of its surf cut (2.4 m on the steepest, 1:3.5; the gentler values are extrapolated) but its core rock only to 1.2 to 2.2 m, and depth-limited breakers exceed the armour value where the bed is deeper than about 3.1 m, consistent with the storm movement of exposed core rock during construction and the repositioning of 18 rocks in 2026 (Sheil, 2026). The Cables rock (1.5 and 3 t) is stable to H_s 3.1 to 6.0 m and the Borth rock (4.5 to 8 t) to 3.3 to 6.1 m. A tube at the break sees 7,200 to 14,400 load cycles a day.

For hypothesis (e), that most wave power acts at the break and lifts the tube up and then down, which is why most tubes tear, the load part is supported: the critical state occurs just

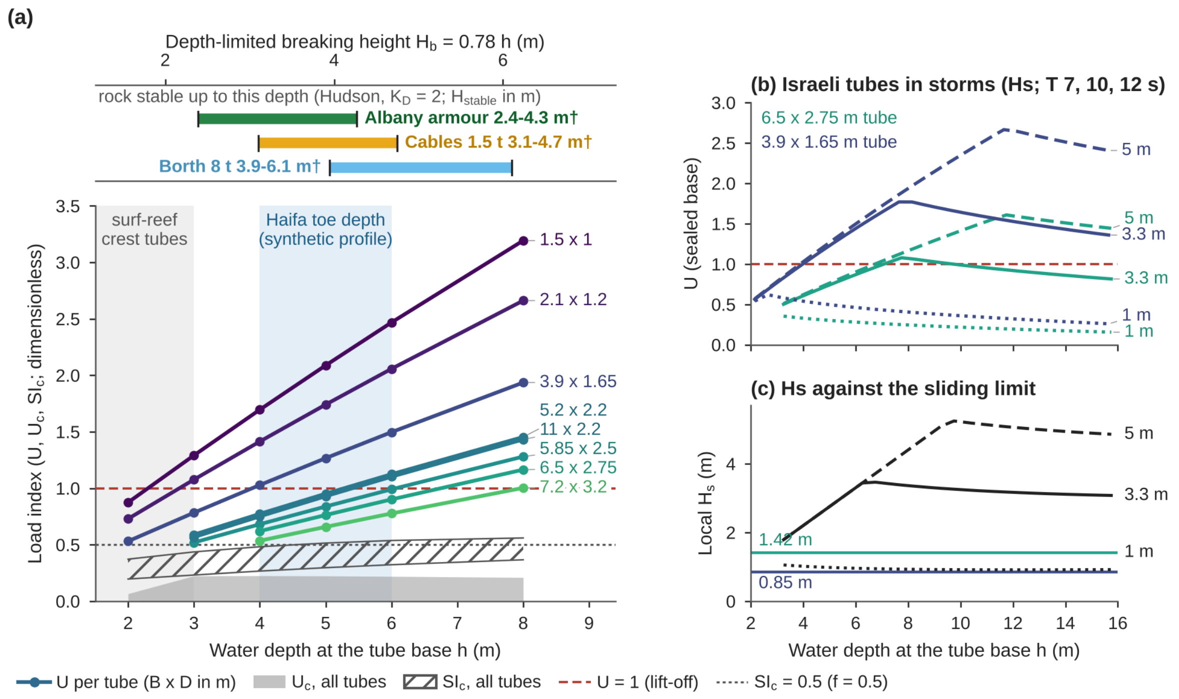


Figure 11. Estimated load on sand-filled geotextile tubes and rock in the breaking zone (an estimate; tearing is not modelled). (a) Sealed-base uplift U, the maximum vertical force over the wave cycle divided by the submerged weight (an upper bound; one line per tube size $B \times D$ in m), under depth-limited breakers ($H_b = 0.78\,h$, maximum over $T = 6$, 9 and 12 s), with the ranges of the uplift U_c if the wave pressure reaches the base (grey, at most 0.22) and of the calibrated sliding index SI_c (hatched, 0.20 to 0.56 over all three periods; 0.37 to 0.56 for the most severe period, the values used in the text); interpolated, tubes of 1.0 to 1.65 m height reach $U = 1$ at base depths of 2.3 to 3.9 m and tubes of 2.2 to 3.2 m at 5.3 to 8.0 m, and the bars above give the depth to which the rock of the Albany, Cables and Borth reefs stays stable under depth-limited breaking (Hudson formula with $K_D = 2$; † extrapolated beyond a 1:5 face). (b) U for the Israeli 6.5 x 2.75 m and 3.9 x 1.65 m tubes in storms of $H_s = 1$, 3.3 and 5 m ($T = 7$, 10 and 12 s; $H_{max} = \min(1.8\,H_s K_s, 0.78\,h)$) and (c) the local H_s, limited to $0.55\,h$, against the single-tube sliding limit of van Steeg et al. (2011), 1.42 and 0.85 m. Forces come from a linear Froude-Krylov model of a rigid superellipse section calibrated against the values read from Bar and Drimer (2023) (factor 1.42 on the sliding index SI in the linear range; 0.94 on SI and 2.24 on SI_c near breaking), not their numerical model; the real behaviour lies between the sealed and the open base, and single tubes are less stable than interlocking mounds (details in Appendix A).

before breaking with simultaneous uplift (Bar & Drimer, 2023), and small or sealed tubes are lifted by ordinary breakers. The tearing part is not modelled by any source found, and "most tubes tear" is not supported, since the reported causes are point damage, undersizing, burial and construction faults (Table 5). Repeated lifting and flapping may fatigue fabric and seams ("the initial cause of failure is masked by the damage suffered during cyclic flapping and deflation of the container"; Jackson et al., 2012, p. 7), but this is a hypothesis that needs fabric and seam tests. Hypothesis (c; Appendix F), that soft structures do not survive long in the breaking zone, is supported by both the record and the stability numbers.

6.4 Ecology and cost

Hypothesis (c) also holds that geotubes support marine life and are the cheapest structure. Colonisation of geotextile is documented (seaweed on Korean geotubes after one year; Oh and Shin, 2006a), and the designer chose containers partly as a "good substrate for marine organisms" (Geosynthetics Magazine, 2023), whereas at Ashkelon a surfers' group stated that "nothing develops near the geotubes" (Rinat, 2019). Rock and concrete are colonised too, and bio-enhanced concrete is reported by its maker to more than double invertebrate species richness on a Haifa port breakwater (ECOncrete, n.d.; company data). No comparative study was found, so the ecological advantage of geotubes is a new, untested observation.

On capital cost the containers are cheaper: A$56 per m^3 at Narrowneck against A$457 for the Cables rock (Blacka et al., 2013, Table 4.4, 2013 AUD), although the small or incomplete GSC reefs cost A$274 to A$545 per m^3. Over the life of the structure the advantage shrinks: renewal (Narrowneck), removal (Pratte's, about $551,000, slightly more than its construction; Blacka et al., 2013, p. 119), plastic debris (Rendle, 2016, p. 122) and a short field life. "Cheapest" is supported for construction cost only; for a reef in the breaking zone, durability decides the choice (Section 7).

7 Implications for the Israeli coast and Haifa

This section applies Sections 4 to 6 to the Israeli Mediterranean coast and to Haifa, where a surf reef is being considered.

7.1 Tide, water level and waves

The semidiurnal tide has a range of about 0.4 m at springs and 0.15 m at neaps along the whole coast (Rosen, 2001; Spanier & Zviely, 2022), smaller than at any built reef (1.1 to 4.3 m; Section 1.3). Weather and season move the water level more: the 1-year low and high levels are -0.38 and $+0.64$ m, and the 100-year high is $+1.10$ m (Rosen, 2001, Table 1). The most common H_s is 0.5 to 1.0 m; summer is calm, winter brings about five storms a year above H_s 3.5 m at Haifa (6 % of the records above 2 m and 1.4 % between 3.0 and 4.5 m off Haifa, 1994-2004) (Bitan & Zviely, 2020, p. 3), and 70 % of waves come from west to west-north-west (Bitan & Zviely, 2020; Spanier & Zviely, 2022); storm waves above 5 m have periods up to 15 s (Carmel et al., 1985). Haifa Bay is sheltered: the nearshore wave exceeded 12 hours a year is 3.0 to 3.3 m against 5.1 to 5.8 m on the open coast, and the closure depth 4.7 to 5.2 m (Bitan & Zviely, 2020). Field observations by the author (about ten weeks, July to October 2025; Appendix G) put the smallest sea that was surfable, borderline or nearly surfable at H_s 0.40 to 0.45 m (H_{max} 0.50 to 0.55 m) at periods of 6.5 to 7.5 s from 284 to 294 degrees at the Shikmona buoy; height alone does not separate the classes, since a short period needs more height (a borderline sea at 4 s had H_s 0.57 m, and a sea of H_s 0.42 m at 5.1 s was not surfable). On days with H_s 0.45 to 0.60 m the waves broke at a water depth of 1.30 to 1.50 m at the peak (1.15 m for the smallest surfable shoulder, 1.70 m for the largest sets), and in borderline seas the take-off was in about 1 m of water, falling to about 0.5 m towards the end of the ride (Appendix G, Table G3). Israeli surf is thus below the Palm Beach window (H_s above 0.8 m, T_p above 7 s; Prenzler et al., 2023) and close to Boscombe's design range ($H_{10} = 1 \pm 0.5$ m, $T = 7 \pm 2$ s; Mead et al., 2010), where a shallow crest built for small waves gave a collapsing wave (Section 4.3); the short periods help, because they soften the breaker on a given ramp.

7.2 A small tide: consistency, not more surf

Insight (h-A) holds that the small tide would make a reef wave more consistent and narrow the range of loads. The first part is supported (Section 4.6, Figure 7): a synthetic Haifa-type reef keeps 0.87 to 0.98 of its no-tide surfable time, and the break depends on wave height alone, although the average surfable time of the built geometries does not rise. The second part is only partly supported. The tidal range of the depth-limited crest wave is 0.31 m in Israel against 0.9 to 3.4 m at the built reefs, but damaging loads come from storms: the storm and seasonal water-level range (1.02 m) is 2.5 times the spring tide, Haifa Bay storm seas of H_s 3.0 to 3.3 m exceed the van Steeg et al. (2011) limit for single tubes by a factor of 2 to 5, and a sealed-base 6.5 x 2.75 m tube is lifted at 7 to 9 m depth (Figure 11). The material, not the tide, decides durability.

7.3 Israeli experience with geotextile tubes

The geotubes that tore in Israel were at Ashkelon North; no geotube was built at Netanya, which is protected by rock breakwaters (State Comptroller of Israel, 2020, p. 71; 2024, p. 48). At Ashkelon North 48 submerged geotubes were laid in five clusters between July 2017 and September 2019 for about NIS 11 million, as temporary cliff protection (State Comptroller of Israel, 2020, pp. 71, 83). Several "were damaged and torn shortly after being laid in the sea, after the winter storms, and had to be repaired" (State Comptroller of Israel, 2020, p. 71; translated), and in early 2019 severe damage of unknown cause halted the work (Rinat, 2019). Twelve tubes were replaced in 2021, maintenance meant "sewing tears and collecting loose sheets", and the protection of one section later collapsed (State Comptroller of Israel, 2024, pp. 29, 48, 53; translated). The official causes are "unprofessional laying, non-optimal filling, external damage for example from propeller blades, anchors or divers' knives" (State Comptroller of Israel, 2024, p. 9; translated), and the agency estimates the life of the unarmoured sleeve used at months to a few years (p. 57). A senior Israeli coastal engineer argues that these failures do not prove geotubes unsuitable, since they can be strengthened and Haifa Bay waves are lower than those of the open coast (N. Drimer, as quoted in Zalul, 2025, p. 23; an expert opinion). The Israeli record agrees with the international one: the fabric is the weak point.

7.4 Sand mobility, burial and the sand-trap idea

The net drift runs northward, from about 400,000 m^3 a year at Ziqim to 80,000 ± 20,000 m^3 a year at the entrance to Haifa Bay (Spanier & Zviely, 2022); the Haifa Port breakwater trapped an average of 66,000 m^3 a year from 1929 to 2004 (Zviely et al., 2009, p. 95). Gross transport is far larger: by the CERC formula (US Army Corps of Engineers, 2002), 1 m breakers at 10 degrees move about 2,700 m^3 a day, the net annual drift of the southern Carmel coast (60,000 to 72,000 m^3; Zviely et al., 2007) in about 22 days (derived; order of magnitude). A reef crest and toe within the 5 m closure depth lie in the active zone. Field observations by the author at the Haifa shore (Appendix G) indicate a seasonal bed-level change of about 1 m, and Google Earth satellite imagery of the coast shows more sand accumulating north of the breakwaters; this is an observation, not a survey, and has not been published, but the seasonal shoreline swing of up to ±20 m (Zviely et al., 2009) on slopes of 1:70 to 1:30 implies 0.3 to 1.3 m (derived). As long as the crest stays above the sand, a seasonal rise of the surrounding bed does not change the break, which is set by the crest and the upper ramp; the wave conditions and crest depths of Section 7.5 therefore hold through the seasons. The risk is to the structure: sand can cover the toe and the lower ramp, reach the crest of a low reef in the high-sand seasons, and, with scour and settlement (geotextile reefs settled by 1 to 2 m; Section 5.5), bury it over time. Burial is therefore a durability and maintenance risk, to be met by a crest kept well above the seasonal high bed and by monitoring, rather than a shift in surf quality; hypothesis (h-B; Appendix F) anticipated it.

A sand-trapping approach is proposed: to trap sand in the high-sand seasons (October and May, from field observations; Appendix G) and to build the reef on it under a geotextile sheet. As a sand source the idea is not supported: at the observed salient ratio a reef holds about 10 to 2,400 m^3, less than 4 % of one year's net drift, and even the Black and Andrews upper bound

(21,000 to 102,000 m^3) is a one-time fill, not a supply (derived). As a construction saving it is plausible: building on a bed 1 m higher under a 70 x 100 m footprint saves about 7,000 m^3, half the fill of a 1:12 reef section (derived). No precedent was found and the sheet would face the uplift of Section 6.3, so the idea remains a proposal that needs a pilot.

7.5 Design envelope for Haifa and next steps

Without Haifa bathymetry, the envelope was screened on a synthetic profile (1:50 bed, crest 10 m wide, 200 m offshore) with the methods of Section 4, the Israeli tide and periods of 6 to 8 s (Figure 12; Table 6). The cells of Figure 12 are computed for Haifa waves and the Israeli tide only; the built reefs are placed in it by their geometry (crest depth and ramp slope) for comparison and are not evaluated under their own wave climates. In the model ($\gamma = 0.78$) the depth at which a wave breaks does not depend on the crest: at 7 s it is about 0.9, 1.0, 1.3, 1.6, 2.2 and 2.8 m for H_0 of 0.5, 0.6, 0.8, 1.0, 1.5 and 2.0 m (derived). A wave is shaped by the reef only if this depth exceeds the crest depth, so the crest depth sets the smallest wave the reef can use: H_0 0.5 m for a crest 0.6 to 0.8 m below MSL, 0.6 m at 1.0 m, 0.8 m at 1.2 m, 1.0 m at 1.5 m and 1.5 m at 2.0 m. Larger waves break farther seaward on the ramp whatever the crest depth, and the largest winter seas (H_s 3.0 to 4.5 m, 1.4 % of the time off Haifa (Bitan & Zviely, 2020, p. 3)) break in about 4 to 6 m of water, near or seaward of the toe, where the reef no longer shapes them. What a shallow crest changes for the larger waves is the water left over the rock: the broken waves cross the crest as bores, a falling surfer meets the rock sooner, and the load on the crest armour grows as the submergence falls. The crest depth is therefore bounded on both sides. On the deep side, the most common seas (H_s 0.5 to 1.0 m) must break on the reef, which by the model requires a crest no deeper than about 1.0 to 1.2 m; a crest at 1.5 m would use only H_0 of 1.0 m or more. On the shallow side, the end of the ride needs about 0.5 m of water (Appendix G), and the water level falls to −0.38 m about once a year (Rosen, 2001, Table 1), so the crest should lie at least about 0.9 m below MSL; a crest at 0.6 m would leave about 0.2 m of water at such lows. The recommended crest is therefore 1.0 to 1.2 m below MSL: by the model it breaks every wave from H_0 0.6 to 0.8 m upward (a surfable fraction of 0.76 to 0.91 over H_0 0.6 to 1.5 m under the Israeli tide), which favours the common surfable winter swells, while the smallest summer seas (H_0 about 0.5 m) would need a crest of about 0.8 m at the cost of that margin. Ramps of 1:8 collapse the wave, 1:12 to 1:16 give an expert wave and 1:20 a mix of advanced and spilling waves. The largest uncertainty is the breaking criterion. The field observations by the author (Section 7.1; Appendix G, Table G3) show waves of H_s 0.45 to 0.60 m (H_{max} 0.6 to 0.8 m) at the buoy breaking at 1.30 to 1.50 m depth, about 0.2 to 0.3 m deeper than the model breaks the same waves when H_{max} is taken as H_0. These single-observer readings are indicative only and do not change the envelope; they suggest that the model errs on the shallow side, so that a crest at 1.0 to 1.2 m would probably also catch the smallest surfable summer seas.

Element	Recommendation	Basis
Crest depth	1.0 to 1.2 m below MSL	model breaking depths: every wave from H_0 0.6 to 0.8 m upward breaks on the reef (the common winter swells); at least 0.5 m of water over the crest at the 1-year low level (−0.38 m); a crest of 0.8 m adds the smallest summer seas (H_0 0.5 m) at the cost of that margin; field breaking depths (Appendix G) indicative only

Element	Recommendation	Basis
Seaward ramp	1:16 to 1:20; 1:30 for a beginner wave; never steeper than 1:12	breaker type and intensity (Section 4.3)
Size	100 m or longer; about 14,000 to 20,000 m^3 per 100 m for a 1.0 to 1.2 m crest (35,000 to 40,000 m^3 at 1:30)	volume and outcome of built reefs (Section 4.5)
Material	rock armour, D_{n50} about 0.7 to 0.85 m (1.0 to 1.6 t), bio-enhanced units optional	Hudson formula, $K_D = 2$ (extrapolated beyond 1:5); record of Section 6.1
Burial margin	crest and upper ramp well above the seasonal high bed (about 1 m change); rock on a prepared bed to limit settlement	durability and maintenance, not surf quality (Section 7.4)
Monitoring	seasonal bed and beach surveys before and after, with control sections; nourishment record	Section 5.3

Table 6. Screening envelope for a Haifa surf reef (derived in this review; Sections 4 to 7).

Before design, the following are needed: (1) multibeam bathymetry of the site and seasonal bed surveys to test the 1 m observation; (2) a west-north-west wave climate giving the days with $H_0 \geq 0.6$ m; (3) a two-dimensional wave model or flume test including refraction and focusing; (4) calibration of the breaking criterion against field breaking depths; (5) seam and fabric tests if any geotextile is retained; and (6) before/after monitoring with controls. A Haifa reef designed this way would combine the features of the reefs that work (a large rock structure with a long, gentle ramp) with the advantage of a small tide, while accepting that it serves intermediate to advanced surfers and acts as a control point for the beach rather than as stand-alone protection.

8 Discussion

This section discusses what the computations and the evidence of Sections 3 to 7 show. The analysis started from eight working hypotheses (Section 1.3); their status is tracked separately in Appendix F, and the discussion below rests on the findings themselves.

8.1 What controls the surf

The ramp, not the crest depth, sets the breaker. The crest depth decides which waves break on a reef: at mean sea level a wave needs H_0 of 0.3 to 1.3 m to break on it, and above 1.6 to 5.2 m it breaks seaward of it (Section 4.2). How the wave breaks is set by the ramp that leads up to the crest. Across the eight modelled reefs with a known outcome, the ramp length ranks with the outcome at Spearman $\rho = 0.93$ (permutation $p = 0.005$), the mean seaward slope at 0.77 ($p = 0.039$) and the median surf similarity at the break at -0.77 ($p = 0.039$), whereas the crest depth does not ($\rho = 0.39$, $p = 0.36$). Under the same 1 m, 10 s wave, Palm Beach and Boscombe break at the same height and depth, but the 89 m ramp at 1:17 gives a plunging breaker ($\xi_b = 0.56$, $Y = 2.05$) and the 13 m rise at 1:3.1 a collapsing one ($\xi_b = 3.00$, $Y = 1.05$; Figure 3). The two built reefs that work, Palm Beach and Albany, both have long, gentle ramps (89 m at 1:17 and 99 m at 1:19.8) and no collapsing cases, while the steep reefs collapse in most cases (Bunbury and Boscombe in all, Mount Maunganui in 91 %; Table 2). For design, this turns the familiar question "how deep should the crest be?" into two: the crest depth for the wave heights to be caught, and a ramp long and gentle enough, 1:16 to 1:20, to make them plunge rather than collapse (Section 7.5).

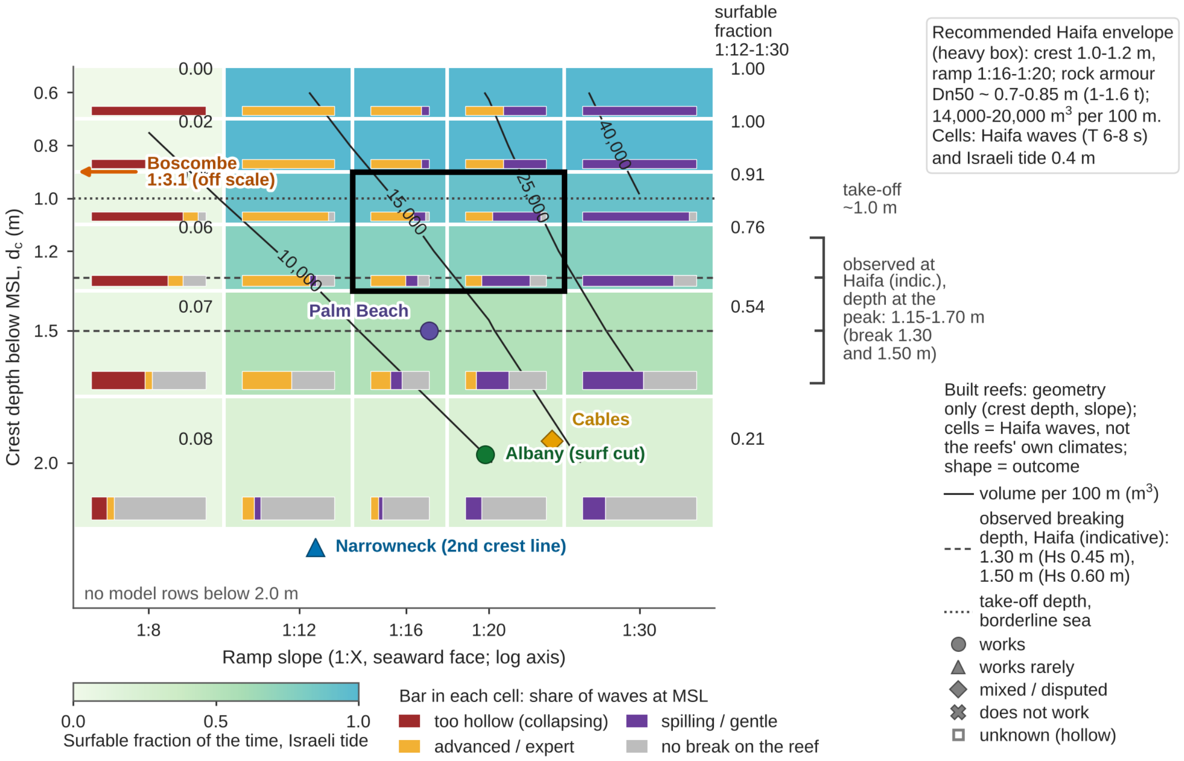


Figure 12. Design envelope for a surf reef at Haifa on a synthetic 1:50 bed (1:70 gives the same result; no Haifa bathymetry is available), as a function of crest depth below MSL and seaward ramp slope (crest 10 m wide, back slope 1:3). All cell values are computed for Haifa waves and the Israeli tide, not for the built reefs' own climates. Cell colour is the fraction of the time the break is surfable under the Israeli tide (spring range 0.40 m; the same for ramps from 1:12 to 1:30 and falling with crest depth), the bar in each cell the share of waves ($H_0 = 0.6$ to 1.5 m, $T = 6$ to 8 s, at MSL) that are too hollow, advanced or expert, spilling or gentle, or do not break on the reef (classes from ξ_b and Y after Battjes, 1974, and Mead and Black, 2001b), and the contours the reef volume per 100 m of length (10,000, 15,000, 25,000 and 40,000 m^3). The heavy box is the recommended envelope (crest 1.0 to 1.2 m, ramp 1:16 to 1:20, about 14,000 to 20,000 m^3 per 100 m, rock armour D_{n50} of about 0.7 to 0.85 m; Section 7.5); the dashed lines (indicative only, not used for the envelope) the water depths at which waves of H_s 0.45 and 0.60 m (H_{max} 0.6 and 0.8 m) were observed to break at Haifa (1.30 and 1.50 m), the bracket the observed depths at the peak from the smallest surfable shoulder to the largest sets (1.15 to 1.70 m) and the dotted line the take-off depth in borderline seas (about 1 m; field observations by the author, Appendix G; Section 7.1); a seasonal rise of the surrounding bed is not shown, because it does not change the break while the crest stays above the sand (Section 7.4). The symbols place the built reefs by geometry only, crest depth and mean seaward slope on their surf cut (Palm Beach, Cables, Albany on its -2 m ridge, Narrowneck second crest line; Boscombe, 1:3.1, is off the scale), coloured and labelled per reef, with the outcome as symbol shape; the cell values beneath them are not their surf performance, which depends on their own swell and tide (Albany, for example, works often under Southern Ocean swell). The model is one-dimensional linear shoaling with $\gamma = 0.78$ and no refraction or focusing, so the envelope is a first screening and needs a Haifa wave climate and a two-dimensional wave model or flume test.

The profile must be read where the wave is ridden. The Albany result depends on where the profile is cut. The automatic cut through the shallowest point crossed the −1 m take-off plateau, where the reef rises 6 m in 22 m (1:4), and put the best wave in the sample among the collapsing reefs; the cut through the long −2 m ridge along which the wave is ridden gives a 99 m ramp and a plunging, “extreme” wave (median $Y = 1.75$), consistent with the reports of a hollow wave for intermediate-to-advanced surfers (Mercer, 2025). The same applies to Cables and Borth (Section 2.4; Figure E1 (Appendix E)). A one-dimensional indicator is meaningful only on the wave path, and on the oblique and refracted swell of a real reef that path is itself uncertain.

The depth after the break matters too. The ramp shapes the wave up to the break; the bed shoreward of the break decides whether the wave keeps its shape. Surfers who rode Pratte’s reported that the wave “hit the shallow reef quickly, then instantly went into deep water again” and would “mush out and disappear”, and that the reef “was shaped like a V instead of a triangle” (Dan Harmon, 2026; grade D). A narrow V leaves deep water between and behind its arms, whereas Albany’s ridge stays nearly flat for about 80 m inside its outer face. Behind a break the wave decays towards a stable height of about 0.4 times the depth and stops breaking where the water deepens (Dally et al., 1985); run on three shore-normal lines across each reef, this decay keeps the wave breaking over the reef for a median of 35 m at Albany and 23 m at Palm Beach against 5 m at Pratte’s and 7 m at Bunbury, and the length ranks with the outcome ($\rho = 0.77$, $p = 0.037$; Section 4.3, Table 3). On a peeling reef the wave also breaks progressively along the arm, so the ride is better measured along the crest: Palm Beach’s oblique crest keeps the wave peeling for a median of 84 m and Albany’s ridge for up to 155 m, against 11.5 m at Pratte’s and 9 m at Bunbury. This crest-line length alone does not rank the outcomes ($\rho = 0.62$, $p = 0.15$), because Boscombe and Narrowneck, which work rarely, also have long shallow ridges: a long arm is needed but the ramp must also give a surfable breaker. The calculation is one-dimensional and idealised, but the testimony, the geometry and the decay model agree: a surf reef should stay shallow for some distance shoreward of the break, with a filled, triangular planform rather than two narrow arms.

Size enables the ramp. Volume ranks with outcome ($\rho = 0.72$ for the stated volume of ten built reefs, 0.93 for the eight models; Figure 6): the three reefs that do not work are 4,900 m^3 or smaller, and both built reefs that work 25,000 m^3 or larger. The mechanism suggested by the profiles is that volume buys ramp length (ramp length $\rho = 0.93$, against 0.62 for relief, $p = 0.15$): on a 4 to 5 m bed a 1:16 to 1:20 ramp needs about 14,000 to 25,000 m^3 per 100 m of reef, depending on the crest depth. Volume is necessary rather than sufficient: Narrowneck, the largest by stated fill, works only rarely, and Burkitts works at 300 m^3 because the natural platform supplies the relief.

Peel angle and skill. Refraction over the reef arms turns designed arm angles of 45 to 81 degrees into peel angles of 16 to 62 degrees at the break, and small waves, which break in shallow water, refract most (Figure 5). Every reef therefore serves intermediate to professional surfers (refracted Hutt ratings 3 to 9, mostly 5 to 9), and only one large-wave case at Borth reaches the angle of a learner’s wave. Spilling breakers, the other beginner condition, are 0 to 29 % of the on-reef cases; a dedicated beginner reef would need a ramp of about 1:30 and 35,000 to 49,000 m^3 per 100 m (Section 4.4). No built reef was intended as, or reported to be, a beginner wave.

Tide and wave height. Even given a suitable swell, the reefs are surfable for 0 to 0.70 of the lunar month (Palm Beach 0.70, Albany 0.62, Cables 0.55, Borth 0.19, the others 0.16 or less), because the crest is in the right depth only over part of the tide. Under the Israeli tide (spring range 0.40 m) the average does not rise (mean change +0.01), but the outcome becomes a function of wave height alone: for a given swell the reef works all day or not at all (Figure 7). Short periods soften the break: the surging share is 0.21 at 6 s against 0.41 at 10 to 12 s, which favours the short-period seas of the Israeli coast.

8.2 Sediment response and the nourishment confound

Standard predictors expect a salient behind every reef (Black & Andrews, 2001; Ranasinghe et al., 2006), but where surveys allow a test the observed response is much smaller: at Boscombe +3.4 m against 84 m predicted, a ratio of 0.02 to 0.06, and at Palm Beach 2 m or less against 114 m (Section 5.3). The reasons are physical and practical. Surfing crests 1 to 2 m below MSL transmit 41 to 80 % of the wave height, and short reefs of 12 to 132 m are bypassed by diffraction; and beach management moves whole frontages by more than any reef signal (Boscombe +72 m^3/m in 2006-07). On the Gold Coast the reef effect cannot be separated from continuous nourishment (Section 5.4), so the claimed widening is not evidence of protection by the reef. The defensible role of a surf reef in coastal management is that of a control point within a nourishment scheme, adding amenity, not that of stand-alone protection (Section 5.6). At the same time, the reefs change the bed close to themselves, through settlement of 0.5 to 2 m, local scour and burial (Blacka et al., 2013, p. 72), which removes relief and degrades the wave.

8.3 Durability by material

Six of the seven geotextile or inflatable reefs suffered major damage or loss within a few years, and the seventh, Narrowneck, survives through renewal of its containers; none of the four rock reefs failed (Fisher exact $p = 0.015$; Section 6.1). The stability estimates explain the pattern: single sand-filled tubes slide at H_s of 0.34 to 1.65 m, heights that depth-limited seas reach inside every surf-reef crest, and a sealed base lets small tubes lift under breaking waves, whereas rock of 1 to 2 t resists the same waves by the Hudson formula (Section 6.3). The tearing mechanism itself is not modelled, and the reported causes are point damage, undersizing and construction faults. The comparison is confounded, because the rock reefs are the newer and better-funded projects, but the direction is consistent between the record and the calculations. Rock is not free of problems: Albany's core rock moved in a storm during construction and 18 rocks were repositioned after injuries (Sheil, 2026).

8.4 Overall rating of the built reefs

To answer the question that motivates the review, which reefs have delivered what they promised, this review rated each reef on three criteria: (A) surf quality, the quality and consistency of the wave reported or computed; (B) safety and longevity, the structural record of the material and any hazard to users; and (C) sediment and storm protection, any measured effect on the beach or on storm damage. Each criterion is scored from 1 (very poor) to 5 (very good), with 3 for mixed or neutral evidence, including “no measurable effect”; the labels are, for A, 5 excellent, 4 good, 3 inconsistent, 2 works rarely and 1 no surfable wave, and, for B, 5 no problems, 4 intact, 3 maintained or renewed, 2 hazard or repairs and 1 failed or removed (as in Table 1); ? means that no usable evidence exists, and * that a signal exists but cannot be attributed to the reef. The reefs are ranked by A, with B as tie-break; C does not affect the rank, because it is mostly unknown. The scores are this review's assessment, based on the evidence cited in Sections 3 to 6 and Appendix C, graded A (measured), B (engineering report or thesis), C (journalism or surfer media) and D (forum or single anecdote) (Table 7).

Rank	Reef	A surf	B safety, longevity	C sediment, storms	Reason (evidence)
1	Albany	5	2	?	best wave, rides of 100-120 m (Raised Water Research, n.d.-d); injuries, 18 rocks repositioned in 2026 (Sheil, 2026); too new for sediment data
2	Cables	3	4	3	rock, intact since 1999; surf-day claims conflict (Blacka et al., 2013; Mead & Borrero, 2011)
3	Burkitts	3	4	?	reshaped natural rock, no maintenance in nearly three decades; about 30 surf days a year is an estimate (Howells & O'Neal, 2024; Raised Water Research, n.d.-b)
4	Palm Beach	3	4	3*	rock, stable; wave only in E or S swell with $H_s > 0.8$ m, $T_p > 7$ s (Prenzler et al., 2023); nourishment confounds the shoreline (Section 5.4)
5	Narrowneck	2	3	2	"rarely surfed" (EcoShape, n.d.); containers settled 1-2 m and were renewed in 2017-18 (Corbett et al., 2023); nourishment confounds
6	Borth	?	4	3*	rock, no damage in 2012-2024 imagery; no surf record (Blacka et al., 2013); +30 m bulge not attributable (Section 5.3)
7	Boscombe	2	1	2	too hollow, mostly bodyboarders (Rendle, 2016); a container lost and the reef closed in 2011
8	Kovalam	2	1	3*	good first season (Collins, 2010), then "most of the waves just roll over the reef" (Raised Water Research, n.d.-c); containers torn and sunk by 2013 (Blacka et al., 2013); beach gain only designer-reported
9	Mount Maunganui	2	1	1	hollow at first, later only in "a very confined set of conditions" (Mead & Borrero, 2011); deflated, removed 2014; council reported a scour hole and more rips (NZ Herald, 2014)
10	Pratte's	1	1	2	too small; 1-5 breaking events a year (Borrero & Nelsen, 2003) and witnesses saw it break almost never (Dan Harmon, 2026, grade D); removed 2008-10
11	Opunake	1	1	?	no expected waves; bags failed, never completed (Reid & Sumner, 2021)
12	Bunbury Airwave	1	1	?	inflatable, tore during installation (ABC News, 2019)
-	Xala	?	?	?	no published information

Table 7. Rating of the built surf reefs by this review: rank, sediment criterion C and the evidence behind each score. Scores 1 = very poor to 5 = very good, with the labels of Table 1, which repeats A and B next to the reef data; ? = no evidence; * = signal not attributable to the reef.

Albany is the only reef that clearly delivers a high-quality wave, but with safety problems: reported injuries led to the repositioning of 18 rocks within a year of opening (Sheil, 2026). The two other reefs classed as working do so only in a narrow set of conditions: Palm Beach needs an easterly or southerly swell above H_s 0.8 m with T_p above 7 s and a suitable tide (Prenzler et al., 2023), and Burkitts gives about 30 surf days a year by a researcher's estimate (Raised Water Research, n.d.-b). On most days, therefore, neither offers a better wave than the surrounding coast, although Palm Beach gives longer rides than the natural surf zone when it works (Hunt et al., 2022). Those longer rides are measured in specific swell conditions; surfer testimony that singles out a "solid" session on the reef as notable (The Surf Eye, 2022), and the scarcity of surf footage from it, suggest that such days are occasional. Hard rock reefs (Cables, Burkitts, Palm Beach, Borth) score best on B, with no structural failure; geotextile and inflatable reefs score worst on B, with six of the seven at the minimum. Criterion C is unknown or confounded for every reef: no score above 3 survives attribution, and the only negative evidence (Mount Maunganui) comes from a press report. The rating thus repeats the central result of this review: a surf reef that works is large, built of rock and given a long ramp, and its benefit is surfing rather than beach protection.

8.5 Limitations and what would change the conclusions

The sample is small (8 to 13 reefs), and the outcome classes are this review's readings of sources that are mostly by the designers; every association is descriptive and confounded, because the large reefs are also the recent, better-funded rock reefs. No reef has a published wave climate at the structure, so the surfable fractions are shares of the time with a suitable swell, not days per year, and the reported surfable days are not comparable (Section 3.6). The geometry is idealised for six of the nine models, and the Albany surface is fitted to a design rule (low confidence). The profile results depend on where the cut is taken: for Albany, Cables and Borth the cut was moved to the surfed part of the reef, and for Borth, where no source locates the break, its position is a judgement (Figure E1, Appendix E). The incidence angle is assumed for five of eight reefs, and the wave model is one-dimensional linear shoaling with $\gamma = 0.78$, normal incidence and no focusing; it stops at the break and so does not test the post-break depth. The Mead and Black (2001b) fit is extrapolated for 32 to 100 % of the on-reef cases at every reef except Palm Beach. Three key sources (Black & Andrews, 2001; Hutt et al., 2001; Mead & Black, 2001a) were used only as reproduced by others. The Haifa analysis rests on a synthetic profile and on the author's unpublished field observations (Appendix G).

The conclusions would change with (1) measured surfable-day counts tied to a wave record at any reef; (2) a small reef that works, or a large rock reef with a long ramp whose wave fails; (3) a controlled before/after survey showing a reef-made salient at a site without nourishment; (4) a geotextile reef that survives a decade in the breaking zone without renewal; and (5) Haifa breaking depths that confirm the early breaking observed in the author's field observations (Appendix G; H_s/d of about 0.35 to 0.40), which would move the recommended crest from 0.6-1.0 m to 1.2-1.5 m below MSL.

8.6 Implications for design

The findings point to one design rule set. A surf reef should be large enough to carry a long, gentle ramp (1:16 to 1:20 for an advanced to intermediate wave; never steeper than 1:12), should stay shallow for some distance shoreward of the break, with a filled planform rather than narrow arms, and should be built of rock. Its crest depth should be chosen for the local wave heights and the local tide, with a margin for settlement and burial, and its success should be judged by surfing, with any coastal benefit tested by before/after surveys with controls and not assumed. For Haifa, the small tide removes the tidal window and leaves wave height as the control, while the mobile sand makes burial the main risk (Section 7).

9 Conclusions

1. **Record.** Of 13 surf reefs built or reshaped since 1997, three work for surfing (Burkitts, Palm Beach, Albany), three work rarely, two are disputed, three do not work and two are unknown (Section 3). Only Albany works well: it gives a high-quality, hollow wave with rides of 100 to 120 m (Raised Water Research, n.d.-d), but it has had safety problems, with injuries and 18 rocks repositioned within a year of opening (Sheil, 2026). The other two "working" reefs do not work in most conditions and are rarely better than the other surf spots of their area: Palm Beach breaks well only in an easterly or southerly swell above H_s 0.8 m with T_p above 7 s (Prenzler et al., 2023), and Burkitts gives about 30 surf days a year by a researcher's estimate (Raised Water Research, n.d.-b). In this review's rating (Section 8.4), rock reefs score best on safety and longevity, geotextile and inflatable reefs worst, and the effect on sediment is unknown or confounded at every reef.
2. **Size.** The reefs that work are large: the three that do not work are 4,900 m^3 or smaller, both built reefs that work 25,000 m^3 or larger (Spearman $\rho = 0.72$ to 0.93; descriptive). Volume buys ramp length: on a 4 to 5 m bed a 1:16 to 1:20 ramp needs 15,000 to 25,000 m^3 per 100 m.
3. **Ramp, not gap, and the depth after the break.** The crest depth decides which waves break; the length and gradient of the ramp decide how. Read on the part of each reef where the wave is ridden, the ramp length ranks with the outcome at $\rho = 0.93$ ($p = 0.005$) and the mean seaward slope at 0.77 ($p = 0.039$), whereas the crest depth does not (0.39, $p = 0.36$). Over similar depth gaps, a long 1:17 ramp gives a plunging, surfable wave and a 1:3 rise a collapsing one (Section 4.2), which explains the Boscombe wave, steep and suited mostly to bodyboarders and advanced surfers; Albany, on its 99 m ridge at 1:19.8, plunges like Palm Beach. The depth just after the break matters too: the reef must stay shallow for some distance shoreward of the break so that the wave keeps breaking instead of re-forming in the deeper water behind it (Section 4.3, Table 3, idealised one-dimensional decay model on three shore-normal lines per reef: a median of 35 m of sustained breaking at Albany and 23 m at Palm Beach against 5 m at Pratte's and 7 m at Bunbury, ranking with the outcome at $\rho = 0.77$, $p = 0.037$). Along the crest, where a peeling wave is ridden, the shallow arm is longer (Palm Beach 84 m, Albany up to 155 m, Pratte's 11.5 m, Bunbury 9 m), but crest length alone does not separate the reefs that work from Boscombe and Narrowneck ($\rho = 0.62$). Surfers who rode Pratte's reported that the wave reached deep water again immediately and "mushed out", and that a triangular rather than V-shaped reef would have held it (Dan Harmon, 2026; grade D testimony); a filled, triangular planform that stays shallow toward the shore is preferable to narrow arms (Sections 4.3 and 8.1).
4. **Surfer level.** Refraction brings every reef arm to peel angles of 16 to 62 degrees at the break, suited to intermediate to professional surfers; no reef offers a beginner's wave, which would need a spilling ramp of about 1:30 (Section 4.4).

5. **Sand.** The observed shoreline response is 0.02 to 0.06 of the standard prediction at the sites that can be tested; on the Gold Coast the reef effect cannot be separated from nourishment. A surf reef can serve as a control point within a nourishment scheme, not as stand-alone protection (Section 5).
6. **Materials.** Six of seven fabric reefs suffered major damage or loss within a few years, none of four rock reefs did (confounded by date and budget). Breaking-zone loads lift small or sealed tubes, but the tearing mechanism is unmodelled (Section 6).
7. **Haifa.** The small tide (spring range 0.4 m) makes a reef wave depend on wave height alone, not on the hour, but does not reduce storm loads. A Haifa reef should be rock, 100 m or longer, about 14,000 to 20,000 m^3 per 100 m, with a crest 1.0 to 1.2 m below MSL, a seaward ramp of 1:16 to 1:20 (never steeper than 1:12), a crest that stays shallow shoreward of the break, and a structure that stays well above the seasonal high bed: sand around the reef does not change the break while the crest is clear, but burial is a durability and maintenance risk. That crest is a compromise set with the model's breaking depths: it breaks the common winter swells (every wave from H_0 0.6 to 0.8 m upward) and keeps about 0.5 m of water over the rock at the yearly low water level, whereas a shallower crest would add the smallest summer seas but leave too little water for larger winter waves; the field breaking depths (Appendix G), about 0.2 to 0.3 m deeper than the model, are indicative only and suggest the model is conservative. Multibeam and seasonal bed surveys, a local wave climate, field calibration of the breaking depth and a two-dimensional wave model or flume test are needed before design, and before/after monitoring with control sections after construction (Section 7.5).

References

Scholarly and technical sources

Alvarez, E., Rubio, R., & Ricalde, H. (2006). Back to the beach in Mexico: Shoreline restored with geotextile tubes as submerged breakwaters. *Geosynthetics Magazine.* https://geosyntheticsmagazine.com/2006/06/01/shoreline-restored-with-geotextile-tubes-as-submerged-breakwaters/

Alvarez, E., Rubio, R., & Ricalde, H. (2007). Beach restoration with geotextile tubes as submerged breakwaters in Yucatan, Mexico. *Geotextiles and Geomembranes*, *25*(4-5), 233–241. https://doi.org/10.1016/j.geotexmem.2007.02.005 [not read].

Antunes do Carmo, J. S., Neves, M. G., & ten Voorde, M. (2011). Designing a multifunctional artificial reef: Studies on the influence of parameters with most influence in the vertical plane. *Journal of Coastal Conservation*, *15*(1), 99–112. https://doi.org/10.1007/s11852-010-0124-1 [not read; metadata checked in Crossref].

Baker, S., Brashear, P., Tschirky, P., Marrone, J., & Larson, M. (2018). Design and physical model studies of innovative living breakwaters. *Coastal Engineering Proceedings*, *36*, papers.59. https://doi.org/10.9753/icce.v36.papers.59 [abstract only].

Bar, D., & Drimer, N. (2023). Preliminary design tools for hydrodynamic aspects of submerged impermeable breakwaters. *Journal of Marine Science and Engineering*, *11*(2), 236. https://doi.org/10.3390/jmse11020236

Barlow, M. (2013). *The effects of anthropometrical, physiological and environmental factors on surfing performance* [Doctoral dissertation]. Leeds Metropolitan University. https://eprints.leedsbeckett.ac.uk/id/eprint/592/ [Table 1.1, p. 12, reproduces the Hutt et al. (2001) skill table].

Battjes, J. A. (1974). Surf similarity. *Proceedings of the 14th International Conference on Coastal Engineering (ICCE 1974)*, 466–480. https://doi.org/10.9753/icce.v14.26 Chapter 26.

Bitan, M., & Zviely, D. (2020). Sand beach nourishment: Experience from the Mediterranean coast of Israel. *Journal of Marine Science and Engineering*, *8*(4), 273. https://doi.org/10.3390/jmse8040273

Black, K. P., & Andrews, C. J. (2001). Sandy shoreline response to offshore obstacles. Part 1: Salient and tombolo geometry and shape; Part 2: Discussion of formative mechanisms. *Journal of Coastal Research.* [Volume and pages not confirmed; a secondary listing gives Special Issue 29, 82-93, unverified. Original not read: criteria and Eqs. 30-31 as reproduced by Blacka et al., 2013, Section 2.5.2, p. 21.]

Blacka, M. J., Shand, T. D., Carley, J. T., & Mariani, A. (2013). A review of artificial reefs for coastal protection in NSW (WRL Technical Report 2012/08, final report of 11 June 2013). https://doi.org/10.4225/53/58e1dc7368c7a [pages cited are printed report pages, = PDF page minus 12].

BMT Western Australia. (2018). Middleton Beach artificial surf reef environmental impact assessment (Report 1345_001/1_Rev2, September 2018). https://epa.wa.gov.au/sites/default/files/Referral_Documentation/Environmental%20Impact%20Assessment_0.pdf

Borrero, J. C., & Nelsen, C. (2003). Results of a comprehensive monitoring program at Pratte's Reef. In K. Black & S. Mead (Eds.), *Proceedings of the 3rd International Surfing Reef Symposium*. Raglan, New Zealand (not verified). [The copy read prints neither venue nor year; both are taken from the publisher record; the content (monitoring to October 2002) is consistent with them, but year and venue could not be confirmed. Not verified.]

Bowen, A. J., Inman, D. L., & Simmons, V. P. (1968). Wave "set-down" and set-up. *Journal of Geophysical Research*, *73*(8), 2569–2577. https://doi.org/10.1029/JB073i008p02569 [not read; cited for the set-down and set-up shown schematically in Figure 3].

Briganti, R., van der Meer, J. W., Buccino, M., & Calabrese, M. (2003). Wave transmission behind low-crested structures. *Coastal Structures 2003*, 580–592. [Original not read; equation as reproduced by Blacka et al., 2013, Eq. 29.]

Carmel, Z., Inman, D. L., & Golik, A. (1985). Characteristics of storm waves off the Mediterranean coast of Israel. *Coastal Engineering*, *9*(1), 1–19. https://doi.org/10.1016/0378-3839(85)90024-9 [abstract only; full text not read].

Channel Coastal Observatory. (2026a). Beach profile surveys, Boscombe frontage, 11 profile lines, 316 surveys 2004-2026 [Data set]. https://coastalmonitoring.org/ (accessed October 2026) [Appendix C cites the same series by one survey date: lines 5f00424 to 429A, survey of 20 April 2010.]

Channel Coastal Observatory. (2026b). Beach profile surveys, Boscombe lines 5f00424 to 429A, survey of 20 April 2010 [Data set]. https://coastalmonitoring.org/ (accessed 6 October 2026).

Chien, A., Tseng, F., & Tang, A. (2013). Geotextile tube application on beach refurbishment in UAE. *WIT Transactions on Ecology and the Environment*, *169*, 231–239. https://doi.org/10.2495/CP130201 [not read; metadata checked in Crossref].

Chien, A., Wu, S., Tseng, F., & Tang, A. (2014). Geotextile tubes application on beach nourishment in UAE. *International Journal of Environmental Science and Development*, *5*(5), 506–509. https://doi.org/10.7763/IJESD.2014.V5.535 [metadata confirmed in Crossref; full text not read].

City of Albany. (n.d.). Southern Ocean Surf Reef (SOSR) [Project pages, including the detailed-design table; page current 24 September 2026]. https://www.albany.wa.gov.au/council/projects/completed-projects/southern-ocean-surf-reef-sosr.aspx

City of Albany. (2023, May). Albany artificial surf reef: Community update and FAQs [Community information document]. [Design claims, not measurements.]

City of Gold Coast. (n.d.). Artificial reef [Feature layer, Open Data, CC BY 3.0]. https://data-goldcoast.opendata.arcgis.com/datasets/c9d0b521374740cc9e9561b04c453736_0

Corbett, B. B., Mulcahy, M. G., Elliott-Perkins, Z., & Hunt, S. (2023). Narrowneck artificial reef renewal. *Australasian Coasts & Ports 2023 Conference*, 663–669. https://doi.org/10.3316/informit.477067242341136 Sunshine Coast, Queensland, 15-18 August 2023.

Corvaro, S., Lorenzoni, C., Mancinelli, A., Marini, F., & Rocchi, S. (2022). Hydraulic performance of geotextile sand containers for coastal defenses. *Journal of Marine Science and Engineering*, *10*, 1321. https://iris.univpm.it/retrieve/dcfda89a-7b3a-4790-

92d7-94e0aa4b9eb7/jmse-10-01321-v2%20%281%29.pdf [open-access copy; DOI not extracted].

Dahm, J., & Gibberd, B. (2013). Mount Maunganui Reef: Assessment of management options (Focus Report No. 131). https://www.boprc.govt.nz/media/558385/mount-maunganui-reef-assessment-of-management-options.pdf

Dally, W. R., Dean, R. G., & Dalrymple, R. A. (1985). Wave height variation across beaches of arbitrary profile. *Journal of Geophysical Research*, *90*. https://doi.org/10.1029/JC090iC06p11917 (C6), 11917-11927 [full text not accessed; used as cited by other sources].

d'Angremond, K., van der Meer, J. W., & de Jong, R. J. (1996). Wave transmission at low-crested structures. *Proceedings of the 25th International Conference on Coastal Engineering*, 2418–2427. [Original not read; equations as reproduced by Blacka et al., 2013, Eqs. 25-27, with the standard signs.]

Davidson, M. (2010). 6-month interim report: Performance of Boscombe surfing reef [Technical report]. [not obtained; cited as reported by Rendle (2016)].

Dean, R. G., & Dalrymple, R. A. (1991). *Water wave mechanics for engineers and scientists*. World Scientific. (Original work published 1984)

EMODnet Bathymetry Consortium. (2024). EMODnet digital bathymetry (DTM 2024) [Data set]. https://doi.org/10.12770/cf51df64-56f9-4a99-b1aa-36b8d7b743a1

Esri, Maxar, Earthstar Geographics, & the GIS User Community. (n.d.). World Imagery and World Imagery Wayback [Satellite and aerial imagery, dated releases cited in the text]. https://livingatlas.arcgis.com/wayback/

Galvin, C. J. (1968). Breaker type classification on three laboratory beaches. *Journal of Geophysical Research*, *73*(12), 3651–3659. https://doi.org/10.1029/JB073i012p03651 [not read; cited through Battjes, 1974].

Garmin Navionics. (n.d.). Navionics SonarChart [Crowd-sourced bathymetric chart, web viewer; not for navigation]. https://webapp.navionics.com/ (accessed October 2026).

GHD. (2021). Tidal inundation monitoring and modelling report (Appendix B, Table 1, "Tidal planes for Port of Bunbury", citing Department of Defence, 2018). https://www.epa.wa.gov.au/sites/default/files/PER_documentation2/App%20B%20-%20Tidal%20innudation%20report.pdf

González Leija, M. B., Chávez, X., Alvarez, E., Mendoza, E., & Silva, R. (2014). Experimental study on geotextile tube applications as submerged breakwaters for beach protection in Yucatan, Mexico. *Coastal Engineering Proceedings*, *34*, posters.25. https://doi.org/10.9753/icce.v34.posters.25

Government Company for the Protection of Mediterranean Sea Cliffs. (2026). Projects list: Ashkelon North geotubes, Netanya breakwaters, Ashkelon geotube maintenance [In Hebrew]. https://www.mccp.co.il/page.php?type=projectsGroups (accessed 15 October 2026).

Haskoning UK. (2011). Borth coastal protection scheme, Phase 1: Drawings 9V5090/1001, 1020, 1021, 1022 and 1023, rev. C1 (for construction, January 2011). http://www.borthcommunity.info/index.php/planning-drawings/65-planning-drawings

Hunt, S., Britton, G., Messiter, D., Prenzler, P., Knight, S., & Watterson, E. (2022). Palm Beach shoreline project: Innovative coastal management solution. *Coastal Engineering Proceedings*, *37*, management.66. https://doi.org/10.9753/icce.v37.management.66 [ICCE 2022; published online 2023].

Hutt, J. A., Black, K. P., & Mead, S. T. (2001). Classification of surf breaks in relation to surfing skill. *Journal of Coastal Research*, *Special Issue 29*, 66–81. http://www.jstor.org/stable/25736206 [original not read; Table 3 used as reproduced by Lewis et al. (2015)].

Jackson, L. A., & Corbett, B. (2007). Review of existing multi-functional artificial reefs. *Australasian Conference on Coasts and Ports 2007*.

Jackson, L. A., Corbett, B., Tomlinson, R., McGrath, J., & Stuart, G. (2007). Narrowneck Reef: Review of seven years of monitoring. *Shore & Beach*, *75*(4), 67–79. http://hdl.handle.net/10072/17995

Jackson, L. A., Tomlinson, R., Corbett, B., & Strauss, D. (2012). Long term performance of a submerged coastal control structure: A case study of the Narrowneck multi-functional artificial reef. *Coastal Engineering Proceedings*, *33*, structures.54. https://doi.org/10.9753/icce.v33.structures.54

Kurian, J. (1995). Collective action for common property resource rejuvenation: The case of people's artificial reefs in Kerala State, India. *Human Organization*, *54*, 160–168. [not read; cited as reported by Mead & Borrero (2011)].

Land Information New Zealand. (n.d.). Standard port tidal levels (Tauranga) and Notes about the Moturiki annual mean sea level data (updated 16 February 2023). https://www.linz.govt.nz/guidance/marine-information/tide-prediction-guidance/standard-port-tidal-levels

Land Information New Zealand. (2010–2011). Bay of Plenty 0.125 m urban aerial photos [Aerial imagery, tile BD37_1000_1314; CC BY 4.0]. https://nz-imagery.s3.ap-southeast-2.amazonaws.com/bay-of-plenty/bay-of-plenty_2010-2011_0.125m/rgb/2193/BD37_1000_1314.tiff

Land Information New Zealand. (2011–2012). Taranaki 0.4 m rural aerial photos [Aerial imagery, tile BJ28_5000_0806]. https://nz-imagery.s3.ap-southeast-2.amazonaws.com/taranaki/taranaki_2011-2012_0.4m/rgb/2193/BJ28_5000_0806.tiff

Land Information New Zealand. (2026). Standard port tidal levels, Port Taranaki, 1 July 2026 to 30 June 2027. https://www.linz.govt.nz/guidance/marine-information/tide-prediction-guidance/standard-port-tidal-levels [values read from a search extract on 2026-10-19].

Lewis, J., Hunt, S., & Evans, T. (2015). Quantification of surfing amenity for beach value and management. *NSW Coastal Conference 2015*. https://www.coastalconference.com/2015/papers2015/James%20Lewis.pdf

Longuet-Higgins, M. S., & Stewart, R. W. (1964). Radiation stresses in water waves; a physical discussion, with applications. *Deep-Sea Research*, *11*(4), 529–562. https://doi.org/10.1016/0011-7471(64)90001-4 [not read].

López, I., Tinoco, H., Aragonés, L., & García-Barba, J. (2016). The multifunctional artificial reef and its role in the defence of the Mediterranean coast. *Science of the Total*

Environment, *550*, 910–923. https://doi.org/10.1016/j.scitotenv.2016.01.180 [abstract only, via the University of Alicante record].

Lyons, M. (1992). Design studies for an artificial surfing reef [Unpublished report]. [Cited as reproduced in Blacka et al. (2013, Fig. A.2); title and venue as given there; not read.]

Maritime Safety Queensland. (2026). Semidiurnal tidal planes 2026. https://www.msq.qld.gov.au/ (file tides/tidal-planes/2026-semidiurnal-tidal-planes.pdf).

Marrone, J., Zhou, S., Brashear, P., Howe, B., & Baker, S. (2019). Numerical and physical modeling to inform design of the Living Breakwaters project, Staten Island, New York. *Proceedings of the Coastal Structures Conference 2019.* Hanover, Germany.

McCowan, J. (1894). On the highest wave of permanent type. *Philosophical Magazine, Series 5*, *38*, 351–358. [Not read; title and volume differ between reference lists (Goda, 2010: ”On the highest waves in water”, vol. 36); the value 0.78 was read in Weggel, 1972, Eq. 6.]

Mead, S., & Black, K. (1999). A multipurpose, artificial reef at Mount Maunganui Beach, New Zealand. *Coastal Management*, *27*(4), 355–365. https://doi.org/10.1080/089207599263767

Mead, S., Blenkinsopp, C., Moores, A., & Borrero, J. (2010). Design and construction of the Boscombe multi-purpose reef. *Coastal Engineering Proceedings*, *32*, structures.58. https://doi.org/10.9753/icce.v32.structures.58 [ICCE 2010; published online 2011; designers].

Mead, S. T. (2000). *Incorporating high-quality surfing breaks into multi-purpose reefs* [Doctoral dissertation]. University of Waikato. https://researchcommons.waikato.ac.nz/handle/10289/13967 [Eq. 6.5, printed p. 115; Table 6.2, printed p. 121].

Mead, S. T., & Black, K. P. (2001a). Field studies leading to the bathymetric classification of world-class surfing breaks. *Journal of Coastal Research*, *Special Issue 29*, 5–20. [Pages conflict (5-20 or 5-10); not read; content as summarised by Blacka et al., 2013, Section 2.6.]

Mead, S. T., & Black, K. P. (2001b). Predicting the breaking intensity of surfing waves. *Journal of Coastal Research*, *Special Issue 29*, 51–65. [Pages conflict (51-65 or 103-130); read as Chapter 6 of Mead, 2000. Letters a/b assigned by title order.]

Mead, S. T., Black, K. P., & Moores, A. (2007). Mount Maunganui Reef: Amalgamating design and the constraints of construction. *Coasts and Ports 2007: 18th Australasian Coastal and Ocean Engineering Conference*, 768–773. https://doi.org/10.3316/informit.816670201626573

Mead, S. T., & Borrero, J. C. (2011). Multi-purpose reefs: A decade of applications. *Coasts and Ports 2011: 20th Australasian Coastal and Ocean Engineering Conference.* https://doi.org/10.3316/informit.715072832934166 [designers].

Mortensen, S. B., Hibberd, S., Kaergaard, K., Kristensen, S. E., Deigaard, R., & Hunt, S. (2015). Concept design of a multipurpose submerged control structure for Palm Beach, Gold Coast. *Australasian Coasts & Ports Conference 2015.* https://www.dhigroup.com/upload/publications/coastsea/Mortensen_2015.pdf

National Tidal and Sea Level Facility. (n.d.). Chart datum and ordnance datum. https://ntslf.org/tides/datum

NOAA Center for Operational Oceanographic Products and Services. (2026). Tidal datums, station 9410840 Santa Monica, CA (epoch 1983-2001) [Data set]. https://api.tidesandcurrents.noaa.gov/mdapi/prod/webapi/stations/9410840/datums.json?units=metric

NOAA National Geophysical Data Center. (2010). Santa Monica, California 1/3 arc-second NAVD 88 coastal digital elevation model [Data set]. https://www.ngdc.noaa.gov/metaview/page?xml=NOAA/NESDIS/NGDC/MGG/DEM/iso/xml/726.xml

Oh, Y. I., & Shin, E. C. (2006a). Application of submerged geotextile tubes for erosion prevention in east coast of Korea. In F. Kuwano & J. Koseki (Eds.), *Geosynthetics: Proceedings of the 8th International Conference on Geosynthetics* (pp. 757–760). Millpress. https://library.geosyntheticssociety.org/wp-content/uploads/resources/proceedings/C.2.03%20Application%20of%20submerged%20geotextile%20tubes%20for%20erosion%20prevention%20in%20east%20coast%20of%20Korea.pdf

Oh, Y. I., & Shin, E. C. (2006b). Using submerged geotextile tubes in the protection of the E. Korean shore. *Coastal Engineering*, *53*(11), 879–895. https://doi.org/10.1016/j.coastaleng.2006.06.005 [paywalled, not read].

Pattiaratchi, C. B. (1999). Design studies for an artificial surfing reef: Cable Station, Western Australia (Report ED1483CP). http://joas.free.fr/studies/bei/g2s/reefdesign.pdf [mirror copy].

Peregrine, D. H. (1983). Breaking waves on beaches. *Annual Review of Fluid Mechanics*, *15*, 149–178. https://doi.org/10.1146/annurev.fl.15.010183.001053 [not read; typical plunging-lip thickness].

Pitt, A. (2009). Critique of Cables Reef. [original not seen; cited as reported by Raised Water Research (2019b)].

Prenzler, P., Hunt, S., Elliott-Perkins, Z., Hamilton, D., Messiter, D., Wharton, C., & Watterson, E. (2023). Monitoring of the Palm Beach artificial reef. *Coastal Engineering Proceedings*, *37*, structures.65. https://doi.org/10.9753/icce.v37.structures.65 [ICCE 2022 extended abstract; proceedings volume dated 2023 in Crossref].

Ranasinghe, R., & Turner, I. L. (2006). Shoreline response to submerged structures: A review. *Coastal Engineering*, *53*(1), 65–79. https://doi.org/10.1016/j.coastaleng.2005.08.003 [abstract read; full text not read].

Ranasinghe, R., Turner, I. L., & Symonds, G. (2006). Shoreline response to multi-functional artificial surfing reefs: A numerical and physical modelling study. *Coastal Engineering*, *53*(7), 589–611. https://doi.org/10.1016/j.coastaleng.2005.12.004 [not read; thresholds as reproduced by Blacka et al., 2013, pp. 22-23, mechanism as summarised by Rendle, 2016, p. 43].

Rendle, E. J. (2016). *The environmental, social and economic impacts of an artificial surf reef: The UK experience* [Doctoral dissertation]. University of Plymouth. https://doi.org/10.24382/4536 [pages cited are printed pages, = PDF page minus 3].

Rendle, E. J., & Davidson, M. A. (2012). An evaluation of the physical impact and structural integrity of a geotextile surf reef. *Coastal Engineering Proceedings*, *33*, structures.21. https://doi.org/10.9753/icce.v33.structures.21

Rigden, T., Stewart, T., Allsop, W., & Johnson, A. (2013). Surf reefs: Physical modelling results, not pipe dreams (HR Wallingford Report HRPP576). https://eprints.hrwallingford.com/931/1/HRPP576_SurfReefs.pdf

Robertson, B., Nistor, I., Hall, K., & Buckham, B. (2014). Remote measurement and prediction of breaking wave parameters. *Coastal Engineering Proceedings*, *34*, waves.41. https://doi.org/10.9753/icce.v34.waves.41

Roelse, A., Granger, H. W., & Graham, J. W. (1975). The adjustment of the Australian levelling survey 1970-1971 (Technical Report 12, 2nd ed.)

Rosen, D. S. (2001). A review of sea level monitoring status in Israel. https://gloss-sealevel.org/sites/gloss/files/publications/documents/israel_2001.pdf

Royal Commission on the Ancient and Historical Monuments of Wales. (2020). Coflein: Coastal defences, Borth (NPRN 424699). https://coflein.gov.uk/en/site/424699 (accessed 6 October 2026).

Royal HaskoningDHV. (2011). West of Wales Shoreline Management Plan 2, Section 4: Coastal Area C introduction (p. 4C.3, water-level table for Aberystwyth). https://www.grwparfordirolgorllewincymru.cymru/sites/default/files/2019-06/4c1%20-%20Section%204%20Coastal%20Area%20C%20Introduction.pdf

Royal HaskoningDHV. (2015). Albany artificial surf reef feasibility study, Middleton Beach: Executive summary (Report PA1039_RP150622, 13 July 2015). https://www.albany.wa.gov.au/documents/435/albany-artificial-surf-reef-feasibility-study-middleton-beach [Option B contour plan used as reproduced in Raised Water Research (n.d.-d)].

Skelly Engineering. (2000). Pratte's Reef [Plan-view design drawing, 110 bags plus 30 optional]. https://raisedwaterresearch.com/wp-content/uploads/2019/07/Pratts-Reef-Diagram.png Dated c. 2000. As reproduced by Raised Water Research at https://raisedwaterresearch.com/wp-content/uploads/2019/07/Pratts-Reef-Diagram.png and as Fig. 4 of Borrero and Nelsen (2003).

Spanier, E., & Zviely, D. (2022). Key environmental impacts along the Mediterranean coast of Israel in the last 100 years. *Journal of Marine Science and Engineering*, *11*(1), 2. https://doi.org/10.3390/jmse11010002

State Comptroller of Israel. (2020). How government authorities deal with the processes of coastal cliff destruction [In Hebrew; chapter of the Local government audit reports 2020, 128-page PDF]. https://library.mevaker.gov.il/sites/DigitalLibrary/Documents/2020/Shilton-2020/2020-Shilton2020-102-Matzok.pdf [page numbers cited are PDF pages].

State Comptroller of Israel. (2024, July). How government authorities deal with the processes of coastal cliff destruction: Follow-up audit [In Hebrew; Local government audit report, 76-page PDF]. https://library.mevaker.gov.il/sites/DigitalLibrary/Documents/2024/Shilton/2024-Shilton-501-Matzok.pdf [page numbers cited are PDF pages].

Sulaiman, D. M., Bachtiar, H., Taufiq, A., & Hermanto. (2015). Beach profile changes due to low crested breakwaters at Sigandu Beach, Central Java. *Procedia Engineering*, *116*, 510–519. https://doi.org/10.1016/j.proeng.2015.08.320 [abstract only; full text not accessible].

Sulaiman, D. M., Junarsa, D., Bachtiar, H., & Nurhabni, F. (2017). Lessons learned from deploying low crested breakwaters in North Coast of Java. *Asian and Pacific Coasts 2017*, 445–454. https://doi.org/10.1142/9789813233812_0041 [not read].

Taranaki Regional Council. (2009). Opunake artificial reef monitoring programme 2005-2009 (Technical Report 2009-33). [Private copy; the council site blocks automated access.]

ten Voorde, M., Antunes do Carmo, J., & Neves, M. G. (2009). Designing a preliminary multifunctional artificial reef to protect the Portuguese coast. *Journal of Coastal Research*, *25*(1), 69–79. https://doi.org/10.2112/07-0827.1 [not read].

US Army Coastal Engineering Research Center. (1984). Shore protection manual (4th ed.) [Not read; K_D value as tabulated by Verhagen, 2012.]

US Army Corps of Engineers. (2002). Coastal engineering manual (EM 1110-2-1100), Part III, Chapter 2: Longshore sediment transport. https://pdhonline.com/courses/c773/Part-III-Chap_2entire.pdf [copy of the public-domain manual; CERC formula, Eq. III-2-7b].

van Steeg, P., Vastenburg, E., Bezuijen, A., Zengerink, E., & de Gijt, J. (2011). Large-scale physical model tests on sand-filled geotextile tubes and containers under wave attack. *Coastal Structures 2011*. https://repository.tudelft.nl/file/File_6363c965-0635-4e41-9e96-1dde8ce20c3b (extended abstract, 2 pp.)

Verhagen, H. J. (2012). CT5308 Breakwaters and closure dams, Chapter 7: Stability of randomly placed rock mounds [Lecture notes]. https://ocw.tudelft.nl/wp-content/uploads/ct530807.pdf

Vieira da Silva, G., Hamilton, D., Strauss, D., Murray, T., & Tomlinson, R. (2021). Sediment pathways and morphodynamic response to a multi-purpose artificial reef: New insights. *Coastal Engineering*, *171*, 104027. https://doi.org/10.1016/j.coastaleng.2021.104027

Walker, J. R. (1975). Surf parameters. *Beach and Surf Parameters Workshop*. https://repository.library.noaa.gov/view/noaa/36614/noaa_36614_DS1.pdf University of Hawaii, 7 November 1975. [workshop summary of Walker's 1974 study (LOOK Laboratory Technical Report TR-30); page cited is the PDF page].

Weggel, J. R. (1972). Maximum breaker height for design. *Proceedings of the 13th International Conference on Coastal Engineering*, 419–432. https://doi.org/10.9753/icce.v13.21

Welsh Government. (2022). LiDAR 2020-22, 1 m DSM and DTM, tile SN6089, flown 19 March 2022 [Data set]. https://dmwproductionblob.blob.core.windows.net/lidar-zips/2020-22/dsm/wg_del_29_260289_20220319dsm.tif

Western Australia Department of Transport. (2009). Two Rocks to Naturaliste lidar 2009 (survey Lidar2009TRBS, file BU2009TRBS_Lidar.bag) [Data set]. https://dotazprdauegisextpubst01.blob.core.windows.net/transport-wa-public/bathymetry/rasters/BU2009TRBS_Lidar.bag

Western Australia Department of Transport. (2020). Bathymetry point data, Fremantle ("FR") package: Surveys of 30 March 1998, 22 November 1999 and 1 February 2002 [Data set, CC BY 4.0; file dated 11 November 2020]. https://catalogue.data.wa.gov.au/dataset/dot-bathy [the December 1995 pre-construction survey used in Section 5 is from the same DoT WA FR package; its listing should be added].

Western Australia Department of Transport, & Bureau of Meteorology. (2024). Tide predictions by location, WA 2024: Albany (datum of predictions LAT). https://transport.wa.gov.au/mediaFiles/marine/MAR_P_2024_Tide_Predictions_by_Location_WA.PDF

Western Australia Department of Transport, & Bureau of Meteorology. (2026). Tide predictions by location, WA 2026: Albany, Bunbury and Fremantle [Predicted high and low waters, heights above LAT]. https://transport.wa.gov.au/getmedia/482d7b83-1a48-4220-9e2c-b07ebfcf8594/MAR_P_2026_Tide_Predictions_by_Location_WA.pdf [downloaded 2026-10-19; annual extremes taken from three of four quarterly blocks, so they are lower bounds].

World Bank. (2026). World Development Indicators: PA.NUS.FCRF Official exchange rate (LCU per US$, period average) and FP.CPI.TOTL Consumer price index (2010 = 100), United States [Data set]. https://api.worldbank.org/v2/country/USA/indicator/PA.NUS.FCRF?format=json (retrieved 15 October 2026).

Zalul. (2025, July 2). Appeal to the appeals sub-committee of the National Planning and Building Council against the decision of the Coastal Environment Protection Committee of 21 May 2025 (plan 304-0860023, Kiryat Haim breakwaters) [In Hebrew; 28 pp.; quotes an expert opinion by N. Drimer of 17 November 2024]. https://zalul.org.il/wp-content/uploads/2025/07/%D7%A2%D7%A8%D7%A8-%D7%A9%D7%95%D7%91%D7%A8%D7%99-%D7%92%D7%9C%D7%99%D7%9D-%D7%A7%D7%A8%D7%99%D7%AA-%D7%97%D7%99%D7%9D-02.07.25.pdf

Zviely, D., Kit, E., & Klein, M. (2007). Longshore sand transport estimates along the Mediterranean coast of Israel in the Holocene. *Marine Geology*, *238*(1-4), 61–73. https://doi.org/10.1016/j.margeo.2006.12.003

Zviely, D., Kit, E., Rosen, B., Galili, E., & Klein, M. (2009). Shoreline migration and beach-nearshore sand balance over the last 200 years in Haifa Bay (SE Mediterranean). *Geo-Marine Letters*, *29*(2), 93–110. https://doi.org/10.1007/s00367-008-0126-2

Media, web and other non-academic sources

4Surfers. (2025). Israeli surf forecast [Hebrew-language wave forecast for the Israeli coast; consulted July to October 2025]. https://4surfers.co.il/

ABC News. (2019, December 16). *Bunbury's Back Beach surfers deflated as artificial reef, Airwave, tears during installation.* https://www.abc.net.au/news/2019-12-16/word-first-surf-reef-tears-during-installation/11803228

ABC News. (2022, September 3). *Millions spent to protect Gold Coast beaches, but climate change poses a huge challenge.* https://www.abc.net.au/news/2022-09-03/gold-coast-no-stranger-to-beach-erosion/101381812 [statements attributed to the City of Gold Coast and its engineers].

ACE Geosynthetics. (n.d.). Beach nourishment, Fujarah, UAE [Case study, 2010-2014]. https://www.geoace.com/case/Marine-and-Coastal-Structures-Construction/Beach-Nourishment,-Fujarah,-UAE [manufacturer page, read].

Bluecoast Consulting Engineers. (n.d.). Artificial reefs for coastal protection and surfing [Project page with the Nearmap aerial of the Palm Beach reef]. https://www.bluecoastconsulting.com.au/artificialreefs

Borth Community. (n.d.). Borth coastal defence overview. http://www.borthcommunity.info/index.php/planning-drawings/235-borth-coastal-defence/46-overview

Bunbury Mail. (2022, March 22). *Airwave founder Troy Bottegal moves towards making Bunbury WA's premier surfing destination.* https://www.bunburymail.com.au/story/7667604/bunbury-still-set-to-become-surfing-destination/ [developer's account].

Collins, S. (2010, April 29). *Artificial reef in Kovalam.* https://surfingindia.net/random-stuff/artificial-reef-kovalam/

Daily Post. (2013). Borth's new coastal defences to transform town into a surfer's paradise. *North Wales Live.* https://www.dailypost.co.uk/news/local-news/borths-new-coastal-defences-transform-2685578

Dan Harmon. (2026, February 17). The $500k surf reef failure [Video; viewer comments retrieved 19 October 2026, evidence grade D]. https://www.youtube.com/watch?v=DK7cqXDzLpM

ECOncrete. (n.d.). ECOncrete case studies [Company document]. https://www.monmouth.edu/uci/wp-content/uploads/sites/58/2021/09/ECOncrete-Case-Studies.pdf [company-supplied evidence].

EcoShape. (n.d.). Artificial reefs: Example cases. https://www.ecoshape.org/en/knowledge-articles/artificial-reefs/example-cases/

Elezovich, J. (2026). Albany 2026: Reef advocates look back on decades-long journey to bring Southern Ocean Surf Reef to life. *Albany Advertiser.* https://www.albanyadvertiser.com.au/news/albany-advertiser/albany-2026-reef-advocates-look-back-on-decades-long-journey-to-bring-southern-ocean-surf-reef-to-life-c-22657778 [source of the final cost of AUD 13 M; read from a search-engine extract, the full page was not accessed].

Fontaine, E. (2008). Sandbagged: After years of unspectacular closeouts, Pratte's Reef is removed from El Segundo. *Surfline.* https://www.surfline.com/templates/article_html.cfm?n=2&id=19261&p=1

Geosynthetics Magazine. (2023, June 7). *Why have most geotextile artificial reefs never really worked?* Geosynthetics Magazine. https://geosyntheticsmagazine.com/2023/06/07/why-have-most-geotextile-artificial-reefs-never-really-worked/ [staff byline ”ATA”; quotes an e-mail from S. T. Mead].

Government of Western Australia. (2024, February 4). Surf's up for Albany artificial reef project [Media statement]. https://www.wa.gov.au/government/media-statements/Cook-Labor-Government/Surf%27s-up-for-Albany-artificial-reef-project-20240204 [project value AUD 11.75 M; State share AUD 4.75 M].

Government of Western Australia. (2025, July 16). Joint media statement: Albany artificial reef makes a splash [Media statement].

https://www.wa.gov.au/government/media-statements/Cook-Labor-Government/Joint-media-statement---Albany-artificial-reef-makes-a-splash-20250716 [funding shares: Australian Government AUD 5 M, State Government AUD 4.75 M; read from a search-engine extract, the full page was not accessed].

Hansford, M. (2011). Borth's big dig. *New Civil Engineer.* https://www.newcivilengineer.com/archive/borths-big-dig-15-09-2011/ [used for rock quantities as an upper bound].

Hochul, K. (2024). Governor Hochul announces completion of $111 million coastal resiliency project on Staten Island [Press release]. https://www.governor.ny.gov/news/governor-hochul-announces-completion-111-million-coastal-resiliency-project-staten-island

Howells, S., & O'Neal, D. (2024). How a sugarcane cutter and his mates created a world-first DIY surf break at Burkitts Reef, Bargara. *ABC News.* https://www.abc.net.au/news/2024-01-28/diy-surf-break-built-by-mates-at-burkitts-reef-bargara/103387450

Indelicato, P. (2026). Unveiling the new reef in Mexico. *Raised Water Research.* https://raisedwaterresearch.com/unveiling-the-new-reef-in-mexico/ [industry website, announcement].

International Coastal Management. (2025). Artificial surf reefs [Company web page, updated February 2025]. https://coastalmanagement.com.au/artificial-surf-reefs [Narrowneck AUD 2.3 M for about 70,000 m^3; Palm Beach AUD 18.3 M, about AUD 700 per m^3, quarried rock about 25,000 m^3].

ISRAMAR. (2025). Shikmona buoy: Significant wave height and period of maximal waves [Data set]. https://isramar.ocean.org.il/isramar2009/station/shikmonaBuoyM.aspx

Mercer, D. (2025). Artificial reef at Middleton Beach transforms Albany's surf scene. *ABC News.* https://www.abc.net.au/news/2025-07-16/southern-ocean-surf-reef-transforms-albany-wa/105530202

Moores, A. (2006). Artificial surf reefs: Coming to a beach near you. *New Zealand Geographic*, *78*. https://www.nzgeo.com/stories/artificial-surf-reefs-coming-to-a-beach-near-you/ March-April issue.

Nettle, S. (2017). Narrowneck artificial reef nears completion (updated). *Swellnet Dispatch.* https://www.swellnet.com/news/swellnet-dispatch/2017/11/07/narrowneck-artificial-reef-nears-completion-updated [read 24 September 2026; source of the south-east swell observation].

Nettle, S. (2019). Narrowneck renewal deemed a success. *Swellnet Analysis.* https://www.swellnet.com/news/swellnet-analysis/2019/08/19/narrowneck-renewal-deemed-success

NZ Herald. (2014). Mount's artificial reef to be removed. *Bay of Plenty Times.* https://www.nzherald.co.nz/bay-of-plenty-times/news/mounts-artificial-reef-to-be-removed/GA44JIIW7LW5P457SBPXO5PKYU/ [the copy read prints no date].

Raised Water Research. (n.d.-a). Bunbury Airwave [Spot page in the artificial-reef database]. https://raisedwaterresearch.com/spot/artificial-reef/australia/western-australia/bunbury-airwave/

Raised Water Research. (n.d.-b). Burkitts Reef [Spot page in the artificial-reef database]. https://raisedwaterresearch.com/spot/artificial-reef/australia/queensland/burkitts-reef/

Raised Water Research. (n.d.-c). Kovalam [Spot page in the artificial-reef database; page footer 2020]. https://raisedwaterresearch.com/spot/artificial-reef/asia/india/kovalam/

Raised Water Research. (n.d.-d). Midds Reef [Spot page in the artificial-reef database; includes the Bluecoast plan diagram and the Royal HaskoningDHV Option B plan, 2015]. https://raisedwaterresearch.com/spot/artificial-reef/australia/western-australia/albany-artificial-surf-reef/

Raised Water Research. (n.d.-e). Boscombe Surf Reef [Spot page in the artificial-reef database]. https://raisedwaterresearch.com/spot/artificial-reef/europe/united-kingdom/boscombe-surf-reef/ [letter provisional; to be set when the reference lists are merged].

Raised Water Research. (n.d.-f). Pratte's Reef [Spot page in the artificial-reef database]. https://raisedwaterresearch.com/spot/artificial-reef/us/california/prattes-reef/ [letter provisional; to be set when the reference lists are merged].

Raised Water Research. (2019a). Cables Reef [Spot page in the artificial-reef database; read 10 October 2026]. https://raisedwaterresearch.com/spot/artificial-reef/australia/western-australia/cables-reef/ [includes the quotation of Pitt, 2009, whose original was not seen].

Raised Water Research. (2019b). Mount Maunganui [Spot page in the artificial-reef database; image "Mount Reef Installed" by ASR Ltd]. https://raisedwaterresearch.com/spot/artificial-reef/new-zealand/north-island/mount-maunganui/

Raised Water Research. (2019c, December 16). *The Bunbury Airwave artificial surf reef tears during installation*. https://raisedwaterresearch.com/the-bunbury-airwave-artificial-reef-tears-during-installation/

Reid, M., & Sumner, B. (2021). The Kiwi scientist and the failed surf breaks. *Newsroom*. https://newsroom.co.nz/2021/12/06/the-kiwi-scientist-and-the-failed-surf-breaks/

Rinat, Z. (2019). Agency halts project to maintain Ashkelon cliff after anti-erosion devices damaged. *Haaretz*. [Read as reposted at https://mideastenvironment.apps01.yorku.ca/?p=14356; the original is paywalled.]

Romeyn, K. (2025). With Xala, Richard Gere has co-founded a luxury Mexican paradise. *The Hollywood Reporter*. https://www.hollywoodreporter.com/lifestyle/lifestyle-news/xala-richard-gere-luxury-mexican-paradise-1236192415/

Royal HaskoningDHV. (n.d.). Holistic beach nourishment to protect Palm Beach, Gold Coast [Project page]. https://www.haskoning.com/en/projects/holistic-beach-nourishment-to-protect-palm-beach-gold-coast (accessed 15 October 2026).

Sheil, M. (2026). Eighteen rocks of Southern Ocean Surf Reef repositioned in successful bid to reduce user injuries. *Albany Advertiser*. https://www.albanyadvertiser.com.au/news/albany-advertiser/eighteen-rocks-of-southern-ocean-surf-reef-repositioned-in-successful-bid-to-reduce-user-injuries-c-21896905

Solmax. (2023). GEOTUBE breakwaters: Shoreline protection at Sigandu Beach, Indonesia [Case study]. https://www.solmax.com/en/apac/case-studies/geotube-breakwaters-shoreline-protection-at-sigandu-beach-indonesia [manufacturer page, read].

South Western Times. (2019). Artificial surfing reef for Bunbury. *South Western Times.* https://swtimes.com.au/news/south-western-times/artificial-surfing-reef-for-bunbury-ng-b881227223z [City of Bunbury sponsorship of up to AUD 75,000].

SunLive. (n.d.). *Time called on artificial reef.* https://www.sunlive.co.nz/news/86837-time-called-on-artificial-reef.html [web news report; date not recorded].

Surfline. (2025). Haifa surf report and forecast [Wave forecast pages, including the former Magicseaweed (MSW) forecast; consulted July to October 2025]. https://www.surfline.com/travel/israel/haifa-surfing-and-beaches/294800

The Daily News. (2004). Surf reef decision has Opunake buzzing at tourism prospects. *Stuff.* [Copy at https://artificialreefs.org/Articles/STUFF%20%20TRAVEL%20%20IN%20NEW%20ZEALAND%20-%20STORY%20%20New%20Zealand's%20leading%20news%20and%20information%20website.htm; author not printed].

The Inertia. (2017). $2 million to be spent on Gold Coast artificial reef renovation; officials expect better surf. *The Inertia.* https://www.theinertia.com/surf/2-million-to-be-spent-on-gold-coast-artificial-reef-renovation-officials-expect-better-surf/ [renewal of Narrowneck Reef, about AUD 2 M].

The Surf Eye. (2022, June 27). Surfing the artificial reef at Palm Beach [Video; viewer comments; filmed 23 June 2022]. https://www.youtube.com/watch?v=mHSiPEh-93M

Appendix A. Methods and sources for the individual reefs

A.0 Common workflow

The author selected and supplied the primary sources (papers, reports, survey data, imagery and surfer testimony collected by the author, for example viewer comments on online videos), defined the research questions and the hypotheses to be tested, and directed the analysis. AI tools assisted with source extraction, data compilation, computation and drafting. Every source added by the AI tools was validated by the author, who also reviewed and confirmed all text, data, calculations and figures, including by comparing the three-dimensional reconstructions with the published figures and imagery.[11]

The same workflow was applied to all 13 reefs. It has six steps, and every step leaves a written record that can be traced from the numbers in the tables (Appendix C) back to the source.

1. Sources. For each reef the published design and monitoring papers, official reports, survey data sets, georeferenced aerial and satellite images, press articles and the pages of reef-specific websites were collected. Each item was recorded with its full citation, link, licence, access date, what it shows and how it was used. Every number in the dataset carries a source, the place where it was read (page, table, figure or file), the method used, and a status: sourced (read in a source), derived (computed in this study), assumed (an assumption, flagged A1, A2, and so on in the sections below) or unknown (Appendix C). Values from secondary compilations whose cited sources could not be fetched and confirmed were not used.

2. Plan shape. The outline of each reef was traced on the best available image, in a fixed order of preference: a measured survey (laser or multibeam) of the structure, then a design or construction drawing of the built version, then a photograph. Satellite images (Esri World Imagery and dated Esri Wayback releases) were georeferenced from their Web-Mercator tile bounds. Drawings and figures were scaled from a printed scale bar or a known dimension and placed by fitting to an image or a survey. Each outline was checked on a second, independent image or raster (overlap measured as intersection over union) and against the dimensions given in the text. Where more than one edge is defensible, the other outlines are kept as named, dated versions. Each outline is expressed in a local frame in metres (origin at the shoreline point nearest the centroid, x alongshore, y offshore), with the shoreline bearing from the same image. Confidence follows a fixed rubric: high if the structure is clearly visible on a georeferenced image or an as-built survey and the dimensions agree with the text to about 10 %; medium if the trace is on a scaled design drawing, only part of the structure is visible, or the dimensions differ by 10 to 25 %; low if no image of the structure exists and the shape is inferred from text or a sketch.

3. Three-dimensional reconstruction. A model was built only where the plan, the crest level and the seabed depth are each sourced. The seabed is taken from the best measured data (single-beam soundings, airborne lidar, multibeam read from a published figure, a coastal digital elevation model) and otherwise from a chart (Garmin Navionics SonarChart read in a web browser, datum tested; Garmin Navionics, n.d.) or from design contours. The reef surface is either the surveyed surface or a loft from the crest level, over the traced toe outline, down to the seabed; its volume is a mound envelope (voids included) and is not the fill volume. All heights are brought to mean sea level (z = 0 at MSL) through an explicit datum chain (Appendix C, note 1). Only local tidal planes of the nearest standard port are used. A model with an invented platform height or bed would look like data; where the three inputs are not all sourced (Burkitts, Opunake, Kovalam, Xala) no model was built.

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[11] Section A.14 gives a worked example of this cross-checking for the reef with the largest number of different source types (Borth), with the numbers of each check and the conflicts that were found.

4. Derived quantities. The depth profile was taken on the shore-normal transect through the shallowest crest cell, except for Albany, Cables and Borth, where it is a surf cut through the part of the reef where the wave breaks and is ridden, perpendicular to the local shoreline, sampled every 1 m by bilinear interpolation on the model grid (Albany: shore-normal through the middle of the -2.0 m AHD ridge, "crest 110 m @ -2 m" in the City of Albany design table, instead of the -1.0 m AHD take-off plateau at the offshore end; Cables: through the footprint centroid on a bearing perpendicular to the fitted 0 m AHD shoreline, 17.5 degrees from the design-frame axis; Borth: shore-normal 10 m south of the former centreline, a judgement, since no source locates the break). The automatic line is kept for reference; the positions of all lines are shown in Figure E1 (Appendix E). For the other six reefs every value is unchanged. The depth gap is the seabed depth at the toe minus the crest depth, the ramp length is the horizontal toe-to-crest distance, and the mean slope is their ratio (definitions and sensitivities in Appendix C, note 2). The breaking length behind the break (Table 3) is read on five lines per reef, sampled every 1 m on the same model grids (Figure E2): the Figure 1 cut; three shore-normal lines at 25, 50 and 75 % of the alongshore extent of the reef footprint (thickness ≥ 0.10 m), counted from its left end looking seaward (Cables: normal to the fitted 0 m AHD shoreline); and the longest crest line, defined as follows: the ridge is the set of footprint cells within 1.0 m of the shallowest reef cell (gaps up to 3 m bridged, largest connected group kept); the line starts at the most seaward ridge cell and ends at the ridge cell farthest from it measured along the ridge, follows the shallowest cells between them, and is smoothed over 5 m (ridge bands of 0.5 and 1.5 m as sensitivity). On every line the wave of breaking height $H_b = 1$, 1.5 or 2 m breaks where it first meets $h \leq H_b/0.78$ at low, mean or high water. On the shore-normal lines the length integrates the energy-flux decay of Dally et al. (1985) shoreward on the 1 m grid (shallow-water linear theory, $K = 0.15$, $\Gamma = 0.4$) until $H \leq \Gamma h$, the shoreline or the end of the reef. On the crest line, where the wave peels along the arm rather than travelling along it, the length is that of the contiguous stretch with $0.05 \text{ m} < h \leq H_b/0.78$ (peel length). Per line the median over the cases that break on the reef is reported.

5. Documentation. For every reef the assumptions (numbered A1, A2, … with their effect on the result), the design choices (D1, D2, …, with their rationale), the validation against independent values, the confidence per element and the unknowns, with what would resolve them, are written in a methods file next to the model. This appendix condenses those files.[12]

6. Tools. Image analysis used threshold and morphological operations, normalised cross-correlation and phase correlation for image registration, distance transforms for lofted surfaces, and piecewise-cubic interpolation for seabed profiles (Python with NumPy, SciPy, Shapely, pyproj and PyMuPDF). The design choices taken in this review are marked D below (see Section 2.6 on the tools used). Readings of values from figures are by eye or by calibrated colour or tick scales, and carry the stated reading error.

Table A1 summarises the status of the 13 reefs. Three-dimensional previews of the models are shown in Appendix B.

#	Reef	Default plan outline	Plan	Seabed	Crest level	3D
A.1	Burkitts	visible platform, Esri 5 Sep 2025 (stage, not the reef)	low	none	none	not built

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[12] The validation tables of the methods files are the record of these checks; Section A.14 reproduces them for one reef.

#	Reef	Default plan outline	Plan	Seabed	Crest level	3D
A.2	Cables	design plan of Lyons (1992) fitted to the 1999 soundings	high	single-beam survey, Mar 1998	surveyed rock surface, Nov 1999	medium
A.3	Narrowneck	visible container field, Esri 8 Aug 2020	medium	2017 multibeam read from a figure; Navionics	measured crest -2.2 m AHD (2018)	medium
A.4	Pratte's	Phase I design layout, Skelly Engineering	medium	NOAA 1/3 arc-second DEM	apex -6 ft MLLW, constant thickness	low
A.5	Mount Maunganui	-2.0 m CD contour of the 2013 multibeam	medium	Navionics 2026, bed level assumed	-0.9 m CD (council report)	medium
A.6	Opunake	consent occupation area only (not the reef)	low	none	design text only	not built
A.7	Boscombe	bag field on Esri image of 28 Sep 2011	medium	DGPS bathymetry, Apr 2011	designers' +0.5 m ACD	medium
A.8	Kovalam	photo trace at assumed camera height	low	none	none	not built
A.9	Borth	laser-survey edge, LiDAR 19 Mar 2022	high	design bed levels, contour figure, EMODnet	LiDAR and drawings	medium
A.10	Palm Beach	toe outline of the City asset polygon and aerial	high	Navionics	-1.5 m MSL (four sources)	medium
A.11	Bunbury	circle of 12 m on a drone photograph	medium	WA lidar 2009	height 1.6-2.0 m (text)	medium
A.12	Albany	Bluecoast design outline	high	2015 design contours	-1.0 m AHD (City design)	low
A.13	Xala	none	not traceable	none	none	not built

Table A1. Plan outline, data used for the reconstruction, and confidence. Plan confidence follows the rubric in step 2. "3D" gives the overall confidence of the three-dimensional model.

A.1 Burkitts Reef, Bargara, Queensland (1997)

Sources. Burkitts Reef is a natural basalt headland platform at the south end of Bargara Beach. In February 1997 about 300 m^3 of boulders (at least 12 large ones) were moved with an excavator at a cost of about AUD 10,000 (Rendle, 2016, Table 2.1; Howells & O'Neal, 2024; Raised Water Research, n.d.-b). No source gives a length, width, crest level or depth. The work zone of a later phase is shown on an annotated oblique aerial photograph (Jackson & Corbett, 2007, Fig. 1). The only georeferenced image is Esri World Imagery of 5 September 2025 (Esri, Maxar, Earthstar Geographics, & the GIS User Community, n.d.). Tidal planes of the nearest standard port, Burnett Heads, are from Maritime Safety Queensland (2026; MHWS 2.94 m, MSL 1.78 m, MLWS 0.61 m above LAT).

Plan shape. The visible platform was traced on the 2025 image (31,485 m^2, a lower bound because the image frame cuts it). The later work zone was georeferenced with three control points on an Esri Wayback image of 31 July 2016 (an ellipse of about 1,750 m^2, accuracy 20-30 m). The traced outline is the platform on which the reef sits and not the reef, because no source locates the 1997 works; the outline is therefore not used in the analysis (confidence low). The assumptions A1-A8 of the shape file are that the platform is the stage, that its edge is the breaker line at an unknown tide, and that nothing has changed since 1997.

3D reconstruction. Not built: crest level, platform height and seabed depth are all missing (the depths quoted for the dive site are site-wide ranges, not a profile).

Unknown and what would resolve it. The reshaped zone, the crest depth, the seabed and the tide at the time of the image; dated low-tide imagery with tide tables, a chart reading or a local account would resolve them.

A.2 Cables Reef, Perth, Western Australia (1999)

Sources. The design plan of Lyons (1992) as reproduced in Blacka et al. (2013, Fig. A.2) gives the V-shaped outline with design levels (RL); the design study (Pattiaratchi, 1999) and the copies of the drawings on Raised Water Research (2019a) give the 45-degree peel angle, the 1:20 ramp and the proposed extension of the southern arm. The measured data are the single-beam soundings of the Western Australian Department of Transport (2020): a pre-construction survey ending 30 March 1998 (14,303 soundings in the window), and surveys of 22 November 1999 and 1 February 2002 (heights in m AHD). The Australian Height Datum (AHD) is defined by Roelse et al. (1975). Tidal range and mean level are from the Fremantle gauge about 4 km south of the reef.

Plan shape. The silhouette of the drawn structure (111 vertices) was scaled from four printed dimensions (6.51 px per metre, sd 0.05). Its position and rotation (9.74 degrees) were found by maximising the share of 1999 rock soundings inside the outline minus the share of bed soundings inside it: 91.0 % of rock and 18.5 % of bed soundings lie inside. With the scale also free, the optimum scale was within 2 % of the drawing scale. The outline coincides with the dark rock patch on four Esri images. Its size (121 by 76 m, 4,049 m^2) differs from the text (140 by 70 m) by -13 % and +9 % (confidence high; extension version 4,355 m^2, unconfirmed).

3D reconstruction. The seabed is the 1998 survey averaged per metre and interpolated, outside the reef the DoT composite grid shifted by +0.085 m, smoothed with a 2 m Gaussian filter (bed at the anchor -3.68 m AHD, accuracy 0.3 m). The default rock surface is the thickness (sounding minus modelled bed) at the November 1999 soundings inside the footprint and a 6 m margin, interpolated linearly to a 0.5 m raster, with negative values set to zero and a 1 m Gaussian smoothing (the rule that a sounding more than 0.6 m above the 1998 bed is rock was used only to classify soundings for the georeference fit); volumes are 4,671 m^3 (1999) and 5,219 m^3 (2002) (accuracy 1,000 m^3), against 3,500-5,500 m^3 in the literature. Design-surface versions loft the printed contours.

Assumptions and design choices. A1: the design RL is Fremantle low water datum, so z_{AHD} = RL - 0.716 m. A12: AHD is taken as MSL (local MSL 1992-2011 is 0.057 m higher). A5: side slopes of the design versions are linear over the drawn bands. A6: the extension is a pyramid scaled to 320 m^3 (unconfirmed). A14-A17: the 1998 bed is the bed under the reef; soundings are rock-top depths, and interpolation between lines of about 5 m smooths boulder-scale relief. A7: high and low water are taken as MSL +-0.4 m.

Confidence. Plan high; seabed medium-high; surveyed rock surface medium; datum medium-high; tides low-medium (range and MSL measured, HAT and LAT not found); overall medium. The surveyed rock lies 0.13-0.22 m above the design ramp, which is not a datum effect.

Unknown. The extension, the rock surface after 2002 (a 2009 critique by A. Pitt, quoted by Raised Water Research, 2019a, puts the crest about 1.5 m below the lowest tide), HAT and LAT, and the sign of a 0.04 m Perth-zone height difference.

A.3 Narrowneck Reef, Gold Coast, Queensland (1999-2000, renewed 2017-18)

Sources. The monitoring papers (Jackson et al., 2007, 2012), the renewal paper (Corbett et al., 2023), the sediment-pathway study (Vieira da Silva et al., 2021), the decade review (Mead & Borrero, 2011) and the NSW review (Blacka et al., 2013). The plan is traced on Esri Wayback release 9812 of 8 August 2020 (Esri, Maxar, Earthstar Geographics, & the GIS User Community, n.d.). The seabed is read from Fig. 3 of Corbett et al. (2023), a multibeam map of 2017 before the renewal, and from Garmin Navionics SonarChart labels. Tidal planes are from Maritime Safety Queensland (2026) for the Gold Coast Seaway; the asset polygon of the City of Gold Coast (n.d.; an envelope of 34,565 m^2) was not used for the outline.

Plan shape. The visible containers were extracted by a threshold on the green channel, closing of 3.7 m and hand-checked polygons (7 polygons, 6,239 m^2), georeferenced from the tile bounds (0.264 m per pixel; edge accuracy 1 m; waterline 15 m). Independent checks: a second tile set of November 2022 agrees to 0.2 m (phase correlation), images of 2016 and 2019 to 3-4 m by eye, and Navionics shoals to 7-25 m. The footprint is a lower bound because the deeper toe is not visible. Four more outlines are selectable: the Option 2 renewal design of 2017 (13,941 m^2), the revised design of 2004 (13,631 m^2), the relief zone of the 2011 survey (4,922 m^2) and the Navionics 4.0 m shoals (537 m^2). **Design choice D7:** the default is the 2020 photograph trace, because it is the only evidence of the state actually built (Option 2 was built only in part on the south reef) and no survey or drawing of the renewed reef exists.

3D reconstruction. The crest is the measured -2.2 m AHD of the 2018 multibeam (Vieira da Silva et al., 2021). The multibeam map of Corbett et al. (2023, Fig. 3) was converted from colour to depth with a linear fit through four tick labels of its colour bar (depth accuracy 0.4 m, position 10 m); the bed is a grey-scale opening (radius 4.65 m) and 6-pixel Gaussian smoothing of that raster, blended into an alongshore-uniform profile (seaward labels 7-10 m from Navionics; shoreward part assumed). The reef surface is the design surface (linear in distance rank between the -2.5 m crest loop and the -6 m envelope) lowered to the measured crest. Volumes are mound envelopes.

Assumptions and design choices. A1: the figure depths are m AHD. A3: the colour scale is linear. A4: the design surface between the contours is linear (offset +0.26 m, sd 0.66 m against 53 multibeam blocks). A6: Navionics depths are below LAT. A7: the nearshore bed (0-190 m) is a smooth monotone profile. A10: the datum link AHD = LAT + 0.760 m of Maritime Safety Queensland is used instead of the +1.00 m implied by the papers; this moves reef and bed by 0.24 m relative to the tidal planes (crest 1.44 m below LAT, or 1.20 m on the papers' rule) but does not change the geometry.

Confidence. Plan medium; crest medium-high; seabed in the window medium (seaward low-medium, nearshore low); reef height and slopes medium-low; tides medium-high; volume low-medium; overall medium.

Unknown. The renewed plan and survey (City of Gold Coast drawings, 2018-19 multibeam), the datum of the original multibeam, the nearshore bed, and the container-level shape.

A.4 Pratte's Reef, El Segundo, California (2000-2001)

Sources. The design plan of Skelly Engineering (2000), reproduced as Fig. 4 of Borrero and Nelsen (2003); the monitoring paper (Borrero & Nelsen, 2003) for the crest datum (apex "approximately -6 ft MLLW"), bag size, dates and the survey window of its Fig. 9; the NSW review (Blacka et al., 2013); the NOAA Santa Monica 1/3 arc-second coastal digital elevation model (NOAA National Geophysical Data Center, 2010) for the seabed; Navionics depth labels as a check; tidal datums from the NOAA Center for Operational Oceanographic Products and

Services (NOAA Center for Operational Oceanographic Products and Services, 2026; Santa Monica, 11.7 km away, epoch 1983-2001).

Plan shape. The solid-outline bags of the design drawing (110 bags, 76 vertices) were traced and scaled from the printed bar (5.155 px per metre; agrees with the known bag size to 3-6 %); two reproductions agree in area to 1 %. The result (435 m^2, 58.0 by 28.6 m) is a design layout, not an as-built plan; the 30 optional bags are excluded. Position: the centre of the survey window of the paper's Fig. 9 (33.92058 N, 118.43335 W, accuracy 150 m).

3D reconstruction. The Phase I reef follows the DEM seabed at a constant thickness of 1.349 m (one 1.2 m bag course) so that the apex face lies at -2.678 m MSL (-6 ft MLLW, $z_{\mathrm{MSL}} = z_{\mathrm{MLLW}}$ - 0.849 m). The bed is the mean of 31 alongshore DEM profiles, checked against Navionics labels and the paper's profiles (agreement 0.3 m). Phase II and the removal in 2008 and 2010 are not modelled.

Assumptions. A2: the reef centroid is 91.44 m from the MSL waterline (+-25 m moves the bed by 0.78 m). A4: the 6 ft depth applies at the apex face, and the top follows the bed at constant thickness. A5: the bag-stack side slope is 1:1 (0.5-2.5), which is also the **assumed ramp and slope** in Appendix C. A9: the footprint is the design layout. A10: no settlement, scour or burial is modelled, although the real reef sank and was buried. A13: the position is the survey-window centre.

Confidence. Plan medium; datums high; seabed medium; stack height and slope low; overall **low** (design footprint, inferred position, assumed height and slope, and a height uncertainty of 0.85 m against a reef height of 1.35 m).

Unknown. The as-built footprint and bag layout, the crest elevation along the reef, the number of courses, and the true position.

A.5 Mount Maunganui Reef, Bay of Plenty, New Zealand (2005-2008)

Sources. The council assessment of management options (Dahm & Gibberd, 2013), whose Fig. 3 is a multibeam survey of 18 July 2013, and Fig. 4 a survey of January 2007; the design papers (Mead & Black, 1999; Mead et al., 2007); Mead and Borrero (2011); the NSW review (Blacka et al., 2013); an ASR image of the completed reef in colour-coded depth (undated, after August 2008, datum not stated; ASR Ltd, via Raised Water Research, 2019b); aerial photographs of 2010-11 (Land Information New Zealand, 2010–2011) and Esri Wayback release 3630 of 15 January 2011; Navionics SonarChart of 2026; tide levels from Land Information New Zealand (Land Information New Zealand, n.d.).

Plan shape. The default is the -2.0 m chart-datum (CD) contour of the 2013 survey (1,071 m^2), digitised on the NZ-grid ticks of Fig. 3 (5.45 px per metre) and confirmed on a second raster (area 1,074 m^2, intersection over union 0.93) and on the 0.125 m aerial (position within 2 m). The second version is the as-built toe outline of the ASR image (1,424 m^2, scale +-6 %), corroborated by the as-built survey in Fig. 3 of Mead and Borrero (2011). The 2013 outline is five years after completion (apex bags deflated, base bags buried) and is a lower bound.

3D reconstruction. The reef is lofted from the traced outline to a crest at -0.9 m CD (z = -2.03 m relative to the present MSL, with MSL = CD + 1.13 m) over the 2026 Navionics seabed shifted to an as-built bed level of -3.6 m CD at the reef centre. The flank is 1:1 (default) or a linear loft from the toe (second version); the inner profile is the median rise curve of the 2013 digital elevation model scaled to the crest.

Assumptions and design choices. A6 and **design choice D1: the as-built bed at the reef is -3.6 m CD**, an assumption, because no source states the 2008-09 bed; every later source gives a bed 0.8-1.2 m shallower, and the stated built volume of 2,800 m^3 is reproduced only with a bed of -3.51 to -3.66 m CD. A bed of -2.8 m CD would give a height of 1.9 m instead of 2.7 m (sensitivity: +-0.5 m is +-22 % in volume). A3: the crest is -0.9 m CD on every bag. A5: the 2026 seabed shape stands for 2008. A7: flank 1:1 (a run of 0.25-2 changes the volume from 2,567 to 3,421 m^3). The scour hole behind the reef (to -4.9 m CD in 2013) is not modelled.

Confidence. Plan medium (default) and medium-low (toe version); crest medium-high; tides and datums high; seabed shape medium; **bed level and height low (assumed)**; volume low-medium; overall medium.

Unknown. The 2008-09 bed at the reef, the datum of the ASR image, a section drawing, and the as-built footprint (historical imagery of 2008-10 could provide it).

A.6 Opunake Reef, South Taranaki, New Zealand (2006-2009)

Sources. The monitoring report of the regional council (Taranaki Regional Council, 2009), with the coordinates of Coastal Permit 6376-1 (31 May 2005) in its Appendix I and an aerial photograph in its Fig. 1; the NSW review (Blacka et al., 2013, Tables 4.2-4.4); Mead and Borrero (2011); a magazine article (Moores, 2006); a newspaper report (The Daily News, 2004); a news feature (Reid & Sumner, 2021); and aerial photographs of 2011-12 (Land Information New Zealand, 2011–2012), which show no structure.

Plan shape. The four published corner coordinates (a local circuit grid, corrected for one misprinted easting) form a rectangle of 189 by 94 m (17,726 m^2), centred at 39.45943 S, 173.85163 E, georeferenced against Fig. 1 (residual 0.3 m, independent check 0.7 m). This is the **legal occupation area, not the reef**. The reef itself is described only by its design size (99 by 30 by 3.5 m, 4,900 m^3; Taranaki Regional Council, 2009, p. 4). Where the reef lies inside the area is unknown (+-47 m along the axis, +-32 m across); the 99 by 30 m rectangle was assumed to be centred and parallel (A7, A8) and is not shown. The earlier size "90 by 20 m" is the 2004 concept quoted in The Daily News (2004) and repeated by later articles.

3D reconstruction. Not built: no bag layout, no per-bag crest levels and no surveyed seabed exist. Depths are known only below chart datum (LAT): the shallowest contour is at 0.0 m and the seabed is 1.8-3.0 m below chart datum (Taranaki Regional Council, 2009, pp. 3-4); an MSL conversion needs the LINZ standard-port levels and was not made. The as-built survey required by the consent within 90 days of completion is held by the council.

Confidence. Position of the occupation area medium-high; reef shape low (envelope only).

A.7 Boscombe Surf Reef, Dorset, United Kingdom (2008-2009)

Sources. The design paper (Mead et al., 2010), the evaluation of the damaged reef with a DGPS bathymetry of April 2011 in its Fig. 9 (Rendle & Davidson, 2012), the doctoral thesis (Rendle, 2016), Mead and Borrero (2011), the NSW review (Blacka et al., 2013), Esri Wayback release 10 of 28 September 2011, beach profiles of 20 April 2010 (Channel Coastal Observatory, 2026b), the chart-datum offset of Bournemouth (National Tidal and Sea Level Facility, n.d.), and Garmin Navionics for validation.

Plan shape. The bag field is clearly visible on the 2011 image and was traced (4,042 m^2, 88.5 by 94.3 m), agreeing with the April 2011 survey zones and with the designers' own plan. The area is about 40 % of the roughly 1 ha repeated in the literature (confidence medium). The outline is of the state after the damage of March-April 2011; the 2009 edge may differ by a few metres on the damaged offshore half.

3D reconstruction. The as-built state of 2009 is an idealised two-layer loft over the outline: a flat crest at the designers' +0.5 m above chart datum, 1:3 flanks, and a 1 m relief at the outline. The seabed is read from Fig. 9 (accuracy 0.2 m), with the beach from the measured 2010 profiles, and spline or EMODnet slopes beyond the survey. Heights are referred to MSL with $z_{\mathrm{MSL}} = z_{\mathrm{ACD}}$ - 1.40 m; tide levels are from the gauge table of the design paper. Sixteen Navionics soundings agree with the model seabed (mean difference +0.06 m, sd 0.49 m), and its shoal areas match the survey zones to 1-2 %.

Assumptions. A3: the zero of the survey plot is chart datum (if it were Ordnance Datum all absolute heights shift by 1.4 m, with relief and volume unchanged). A6: the built crest is flat at +0.5 m ACD (+-0.3 m is +-4 % in volume). A7: flanks are 1:3 (1:2 or 1:4 change the volume by +8 % and -7 %); this flank is the **assumed slope** behind the ramp of Appendix C. A8: the seabed under the reef is a spline of its surroundings. A12: the 54 containers are not modelled individually.

Confidence. Plan medium; seabed medium-high (measured, +-0.2 m); crest medium (designer value, checked against October 2009 profiles); datum assumption strongly supported but not stated by the authors; overall medium.

Unknown. The datum of the Fig. 9 plot (to be confirmed from the thesis or from survey metadata), the as-built plan (a 2009-10 aerial image), and the crest height along the whole reef.

A.8 Kovalam Reef, Kerala, India (2009-2010)

Sources. The ASR Ltd oblique aerial photograph of the completed reef from the lighthouse end (reproduced in Blacka et al., 2013, Fig. 4.5, where it is credited to ASR Ltd), a second oblique photograph with a hand-drawn break line (Raised Water Research, n.d.-c), Esri Wayback release 10 of 28 February 2011 for the shoreline, the NSW review tables (Blacka et al., 2013, Tables 4.2-4.4, pp. 66-70), Mead and Borrero (2011, Section 3.1.7) and a report by a surfer (Collins, 2010).

Plan shape. The five or six sand-container strips were traced in photograph pixels. Because the photograph has no scale and the camera height is not fixed, a camera model was fitted to the 2011 swash line (residual 2.2-2.8 px for heights of 30-80 m). The metric outline is therefore a family: 640 m^2 (35 m), 1,216 m^2 (60 m, the default) and 1,681 m^2 (80 m). The alongshore length agrees with a secondary table (65 m), but the cross-shore size does not (41 m against 85 m). Confidence low; the shape is hidden in the survey figures.

3D reconstruction. Not built: the sources give only a crest submergence range (0.2-3.4 m, datum not stated, not tied to a position), a water depth range of 1.3-3.7 m and a tide range of about 1 m, with no profile, contour or datum, and the reef later sank, so no as-built survey exists.

Unknown. The true size and position, crest and seabed depth, and tidal planes. A navigation-chart reading, historical commercial imagery for February 2010 to 2012, or the designers' original design report would resolve these.

A.9 Borth reef, Ceredigion, Wales (2011-2012)

Sources. The Phase 1 construction drawings 9V5090/1020-1023 and 1001, rev C1 of January 2011 (Haskoning UK, 2011), with plan rings, 15 setting-out points on the Ordnance Survey grid, crest levels, layer thicknesses and slopes; the laser survey (Welsh Government, 2022; 1 m DSM and DTM, flown 19 March 2022); Esri imagery of 2012-2024; the physical-modelling paper with the pre-construction contours (Rigden et al., 2013, Fig. 2); the tide table of the shoreline management plan (Royal HaskoningDHV, 2011); EMODnet bathymetry (EMODnet Bathymetry Consortium, 2024); a trade-press article for the rock quantities (Hansford, 2011); the national heritage record (Royal Commission on the Ancient and Historical Monuments of Wales [RCAHMW], 2020).

Plan shape. Two mounds, a northern boot-shaped surf reef and a southern oval **shore-parallel breakwater** (not a surf reef), are bare rock on four dated Esri images (2012-2024). The outlines agree with the georeferenced design plan in position (0.3-4 m), orientation (1-3 degrees) and length (-4 % north, +0.4 % south). Three outlines are selectable: the **laser-survey edge (default; 8,675 m^2)**, the design rock-layer foot (6,989 m^2) and the visible rock edge in the photograph of 17 September 2024 (5,743 m^2, used for the plan area in Appendix C). The OS-grid to local-frame transformation has a residual of 0.005 m; the absolute position is limited by the datum transformation (2-5 m).

3D reconstruction. The surface is the 1 m LiDAR surface above the exposed level (about -2.3 mODN) plus the design toe berm and blanket of the sections below it. The seabed is the design bed level under the rock (about -4.0 mODN, “bed levels vary”), the hand-read -3.0 and -4.0 mODN contours of the physical-modelling figure around it, and an EMODnet gradient of -1.1 % beyond the contours. The model rock volume is 30,119 m^3 (51,200 t at an assumed 1.7 t m^{-3}, +-15 %). Heights: $z_{\mathrm{MSL}} = z_{\mathrm{ODN}}$ - 0.31 m; the tide slider stops at MHWS (+2.56 mODN), because HAT of Aberystwyth was not found at a primary source.

Assumptions and design choices. A2: MSL = +0.31 mODN, the mean of the four tidal planes (+-0.15 m). A3: LAT = -2.44 mODN. A4: the LiDAR datum is ODN. D6: the -4.0 mODN contour of the physical-modelling figure became a control after the seabed seaward of the reef was found to be 0.89 m deeper than that contour. Beyond y = 335 m the model is 0.7-1.2 m deeper than the EMODnet cells.

Confidence. Plan high; crest levels high (design drawing, LiDAR and section drawings agree within 0.3 m); tide offsets medium; seabed medium-low (the bed level under the rock is the first-order unknown for height and volume); overall medium.

Unknown. The bed under the rock (no survey), HAT, the LiDAR datum (+-0.15 m on the crest), and any reef-only rock quantity.

A.10 Palm Beach Reef, Gold Coast, Queensland (2019)

Sources. The asset polygon of the City of Gold Coast (City of Gold Coast, n.d.; boulder reef, 160 by 80 m, 11,942 m^2), Esri World Imagery of 1 December 2025, a vertical Nearmap aerial with survey contour lines and a 0-25-50 m scale bar (Bluecoast Consulting Engineers, n.d.), the concept design (Mortensen et al., 2015), the project paper with the final multibeam survey as an oblique image (Hunt et al., 2022, Fig. 4), Garmin Navionics SonarChart, and tidal planes of Maritime Safety Queensland (2026).

Plan shape. The outer contour of the aerial (toe outline) was traced and placed by the north arrow and scale bar; it matches the City polygon to 0.55 m (11,972 against 11,942 m^2) and an independent drawing of the same ICCE paper (about 45 degrees to the shore). The crest slot (59.2 by 5.9 m) is the innermost contour. Confidence high.

3D reconstruction. The ten unlabelled contour lines of the aerial were taken as the surveyed rock surface at an assumed interval of 0.5 m (a Navionics-derived interval of 0.46 m supports this), and the seabed as a smooth surface through the Navionics iso-lines outside 25 m of the reef and the toe elevations. The crest is at -1.5 m MSL (four sources). The two inferences agree to 0.27 m on average (sd 0.55 m, 427 points); no surveyed depth value was read.

Assumptions. A3: the contour interval is 0.5 m. A5: Navionics depths are below LAT and not tide corrected (LAT tested best, MSL rejected). A6: the natural seabed is smooth between iso-lines. A8: one rock surface with voids included. A12: the aerial lines show the as-built surface. A13: the Navionics seabed represents the 2019 bed. A10 and A11: the tidal planes of the Gold Coast Seaway apply (20 minutes earlier), with AHD 0.12 m below MSL.

Confidence. Plan high; crest level high; seabed and reef height +-0.5 m; overall medium. The City's asset register (height 5 m, volume 33,000 m^3, units unknown) describes a reef 20-33 % larger than the model (crest height 3.9 m, envelope about 22,000 m^3); the discrepancy is unresolved.

Unknown. A surveyed as-built seabed or crest value, the datum of Navionics, the interval and date of the aerial contours, and the units of the register entries.

A.11 Bunbury Airwave, Western Australia (December 2019)

Sources. A news report (ABC News, 2019), the developer's account in the local press (Bunbury Mail, 2022), a spoken interview with the founder (T. Bottegal, Airwave founder, pers. comm., 8 October 2026; an interested party), a drone photograph (Raised Water Research, 2019c) and a pre-installation plan graphic (Raised Water Research, n.d.-a), a video frame of the installation (ABC News, 2019), the airborne lidar of 2009 (Western Australia Department of Transport, 2009; 10 m cells, vertical datum AHD), Garmin Navionics SonarChart contours counted in a browser, and the tidal planes of the Port of Bunbury (GHD, 2021).

Plan shape. The bladder is a circle of 12 m diameter, traced with 29 vertices on a near-nadir drone photograph (112 m^2); a second video frame and the plan graphic agree. Neither image has a scale or a georeference, so the size comes from the stated diameter and the position (centre 37.5 m from the waterline, text range 30-45 m) from the text. The coordinates are an estimated centre (assumed). Confidence medium.

3D reconstruction. A symmetric spherical cap of height 2.0 m (range 1.6-2.0 m, volume 93-117 m^3) stands in for the real surface, on the lidar seabed (bilinear, resampled to 5 m). Lidar and Navionics agree (RMS 0.32 m once Navionics depths are read below LAT). The seabed in the text ("about 1 m under the surface at low tide") and the lidar agree to 0.22-0.62 m.

Assumptions. A2: the 2009 0 m AHD contour is the zero of "distance offshore" (+-10 m is +-0.41 m of seabed depth). A4: height 2.0 m. A5: spherical cap. A6: "low tide" is MLLW. A8: the 2009 seabed is unchanged in 2019 (sd 0.30 m, a judgement). A10: the lidar datum is AHD and its uncertainty layer (0.47 m) is one standard deviation. A11: Navionics datum is LAT with a 0.5 m interval.

Confidence. Seabed sourced and independently supported; tides sourced and cross-checked (0.03 m); surface shape assumed; overall medium. The verdict "failed" rests on all dated 2019 reports; the developer's page dates the trial to 2018 and calls it not a failure, a claim contradicted by the dated reports.

Unknown. The as-installed surface (which flank was steep, fill level, the torn quadrant), the exact position, seabed change since 2009, and any rideable-wave record.

A.12 Southern Ocean Surf Reef, Albany, Western Australia (2025)

Sources. Esri World Imagery of 20 January 2026 (the reef is a dark wedge, absent on 16 January 2025) and Esri Wayback release 13192 (Esri, Maxar, Earthstar Geographics, & the GIS User Community, n.d.); the Bluecoast final-design plan with contours, via Raised Water Research (n.d.-d), and the City of Albany design table (City of Albany, n.d.); the Option B plan with depth contours of Royal HaskoningDHV (2015, as reproduced by Raised Water Research, n.d.-d) for the seabed; the environmental impact assessment (BMT Western Australia, 2018) for soundings; tide predictions for Albany (Western Australia Department of Transport, & Bureau of Meteorology, 2024); and construction reports (Mercer, 2025; Sheil, 2026).

Plan shape. The design outer contour (13,104 m^2, 127 by 153 m) was scaled from its bar and placed on the 2026 image; the satellite-visible rock patch (10,918 m^2) is the second version.

The two outlines overlap by 81 %, with orientations within 3 degrees, and the length agrees with the City's 165 m (design plan 165 m, satellite outline 157 m). Confidence high. The default is the design outline, the volume of which (34,000 m^3 on 2017 bathymetry) is stated by the City.

3D reconstruction. The seabed is one shore-normal profile read from the colour contours of the Option B plan along eight transects (the plan was georeferenced on four control points, residual 3-6 m; nodes -1 m at 36 m to -8 m at 330 m from the beach-face line), assumed alongshore uniform. The reef is lofted between a crest ridge, offset 9 m inside the shoreward edge and at -1.0 m AHD for the first 15 m, falling to -2.0 m at 40 m, and the toe outline: $z = z_c + (z_b - z_c)(d_c/(d_c + d_t))^N$, with slopes limited to 1:2. The exponent N = 2.73 is fitted so that the design outline holds 34,000 m^3, so the volume is not an independent check. Rock mass uses 1.847 t m^{-3} (62,800 t / 34,000 m^3).

Assumptions. A2: AHD is MSL (LAT is 0.74 m below AHD, from the mean of 1,043 predicted high and low waters of 2024). A4: the seabed is alongshore uniform. A6 and A7: the crest ridge offset and the crest levels (the City gives -1.0 m "shallowest" and 110 m at -2 m, but not where). A9: N is fitted. A10: bulk density. The datum of the 2015 contours is not stated and is taken as AHD; it conflicts with a 2018 sounding by 1.6 m near the shore.

Confidence. Plan high; crest medium-low (design values, no as-built survey); seabed low; pile height and slopes low (height known to about +-1 m); tides low-medium; overall **low**.

Unknown. The Bluecoast detailed design report with the 2017 bathymetry, the completion survey of 2025, and the rock movements of May 2025 and February 2026 (not modelled). Depth readings, or a survey by the Department of Transport (Middleton Bay), would fix seabed and heights.

A.13 Xala Reef, Jalisco, Mexico (2025-2026)

Sources. An announcement by an industry website on 31 March 2026 (Indelicato, 2026), reporting a boulder reef designed by S. B. Mortensen (DHI Seaport), and a feature reporting "3,000 truckloads" of quarried boulders (Romeyn, 2025). The two images available show a surfer and a drone shot of a peeling wave; the rock reef is submerged and nothing is measurable.

Plan shape and 3D. Not traced and not modelled. No image of the structure, no scaled drawing, no dimension and no coordinate exist (the location was withheld until a reveal event billed for 16 April 2026), and an area pin found by the search is explicitly not the reef. A text-derived envelope of 160 by 80 m, placed 230-310 m offshore by analogy to Palm Beach Reef, is not a measurement and was not used.

Unknown. All geometry, cost, contractor, construction dates (2025 or 2026) and performance.

A.14 Example of source cross-checking: Borth reef

Choice of the example. Each source that contributed a value to a model, or that was used to check one, was assigned to a source type, and types, not documents, were counted. Eleven types were distinguished: (1) published paper or technical report (design, monitoring, review, consultancy); (2) thesis; (3) design or construction drawing; (4) measured survey or survey-derived grid (airborne lidar, multibeam, single-beam or DGPS soundings, coastal elevation models); (5) imagery (satellite, aerial, drone photographs and video frames, any number counting once); (6) nautical chart (Garmin Navionics); (7) tide and datum tables of an authority; (8) government or council record or announcement; (9) press, including trade press; (10) developer or reef-database web page; (11) interviews and online testimony. Sources that were read but fed nothing (context only, or rejected) were not counted. Table A2 gives the result for the nine reefs for which a three-dimensional model was built.

Reef	Source types counted	Count
Boscombe	paper, design plan, DGPS survey, satellite imagery, chart, tide and datum table	6
Narrowneck	paper, design contours, multibeam read from figures, imagery, chart, tide table	6
Cables	paper, design plan, single-beam soundings, imagery, tide table, web page	6
Mount Maunganui	paper and council report, multibeam survey, imagery, chart, tide table, thesis (size comparison)	6
Pratte's	paper, design drawing, agency record, DEM, chart, tide table	6
Palm Beach	paper, council record, aerial and satellite imagery, press, chart, survey and grid data, tide table	7
Bunbury	imagery, developer web page, press, lidar, chart, tide table	6
Albany	paper and report, design plan, council record, imagery, soundings, tide table	6
Borth	**paper, construction drawings, lidar and bathymetry grid, imagery, chart, tide and datum table, trade press, heritage record**	**8**

Table A2. Number of different source types behind the three-dimensional models. Counts follow the rule above and were read from the source registers of the model files; a difference of one type between neighbouring reefs is within the judgement involved in classifying a source.

Borth reef (Ceredigion, Wales) has the largest number, eight types, and is therefore described in detail. The eight types are: the Phase 1 construction drawings (Haskoning UK, 2011); the airborne laser survey of 19 March 2022 (Welsh Government, 2022) and the EMODnet depth grid (EMODnet Bathymetry Consortium, 2024); the physical-modelling paper with the pre-construction contours (Rigden et al., 2013) and the NSW review (Blacka et al., 2013); Esri images of 2012, 2013, 2022 and 2024 (Esri, Maxar, Earthstar Geographics, & the GIS User Community, n.d.); the water-level table of the shoreline management plan (Royal HaskoningDHV, 2011) with the chart-datum offsets of the National Tidal and Sea Level Facility (n.d.); the Navionics chart (Garmin Navionics, n.d.); the trade-press article on the rock quantities (Hansford, 2011); and the national heritage record (Royal Commission on the Ancient and Historical Monuments of Wales, 2020).[13]

Plan shape. The construction drawing 9V5090/1020 was georeferenced on its 15 setting-out points on the Ordnance Survey grid (residual rms 0.005 m, maximum 0.02 m; the fit of the PDF itself has a maximum residual of 0.02 m), and the foot ring reproduced from the drawing has an intersection over union of 0.9998 with the stored outline. The rock edge was traced independently on the Esri images of 2012, 2022 and 2024. The 2022 and 2024 traces overlap with an intersection over union of 0.84, and the lidar edge of 2022 lies within 0.8 m of the 2024 trace; the 2012 image is displaced 5 m, which is the registration error of that image and not a movement of the rock. Against the drawing, the visible mounds agree in position (0.3-4 m), orientation (1-3 degrees) and length (-4 % north, +0.4 % south). The absolute position is limited by the datum transformation from the national grid (2-5 m). The three edges (laser survey 8,675 m^2, design foot 6,989 m^2, 2024 photograph 5,743 m^2) differ because they sit at different heights (-2.3, -2.15 and -1.3 m ODN) on flanks of 1:4 to 1:5; the laser edge is the default because it is the highest-ranking evidence (survey, then drawing, then photograph), and the other two remain selectable.

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[13] Counts for the other reefs: Pratte's and Mount Maunganui each have a source that was only compared (a thesis and a reef-database page); Bunbury additionally has an interview with the developer, which was recorded as a disagreement and changed no geometry, and a rejected concept graphic. Neither changes the ranking. A footprint proposed by an AI tool was compared with the sources and rejected for Borth (200 by 200 m, 110,000 t; it matched neither drawings, lidar nor imagery), and nothing from it was used.

Crest level. The drawings give crest levels of +0.50 m (northern arm), +1.00 m, +0.00 m and +1.50 m ODN (oval), and the sections 1021-1023 give layer thicknesses (armour layer 2.70 m) and slopes. The laser surface inside the design crest strips has a median of +0.58 m (90th percentile +0.91 m) on the northern arm and +1.55 m (+1.75 m) on the oval; on the cells above -2.0 m ODN the model surface exceeds the design surface by 0.14 m on average (standard deviation 0.28 m). Three independent readings (drawing, lidar, section drawings) therefore agree within 0.3 m. The as-built flanks of the oval are 8-10 m wider than designed (slope 1:4.0-1:4.1 against 1:3).

Volume and mass. The rock volume above the modelled seabed is 30,119 m^3 (+-4,556 m^3, mainly from the +-0.4 m uncertainty of the bed under the rock), or 51,200 t at an assumed 1.7 t m^{-3} (48,200-54,200 t for 1.6-1.8 t m^{-3}). No reef-only quantity was found at a primary source, so the check is a bound: the armour layer in the three versions is 26,100-36,100 t, below the 42,000 t of the same rock class that Hansford (2011) gives for the whole of Phase 1 (breakwaters, groynes and revetment included). The default and design versions agree in total volume within 4 %; the photograph version is 15 % lower because its edge is high.

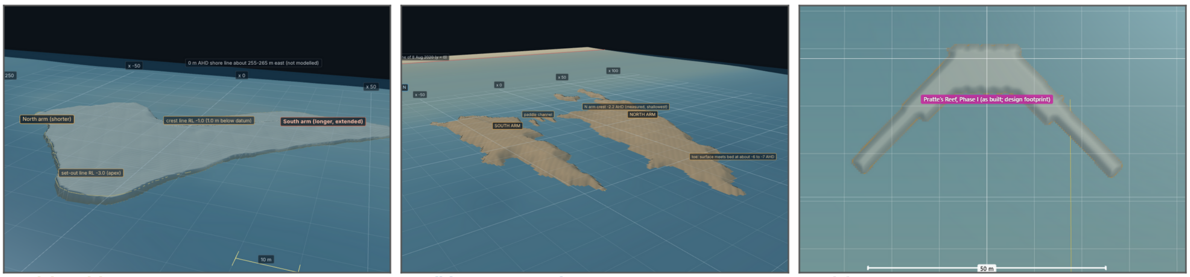
Seabed and conflicts. The pre-construction contours of Rigden et al. (2013, Fig. 2) served as an independent seabed check. The first model was 0.89 m deeper than the -4.0 m contour (standard deviation 0.14 m, 13 points), because a -1.1 % gradient taken from the EMODnet grid had been continued past the contour and the design bed levels under the arm head had been ignored. This was resolved by design choice D6: the -3.0 and -4.0 m contours became controls, after which the model minus contour is 0.00 m (standard deviation 0.01 m, 24 points, and 0.00 m, 13 points); the -2.0 m contour, which was not a control, is reproduced to -0.14 m (24 points), and the rock volumes fell by 3.4-3.9 %. Two sources still disagree with the model offshore: the EMODnet cells are shallower by 0.87 m on average (y = 368-513 m), but the cells in the reef zone are interpolated from a coarse global grid and only the gradient is used; and the Navionics chart implies a zero of -3.58 m (-3.75 m from the model) when its LAT datum is assumed. These conflicts were not removed but are reported, and they are the reason for the lower seabed confidence below. A further inconsistency lies inside the drawing set: the section drawing dimensions the oval as 46 m (2 x 23,000 mm) whereas the setting-out table puts the two end points 38 m apart. The model uses the table (38 m), which affects about 2 % of the oval volume, and the question remains open.

Datums. Heights are referred to mean sea level through the chain $z_{\mathrm{MSL}} = z_{\mathrm{ODN}} - 0.31$ m, where MSL is the mean of the four tidal planes of the shoreline management plan (+0.31 m ODN, +-0.15 m); the tide planes printed on drawing 1021 (MHWS +2.56 m, MLWS -1.74 m) agree with that table. LAT is -2.44 m ODN from the neighbouring ports of Barmouth and Fishguard (Aberystwyth is not listed; National Tidal and Sea Level Facility, n.d.). The vertical datum of the laser survey is not stated and was taken as ODN (+-0.15 m on the crest); the water edge at the time of the flight (about -2.3 m ODN) sets the level of the exposed surface. The highest astronomical tide was not found at a primary source, and the tide slider therefore ends at MHWS.

Confidence. The plan is rated high (drawing and laser survey agree to one pixel and 0.3 m), the crest levels high (three readings within 0.3 m), the tide offsets medium, and the seabed medium-low; the overall confidence is medium. The first-order unknown is the bed under the rock, for which no survey exists (uncertainty +-0.4 m, giving a crest-to-bed height of 4.4-4.5 m +-0.5 m on the northern arm).

Appendix B. Three-dimensional reconstructions

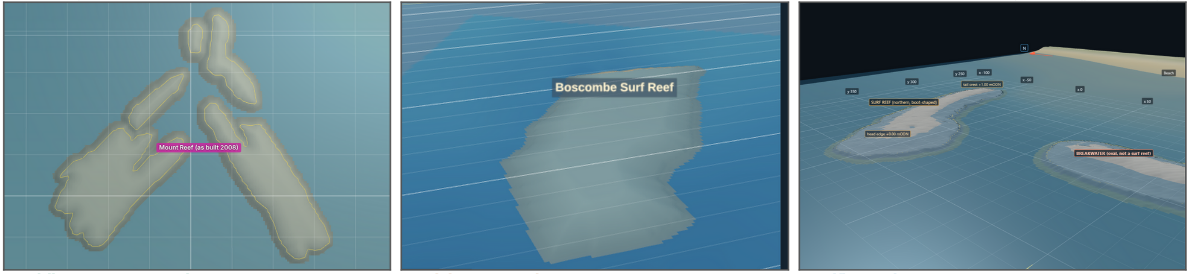
Nine of the 13 reefs had a sourced plan, crest level and seabed depth, and were reconstructed as three-dimensional models of the reef, the surrounding seabed and the local water levels (Section 2.3). The models show the as-built state; later changes such as settlement, renewal or removal are described in Section 3 and Appendix A, not drawn. All heights are relative to mean sea level through the datum chain of each reef (Appendix C, note 1), and only the tidal planes of the nearest standard port are used. The depth profiles of Figure 1, the computations of Sections 4 and 5 and the volumes marked "model" in Table 2 are taken from these models. Figure B1 shows the default version of each; the inputs, sources, assumptions and confidence of every model are listed per reef in Appendix A.


0 m AHD shore line about 255-265 m east (not modelled)
x -50
x 0
x 50
250
North arm (shorter)
crest line RL -1.0 (1.0 m below datum)
South arm (longer, extended)
set-out line RL -3.0 (apex)
10 m
N
x -50
x 0
x 50
x 100
paddle channel
SOUTH ARM
N arm crest -2.2 AHD (measured, shallowest)
NORTH ARM
toe: surface meets bed at about -6 to -7 AHD
Pratte's Reef, Phase I (as built; design footprint)
50 m

◆ **(a) Cables, 1999**
surveyed surface, Nov 1999

▲ **(b) Narrowneck, 1999–2000**
renewed reef, 2020 photo outline

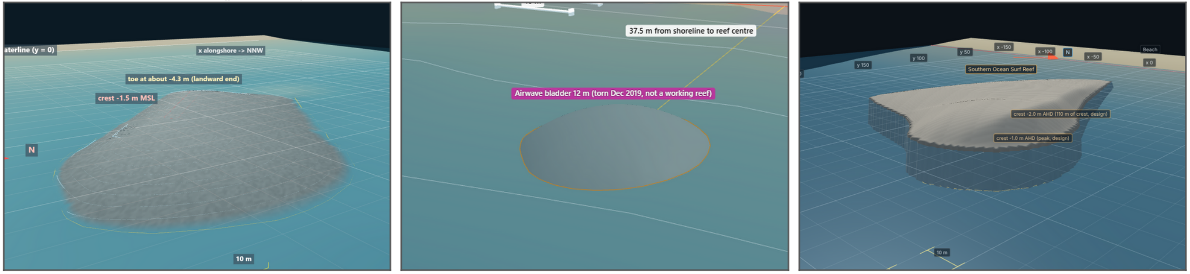
✖ **(c) Pratte's, 2000–01**
Phase I as installed, Sept 2000 (plan view)


Mount Reef (as built 2008)
Boscombe Surf Reef
N
y 350
y 300
y 250
x -100
x -50
x 0
x 50
tail crest +1.00 mODN
SURF REEF (northern, boot-shaped)
head edge +0.00 mODN
BREAKWATER (oval, not a surf reef)
Beach

▲ **(d) Mt Maunganui, 2005–08**
2013 multibeam outline (plan view)

▲ **(e) Boscombe, 2008–09**
as built 2009, idealised loft

□ **(f) Borth, 2011–12**
2022 LiDAR outline


aterline (y = 0)
x alongshore -> NNW
toe at about -4.3 m (landward end)
crest -1.5 m MSL
N
10 m
37.5 m from shoreline to reef centre
Airwave bladder 12 m (torn Dec 2019, not a working reef)
y 150
y 100
y 50
x -150
x -100
N
x -50
Beach
x 0
Southern Ocean Surf Reef
crest -2.0 m AHD (110 m of crest, design)
crest -1.0 m AHD (peak, design)
10 m

● **(g) Palm Beach, 2019**
as built 2019

✖ **(h) Bunbury Airwave, 2019**
Dec 2019 bladder, cap stand-in

● **(i) Albany, 2025**
Bluecoast final design (VE 2x)

Figure B1. Three-dimensional reconstructions of the nine modelled reefs (default version of each model; screenshots of the interactive viewer, panels cropped away). These are reconstructions, not photographs or surveys: each combines a traced or designed outline, a surveyed or inferred seabed, crest levels from design or survey documents and the local tidal planes, and its confidence differs by reef; the inputs, sources and assumptions of every model are listed per reef in Appendix A. Tiles are not at a common scale (use the scale bars and labels inside each tile); views are oblique except (c) and (d), which are plan views because the oblique preview was cut by the viewer panels; vertical scale is true except (i) with 2x exaggeration. The symbol before each name gives the surf outcome as in Figure 1 (circle works, triangle works rarely, diamond mixed or disputed, cross does not work, small hollow square unknown); the name is written in the reef's colour used in all figures.

Appendix C. Reef data table

This appendix reproduces the master data table of the review in condensed form: 13 surf reefs, ordered by the year in which construction started, with 38 data fields per reef. The reef numbers 1 to 13 are those of the sections A.1 to A.13 of Appendix A. The table is compiled from the project dataset (available from the author on request). In the dataset every field is a record with a value, a unit, a status, the full citation and the place where the value was read (page, table, figure or file); only the status is reproduced here, and the principal sources and the methods behind each reef are given in Appendix A. The sources of the outcome and durability entries are cited in Table C4.

Key to the status markers. A marker follows each entry and applies to all numbers in the cell.

Marker	Meaning
^{s}	**sourced**: a number or statement read in a cited source
^{d}	**derived**: computed in this review from sourced inputs, or from a traced plan shape or a reconstructed 3D model; the method is in Appendix A
^{a}	**assumed**: an assumption of this review, flagged in Appendix A (A-numbers)
?	**unknown**: not found in any source examined; no value is given

The outcome classes (works, works rarely, does not work, mixed or disputed, unknown) are this review's reading of the cited sources, marked ^{s} because the underlying statements are sourced; they are not measurements. Units: depths and heights in metres, depths positive downward; areas in m^2; volumes in m^3.

Reef	Lat., lon. (deg)	Built (completed)	Purpose	Material and size	Volume (m^3)	Cost, local currency (price year)	Cost (USD M, 2024 prices)
1. Burkitts Reef, Bargara (AUS)	-24.816, 152.467^d	1997^s	surf^s	natural basalt boulders and platform, reshaped (at least 12 large boulders moved)^s	300^s (rock moved)	AUD 10,000^s (1997)	0.015^d
2. Cables Reef, Perth (AUS)	-32.016, 115.748^d	1999^s	surf^s	granite boulders, 1.5 t body and 3.0 t perimeter stones, placed from a barge^s	5,500^s (rock)	AUD 1.8 M^s (1999)	2.19^d
3. Narrowneck Reef, Gold Coast (AUS)	-27.987, 153.434^d	1999^s (2000)	multi-purpose^s	geotextile sand containers (about 20 m long, 3-4.5 m diameter, 160-300 t); about 400 initially, 534 placed by 2018^s	70,000^s (sand fill of the containers; exposed mound 13,495^d in the 3D model over the visible field; council reef polygon 34,565 m^2)	AUD 2.8 M^s (2000*; incl. top-ups and maintenance to about 2006; build alone AUD 2.3 M, 1999; renewal 2017-18 about AUD 2 M not included)	2.96^d (lower bound)
4. Pratte's Reef, El Segundo (USA)	33.921, -118.433^d	2000^s (2001)	surf^s	sand-filled polypropylene geotextile bags, 3.0 x 2.1 x 1.2 m (110 in Phase I, +90 in Phase II)^s	1,350^s (sand, Phases I + II)	USD 550,000^s (2000)	1.00^d
5. Mount Maunganui Reef (NZ)	-37.645, 176.203^d	2005^s (2008)	multi-purpose^s	24 sand-filled geotextile containers, up to 50 x 7.2 x 3.2 m^s	2,800^s (as built)	NZD 1.5 M^s (2008*)	1.54^d
6. Opunake Reef (NZ)	-39.459, 173.852^d	2006^s (never completed)	surf^s	sand-filled geotextile bags on a thin sand layer over rock strata (27 bags in place April 2008)^s	4,900^s (design; about 3,400^d built)	at least NZD 1.7 M^s (2009; up to NZD 2.1 M if the further NZD 0.4 M sought was spent)	1.55-1.92^d
7. Boscombe Surf Reef (UK)	50.718, -1.839^d	2008^s (2009)	multi-purpose^s	sand-filled non-woven geotextile containers, two layers (32-55 containers, sources differ; each up to 60-70 m long)^s	12,800^s (sand fill, designers)	GBP 3.2 M^s (2009)	7.29^d
8. Kovalam Reef (India)	8.387, 76.975^d	2009^s (2010)	multi-purpose^s	sand-filled geotextile container strips, about 30 m long (5-6 strips fanning out)^s	4,000^s (basis not stated)	INR 75 M^s (2010*)	2.36^d

Reef	Lat., lon. (deg)	Built (completed)	Purpose	Material and size	Volume (m^3)	Cost, local currency (price year)	Cost (USD M, 2024 prices)
9. Borth reef (UK)	52.483, -4.056[d]	2011[s] (2012)	multi-purpose[s]	quarry rock armour (Types 1, 3, 4, 6; 0.005-10 t); about 51,000 t in the two mounds (30,119 m^3 x 1.7 t m^{-3}, derived)[s]	30,119[d] (rock, survey edge)	GBP 13 M[s] (2012*; whole Phase 1 scheme, not the reef)	(28.06)[d]
10. Palm Beach Reef, Gold Coast (AUS)	-28.107, 153.471[d]	2019[s]	multi-purpose[s]	quarried basalt/greenstone boulders, about 60,000 t[s]	25,000[s] (rock, as stated; 50,000-60,000 t also reported, i.e. 28,000-35,000 at 1.7-1.8 t m^{-3}; council register 33,000; 3D envelope 22,020[d])	AUD 18.2 M[s] (2019; announced project value; 2017 nourishment not included)	15.52[d]
11. Bunbury Airwave (AUS)	-33.328, 115.628[a]	2019[s] (installation failed)	surf[s]	inflatable Hypalon rubber bladder, 12 m diameter, ballasted with 140-150 t of beach sand[s]	105[d] (spherical-cap stand-in)	approx. USD 400,000[s] (2019; USD 350,000 in another source; City share AUD 75,000)	approx. 0.49[d]
12. Southern Ocean Surf Reef, Albany (AUS)	-35.018, 117.921[d]	2025[s]	surf[s]	quarried rock in three grades (belly, core, armour), 62,800 t in total[s]	34,000[s] (rock, design)	AUD 13 M[s] (2025*; reported final cost; committed funds AUD 11.75 M)	8.38[d]
13. Xala Reef, Jalisco (Mexico)	? (undisclosed)	2025[s] (completion not stated)	multi-purpose[s]	quarried boulders (“3,000 truckloads”)[s]	?	?	?

Table C1. Identification, construction, material and cost. Country codes: AUS Australia, NZ New Zealand, UK United Kingdom, USA United States. Price year marked * was not stated by the source and is assumed (the completion or construction year).

Reef	Plan area (m^2)	Along × cross-shore (m)	Height (m)	Crest depth, MSL (m)	Crest depth, LAT (m)	Seabed at toe, MSL (m)	Depth gap (m)	Ramp length (m)	Mean slope (1:n)
1. Burkitts Reef	?	?	?	?	?	?	?	?	?
2. Cables Reef	4,049^{d}	121 × 76^{d}	?	1.55^{s}	0.83^{a}	6.51^{d}	4.96^{d}	41.5^{d}	0.119 (1:8)^{d}
3. Narrowneck Reef	6,239^{d}	113 × 157^{d}	2.2^{d}	2.32^{d}	1.44^{d}	6.13^{d}	3.81^{d}	51.0^{d}	0.075 (1:13)^{d}
4. Pratte's Reef	435^{d}	58.0 × 28.6^{d}	1.4^{d}	2.68^{d}	1.21^{d}	4.03^{d}	1.35^{d}	1.4^{a}	1.000 (1:1)^{a}
5. Mount Maunganui Reef	1,071^{d}	67.9 × 60.4^{d}	2.7^{a}	2.03^{d}	0.87^{d}	4.73^{a}	2.70^{d}	5.1^{d}	0.530 (1:2)^{d}
6. Opunake Reef	2,970^{s}	30 × 100^{s}	3.5^{s}	?	0.00^{s}	3.00^{s} †	3.00^{d} †	?	?
7. Boscombe Surf Reef	4,042^{d}	88.5 × 94.3^{d}	3.2^{d}	0.90^{d}	-0.56^{d}	5.46^{d}	4.56^{d}	14.0^{d}	0.326 (1:3)^{d}
8. Kovalam Reef	1,216^{d}	59.0 × 41.0^{d}	?	?	?	?	?	?	?
9. Borth reef	5,743^{d}	196 × 140^{d}	4.5^{d}	-0.19^{d}	-2.94^{d}	4.26^{d}	4.62^{d}	38.0^{d}	0.115 (1:9)^{d}
10. Palm Beach Reef	11,972^{d}	137 × 149^{d}	3.9^{d}	1.50^{s}	0.62^{d}	6.67^{d}	5.17^{d}	94.0^{d}	0.055 (1:18)^{d}
11. Bunbury Airwave	112^{d}	12.1 × 12.1^{d}	2.0^{s}	0.68^{d}	-0.02^{d}	2.92^{d}	2.24^{d}	6.0^{d}	0.373 (1:3)^{d}
12. Southern Ocean Surf Reef	13,104^{d}	127 × 153^{d}	2.6^{d}	1.00^{s}	0.26^{d}	7.33^{d}	6.33^{d}	20.0^{d}	0.317 (1:3)^{d}
13. Xala Reef	?	?	?	?	?	?	?	?	?

Table C2. Plan shape and vertical geometry. All derived values come from the traced plan shapes and the reconstructed 3D models (Appendix A); the reef number is the same as in Table C1. Crest depth is the depth of the shallowest crest point below mean sea level (MSL) or lowest astronomical tide (LAT); a negative value is a crest above that datum. Seabed at toe is the depth of the bed (not of the reef surface) at the seaward edge of the traced outline on the crest transect. Depth gap = seabed at toe minus crest depth (MSL). Mean slope = depth gap divided by the horizontal ramp length from toe to crest. The dagger marks values referred to chart datum (LAT), not MSL (Opunake). Empty cells ("?") mean that no source gives the value and the reef could not be reconstructed (Burkitts, Xala), or that no crest or seabed depth is known (Kovalam).

Reef	Crest distance offshore (m)	Arm angle to shoreline (deg)	Tide range, spring / mean (m)	Typical swell Hs (m)	Typical Tp (s)	Surf outcome	Surfable days per year
1. Burkitts Reef	?	?	2.33[d] / 1.74[d]	0.5-2 (wave face 2-6 ft)[s]	7-10[s]	works[s]	30[s]
2. Cables Reef	268[d]	27 / 60[d]	1.14[d] (annual) / 0.80[s]	1-2 (summer), 1.5-2.5 (winter)[s]	?	mixed/disputed[s]	142[s]
3. Narrowneck Reef	212[d]	72[d]	1.31[d] / 1.02[d]	1.0[s]	?	works rarely[s]	?
4. Pratte's Reef	103[d]	45 / 44[d]	1.65[d] (great diurnal) / 1.15[d]	?	?	does not work[s]	?
5. Mount Maunganui Reef	308[d]	50 / 55[d]	1.75[d] / 1.42[d]	1.5[s]	11[s]	works rarely[s]	?
6. Opunake Reef	300[s]	?	3.16[d] / 2.35[d]	1.8-2.5 (design wave face)[s]	?	does not work[s]	1-in-8 to 1-in-3 days (design aim)[s]
7. Boscombe Surf Reef	239[d]	55[d]	1.76[s] / 1.13[d]	0.5[s]	4[s]	works rarely[s]	77 (design baseline) to 154 (design target)[s]
8. Kovalam Reef	90[s]	38[d]	? / 1.00[s]	1-2.5[s]	11-15[s]	mixed/disputed[s]	?
9. Borth reef	330[d]	64[d]	4.30[d] / 3.00[d]	?	?	unknown[s]	?
10. Palm Beach Reef	243[d]	45[d]	1.31[d] / 1.02[d]	?	?	works[s]	?
11. Bunbury Airwave	38[d]	?	1.20[d] (annual) / 0.25[d]	?	?	does not work[s]	?
12. Southern Ocean Surf Reef	189[d]	71[d]	1.38[d] (annual) / ?	?	?	works[s]	146[d]
13. Xala Reef	?	?	? / ?	?	?	unknown[s]	?

Table C3. Setting, typical conditions and outcome. Distance offshore is that of the crest from the shoreline definition used for each reef (see note 4 below). Arm angle is the angle between the reef arm (or the long axis of the outline) and the shoreline; two values are the two arms. For Narrowneck the value is the northern arm (the southern arm is 90.0 deg). For Pratte's Reef the angles hold if the V axis is shore-normal. Tide range: spring range (MHWS - MLWS: Narrowneck, Palm Beach, Mount Maunganui, Boscombe, Borth, Burkitts at Burnett Heads, Opunake at Port Taranaki) / mean range ((spring + neap)/2). At the diurnal and mixed sites the first value is not a spring range: it is the great diurnal range MHHW - MLLW at Pratte's (NOAA GT 1.654, MN 1.145 m) and the predicted annual range, about LAT to HAT, of the 2026 predictions of the Western Australia Department of Transport and Bureau of Meteorology at Cables (Fremantle, 1.14 m), Bunbury (1.20 m) and Albany (1.38 m) (Western Australia Department of Transport, & Bureau of Meteorology, 2026); the Port Taranaki and Burnett Heads levels are from Land Information New Zealand (2026) and Maritime Safety Queensland (2026); these three are lower bounds (three of four quarterly blocks parsed) and micro-tidal in the sense of a range under 2 m. The Bunbury mean range (0.25 m, from a consultant's tidal planes) and the Cables mean range (0.80 m, NSW review, no definition) are probably too small and were not recomputed; the Kovalam entry carries no definition in its source. Typical swell is as stated in the source and is mostly undefined (Narrowneck 1.0 m, Mount Maunganui 1.5 m and 11 s, Boscombe 0.5 m and 4 s, Kovalam 1-2.5 m and 11-15 s: Blacka et al., 2013, Table 4.3; Burkitts is wave-face height; Opunake is the designed breaking-wave face height).

Reef	Durability	Shoreline and sediment outcome
1. Burkitts Reef	No reported maintenance, storm damage or rework in nearly three decades (status January 2024); still surfed (Howells & O'Neal, 2024).	? Not assessed in any source found.
2. Cables Reef	No structural damage reported. Surveyed rock volume 4,671 m^3 (November 1999) and 5,219 m^3 (February 2002) differ by less than the survey error (this review, from Department of Transport WA soundings); no information on long-term stability (Blacka et al., 2013, p. 93).	None relevant: not designed for coastal protection; stable limestone shoreline, negligible longshore transport (Blacka et al., 2013, p. 67 and p. 92).
3. Narrowneck Reef	Containers stable in waves up to 8-10 m; settlement of 1-2 m locally; damage by anchors, spears, knives and boards; renewal in 2017-18 with 84 containers (Blacka et al., 2013, pp. 97-98; Corbett et al., 2023).	Mixed or disputed. A published 20-50 m salient (Mead & Borrero, 2011, p. 4) is read by Blacka et al. (2013, pp. 98-100) as a probable misreading of an arbitrary dune reference line; sand increases are mostly in the offshore bars (Blacka et al., 2013; Nettle, 2019).
4. Pratte's Reef	Bags moved, sank and were buried; the polypropylene largely disintegrated within about two years; crest dropped 1.5 m in the year after Phase II; removed in 2008 and 2010 (Blacka et al., 2013, pp. 116-119; Borrero & Nelsen, 2003; Fontaine, 2008).	No discernible effect on bathymetry or beach in monthly profiles and quarterly surveys (Blacka et al., 2013, p. 117).
5. Mount Maunganui Reef	Containers ruptured and lost sand during filling; one of the largest was damaged beyond repair; apex bags deflated to -2.0 to -2.4 m CD by 2013; reef removed 25 September to November 2014 (Mead & Borrero, 2011, p. 3; Dahm & Gibberd, 2013).	Weak or disputed. About 25 m of beach widening and an influence of about 150 m to the south-east (Mead & Borrero, 2011, p. 5), read by Blacka et al. (2013, p. 103) as subtle, with protection rated "Negligible".
6. Opunake Reef	Structure began to break apart and pieces washed ashore; designers called bag failure the "fatal flaw" of the technology (Reid & Sumner, 2021); 27 bags in place in April 2008; no formal removal reported.	? Not found.
7. Boscombe Surf Reef	Containers failed from March-April 2011; reef closed 31 March 2011; repairs aborted; operator liquidated in 2012; structure still present (Rendle and Davidson, 2012; this review, Appendix A.7).	Designers: beach in the lee accreted by up to 60 m within six months (Mead & Borrero, 2011, p. 5); later monitoring found no noticeable long-term effect on the 0 m contour (Rendle & Davidson, 2012); protection rated "Minor" (Blacka et al., 2013).

Reef	Durability	Shoreline and sediment outcome
8. Kovalam Reef	Containers torn, deflated and sunk; damage hampering the protective effect by 2013 (E. Rendle, pers. comm., 2013, in Blacka et al., 2013, p. 70); degraded in place, no formal removal (Raised Water Research, n.d.-c).	Designers: beach 10-15 m wider in the monsoon and up to 30 m in the non-monsoon season; fewer than three years of monitoring; independent assessment pending (Mead & Borrero, 2011; Blacka et al., 2013, p. 70).
9. Borth reef	No damage, deflation or removal found (imagery 2012-2024, LiDAR 2022); rock armour sized with three-dimensional physical modelling; defences withstood the December 2011 storms (Daily Post, 2013; Hansford, 2011).	Flood protection reported effective from December 2011; the scheme combines reef, groynes, breakwater and 84,500 t of shingle nourishment; no quantified shoreline response (Blacka et al., 2013, p. 70: performance “yet to be determined”).
10. Palm Beach Reef	Stable, with no significant movement or settlement (Hunt et al., 2022, monitoring abstract); no damage or removal found.	About 550,000 m^3 of sand retained over the first two years (ABC News, 2022); a journalistic statement, not a monitoring report.
11. Bunbury Airwave	Bladder torn along a glued seam during installation (December 2019); removed within about three days (ABC News, 2019).	? Not applicable (never operational).
12. Southern Ocean Surf Reef	Built with graded rock; a storm on 26 May 2025 moved some core rock; about 18 high rocks repositioned in February 2026 after user injuries; no other damage reported (Sheil, 2026).	? Not found.
13. Xala Reef	? Unknown (announced March-April 2026).	? Unknown.

Table C4. Durability and shoreline or sediment outcome as reported. The entries condense the dataset fields `durability` and `sediment_shoreline_outcome`; each carries its source. A question mark means that nothing was found.

Reef	s	d	a	?	Fields
1. Burkitts Reef	16	6	0	16	38
2. Cables Reef	20	15	1	2	38
3. Narrowneck Reef	16	19	0	3	38
4. Pratte’s Reef	15	18	2	3	38
5. Mount Maunganui Reef	16	16	3	3	38
6. Opunake Reef	21	7	1	9	38
7. Boscombe Surf Reef	19	19	0	0	38
8. Kovalam Reef	18	8	0	12	38
9. Borth reef	13	20	0	5	38
10. Palm Beach Reef	16	18	0	4	38
11. Bunbury Airwave	12	16	2	8	38
12. Southern Ocean Surf Reef	14	18	0	6	38
13. Xala Reef	10	0	0	28	38

Table C5. Number of dataset fields (of 38) in each status class. The fields counted include the identification fields (name, country, coordinates), so the totals show how much of the record is measured, derived, assumed or missing. s = sourced, d = derived, a = assumed, ? = unknown.

Notes to Tables C1-C5

1. **Datums.** Each 3D model is referenced to mean sea level (z = 0 at MSL). Conversions applied: Narrowneck and Palm Beach $z_{\mathrm{MSL}} = z_{\mathrm{AHD}}$ - 0.12 m (Maritime Safety Queensland, Gold Coast Seaway; the rule used in the papers would shift this by 0.24 m); Cables, Bunbury and Albany: Australian Height Datum (AHD) taken as MSL (Cables low water datum is 0.716 m below AHD; Albany LAT is 0.74 m below AHD); Mount Maunganui $z_{\mathrm{MSL}} = z_{\mathrm{CD}}$ - 1.13 m (Land Information New Zealand, Tauranga, present MSL); Boscombe $z_{\mathrm{MSL}} = z_{\mathrm{ACD}}$ - 1.40 m (survey zero assumed to be chart datum); Borth $z_{\mathrm{MSL}} = z_{\mathrm{ODN}}$ - 0.31 m; Pratte's $z_{\mathrm{MSL}} = z_{\mathrm{MLLW}}$ - 0.849 m. Opunake and Kovalam have no usable datum.
2. **Ramp and slope.** The ramp is the horizontal toe-to-crest distance on the shore-normal transect through the shallowest crest cell; the crest edge is the seaward-most cell within 0.25 m of that column's maximum (the dataset gives the sensitivity to 0.10 and 0.50 m). Ramp and slope are derived from the reconstructed 3D models for Narrowneck, Cables, Albany, Palm Beach and Borth (median over the columns of the long diagonal mound), from the idealised loft of the 3D model for Boscombe (the 1:3 flank is an assumption), by hand from the 2013 digital elevation model for Mount Maunganui, and from the spherical-cap stand-in for Bunbury. **Pratte's ramp and slope are assumed (1:1 bag face).** Where a sourced crest depth exists (Cables, Palm Beach, Albany) the depth gap uses it. Only Narrowneck, Cables, Albany, Palm Beach, Borth and Boscombe have a ramp from audited geometry; relations should not be fitted through the whole set without the status markers. Table C2 keeps the master-table definition (shallowest crest point); the surf analysis of Section 4 (Table 2) reads Albany, Cables and Borth on a surf cut through the ridden part of the reef (Appendix A, item 4; Figure E1 in Appendix E), so its crest depth, gap, ramp and slope differ for these three reefs (Albany 1.97 m, 5.40 m, 99 m, 1:19.8; Cables 1.92 m, 4.59 m, 45 m, 1:23.7; Borth crest 0.47 m above MSL, 4.77 m, 40 m, 1:9.5).
3. **Plan areas are definitions, not always the reef.** Narrowneck: visible container field on the aerial of 8 August 2020, a lower bound. Cables: original design plan. Pratte's: Phase I design layout. Mount Maunganui: the -2.0 m CD contour of the 2013 multibeam survey (the 2008 toe outline is 1,424 m^2). Opunake: design bounding rectangle 99 x 30 m, not the traced reef; the bag layout is not published, and the along-shore and cross-shore values in Table C2 are the published design size with unknown orientation. Boscombe: outline traced on the image of 28 September 2011. Kovalam: photo trace at an assumed camera height of 60 m (550-1,700 m^2 for 30-80 m), low confidence. Borth: **area is the visible rock edge on the photograph of 17 September 2024; the volume (30,119 m^3) belongs to the laser-survey edge of 2022 (8,675 m^2)**, the default of the 3D model. Palm Beach: toe outline. Albany: Bluecoast design outline (the dark rock patch visible in 2026 is 10,918 m^2). Bunbury: circle of 12 m diameter traced on a drone photograph. Narrowneck volume (70,000 m^3) is the sand fill stated by the designers; the 3D mound envelope is a different quantity.
4. **Distance offshore** is measured from different shoreline definitions: the MSL waterline of the image used (Narrowneck 8 August 2020, Palm Beach 1 December 2025, Pratte's), the wet/dry-sand edge (Mount Maunganui, Albany beach-face line), the back-of-beach defence

line (Borth), the pre-reef shoreline (Kovalam, as stated in the NSW review) and a survey surf line (Boscombe). Differences below 30 m are noise.

5. **Costs.** Costs are in nominal local currency as stated by the source, for what each source calls the reef; they may include studies. What each total covers differs: the Borth figure (GBP 13 M) is for the whole Phase 1 defence scheme and is not a reef cost, so it is bracketed and left out of any cost per m^3; Narrowneck includes the top-ups and maintenance to about 2006 but not the 2017-18 renewal (about AUD 2 M; The Inertia (2017)) or the beach nourishment (Blacka et al., 2013, p. 101); the build alone was AUD 2.3 M (International Coastal Management, 2025); Opunake is the spend known by 2009 and is a lower bound (Reid & Sumner, 2021); Palm Beach is the announced project value without the 2017 nourishment; Bunbury is an approximate figure from a reef database and the surf press (ABC News, 2019; Raised Water Research, n.d.-a); Albany is the reported final cost (Elezovich, 2026), against AUD 11.75 M announced in 2024 (City of Albany, n.d.; Government of Western Australia, 2024). The cost per m^3 of Table 1 divides the 2024-price cost by the reported volume; its outliers are discussed below Table 1. Volumes also differ in kind: Narrowneck's 70,000 m^3 is the sand inside the containers, whereas the exposed mound of the 3D model is about 13,500 m^3; the Palm Beach tonnage and volume imply a solid-rock density (2.4 t m^{-3}), so at least one of them is uncertain; Opunake is the design volume, of which about 3,400 m^3 were built (Mead & Borrero, 2011, p. 3; Blacka et al., 2013, Table 4.4). USD = local amount divided by the World Bank annual-average official exchange rate of the price year; USD (2024) = nominal USD multiplied by the US consumer price index of 2024 divided by that of the price year (2025 uses the 2024 index, the latest in the series) (World Bank, 2026). The price year is the construction or completion year; for spending over several years (Narrowneck, Opunake, Boscombe, Borth) a single year is used, so these values are approximate. Both are derived and inherit the uncertainty of the price year.
6. **Years.** "Built" is the start of construction and "completed" the end of the main phase. Reefs with soft structures that failed or were removed are listed with their original construction years.
7. **Coordinates** are the centres of the traced outlines (derived); Bunbury is an estimated centre (assumed); Pratte's is the centre of the survey window of Borrero and Nelsen (2003) (position +-150 m); Opunake is the centre of the consent area; Burkitts is the platform centre; Xala is undisclosed.
8. **Not tabulated** (available in the dataset): seabed depth at the crest, mean tide range definitions, design-wave statements, swell direction, arm angle to the swell, and the alternative values (other published or derived values for the same quantity, each with its own status and source).
9. **Cautions.** Typical swell (Hs, Tp, direction) is unknown for most reefs because the design reports of the original designers were not read in full; the Haifa wave data are not part of this table. Cost conversions assume the price year given. Values of Mount Maunganui (assumed bed), Bunbury (stand-in surface) and Pratte's (assumed slope) are weak and are marked.

Appendix D. Other reefs and geotextile structures named by Bar and Drimer (2023) and Israeli geotube projects

This appendix lists the structures that are not among the 13 surf reefs of the survey (Section 3, Appendix C) but are named in the survey paragraph of Bar and Drimer (2023, p. 2) or are relevant to the Israeli coast. None of them was built as a surf reef. Of the eleven projects and studies named in Bar and Drimer (2023), three (Narrowneck, Pratte's Reef and Mount Maunganui) are among the 13 reefs of this review; the other eight are D.1 to D.8: four (D.1 to D.4) are geotextile-tube (Geotube) shore-protection breakwaters, one (D.8) is a cliff-foot geotube scheme, one (D.5) is a nature-based breakwater, and two (D.6, D.7) are design studies; D.9 to D.11 are further Israeli geotube projects. Each entry gives type, year, place, purpose, material and size, outcome, and the verification level of this review. The levels are: READ (the cited text was opened and the number checked), ABSTRACT (only an abstract or catalogue record was reached), and VENDOR (a manufacturer page, treated as promotional). Nothing is stated here that is not in the cited source; where the source is silent the entry says "not found". All sources were read or checked on 15 October 2026. Page numbers of conference papers and of the Israeli audit reports are PDF pages.

Bar and Drimer (2023) is not a survey. Their review of built projects is one paragraph; the remainder of the paper develops and validates a two-dimensional boundary-element wave-flume model for geotextile tubes as submerged impermeable structures. They borrow the statement that "most of the projects have failed" from the Raised Water Research website, without a list or a definition of failure (Bar & Drimer, 2023, p. 2). Of the eight built projects they name, only three (Narrowneck, Pratte's Reef and Mount Maunganui) are among the 13 reefs of this review; ten of the 13 reefs are not mentioned by them.

Entry	Place	Type	Built	Outcome reported	Level
D.1 Young-Jin	Gangneung, Korea	submerged geotube detached breakwaters	about 2004-05 (d)	shoreline +2.4 to 7.6 m in 12 months; local re-erosion	READ
D.2 Yucatan	Progreso area, Mexico	submerged geotube breakwaters	2005-07	mixed; downdrift shadow effect	READ
D.3 Al Aqah	Fujairah, UAE	geotube groins and submerged breakwater	2010-14	"well controlled" (manufacturer)	VENDOR
D.4 Sigandu	Batang, Java, Indonesia	low-crested geotube breakwaters	from Sep 2014	salients and tombolos within a month (port jetty confounds)	READ
D.5 Living Breakwaters	Staten Island, USA	armour-stone and eco-concrete breakwaters	2021-24	too recent; model results only	READ
D.6 Portuguese MFAR	Portugal	multifunctional reef, design study	not built (as far as found)	none (design study)	READ
D.7 Babilonia MFAR	Guardamar, Spain	multifunctional reef, design study	not built	none (design study); predicted +33 m beach	READ

Entry	Place	Type	Built	Outcome reported	Level
D.8 Ashkelon North	Ashkelon, Israel	cliff-foot geotubes (48 tubes, 5 clusters)	2017/18-2019	several tubes torn in first winter; later collapsed	READ
D.9 Ashkelon South	Ashkelon, Israel	armoured geotube pilot	from Sep 2023	pilot, not built as a structure	READ
D.10 Hof HaTzuk	Tel Aviv, Israel	submerged geotube breakwater	planned 2019, stopped 2022	not built	READ/secondary
D.11 Ashkelon trial	Ashkelon, Israel	geotube trial, about 20 m tubes	2005-06	storm damage and vandalism	secondary

Table D1. Structures named by Bar and Drimer (2023) that are not in the 13-reef survey, and Israeli geotube projects.

D.1 Young-Jin beach, Gangneung, Gangwon-do, east coast of Korea

Type: submerged double-lined geotextile-tube detached breakwaters for shore protection; an example of submerged geotube shore protection, not a surf reef (Oh & Shin, 2006b, p. 879). *Year:* not stated; the paper was received in March 2005 and reports 12 months of monitoring, so construction was about 2004-2005 (derived; assumption). *Purpose:* severe erosion threatened scenery and public property; a tetrapod breakwater would have needed heavy equipment, so geotubes were chosen as a cheaper and faster alternative (Oh & Shin, 2006a, pp. 1-2). *Material and size:* woven geotextile; sand fill; tubes filled by cutter-suction dredge. Two lines of tubes as detached breakwaters in about 3.0 m of water, 90-100 m from the shoreline; each tube 9.5 m in circumference (3.5 m diameter), 50 m long and 1.8 m effective height (85 % fill), eight tubes in four twin units covering 240 m of shoreline, with a scour apron. Model tests favoured zero water depth above the crest, and the tubes were installed 1.0 m below the water surface, as deep as possible to spare the scenery and reduce navigational risk (Oh & Shin, 2006b, pp. 892-893). *Outcome:* the shoreline advanced about 2.4-7.6 m seaward within 12 months and sand accumulated around the tubes (up to 10.4 cm at the monitored point); at benchmark BM1 the shoreline advanced 5.3 m in three months and then re-eroded, which the authors attribute to changed tidal flow; seaweed colonised the tubes after one year; scour developed on the seaward side and between the tubes (Oh & Shin, 2006b, pp. 879, 893–894). The model tests found the two-line zero-freeboard layout stable up to 5.0 m significant wave height (Oh & Shin, 2006b, p. 894). *Cost and contractor:* not stated. *Relevance:* a zero-freeboard soft structure at about 3 m depth with local re-erosion and no durability record beyond 12 months. *Citations:* Oh and Shin (2006b) (journal version, read in full; journal pages) and Oh and Shin (2006a) (conference version). *Level:* READ.

D.2 Yucatan coast (Progreso area), Mexico

Type: geotextile tubes as low-crested or semi-submerged breakwaters for beach restoration. *Year:* field installation 2005 (4 km of tubes by September 2005), further sites 2005-2007 (Alvarez et al., 2006; González Leija et al., 2014). The paper that Bar and Drimer (2023) cite is a laboratory flume study, not the field project. *Purpose:* shoreline retreat of about 1 m/yr and more over 15 years; federal beach rehabilitation programme from 2001 (Alvarez et al., 2006). *Material and design inputs:* woven polypropylene tubes with a scour apron; breaking wave height below 1.0 m (93 % of the time), T_z 4-8 s, bed slope 0.1-0.5 %, tide range 0.90 m, littoral drift 60,000 m^3/yr, D_{50} = 0.30 mm; the designers computed the Iribarren number, found spilling breakers, and set the crest so that high-tide freeboard avoids reflection (Alvarez et al., 2006). *Outcome:* mixed. "In some cases shore protection and beach stabilization was achieved … but in other places the results have been far from the expected", with a downdrift shadow effect; tubes were set at about -0.3 m relative to MSL (González Leija et al., 2014). The laboratory study found that the transmission coefficient depends on the ratio of submergence depth to incident wave height, and that tubes crowned well below mean water level make waves "steepen and break more violently". *Cost and contractor:* not found. *Citations:* Alvarez et al. (2006), González Leija et al. (2014) read; Alvarez et al. (2007) not read. *Level:* READ (magazine article and one-page poster).

D.3 Al Aqah Beach, Fujairah, UAE

Type: geotextile-tube groins plus a seaward submerged breakwater, for beach nourishment and erosion control; not a surf reef. *Year:* after cyclone Gonu (2007); manufacturer's case-study timeline 2010-2014. *Purpose:* erosion over "several kilometers"; rip-rap groins had sunk under repeated wave action (ACE Geosynthetics, n.d.). *Material and size:* geotextile tubes of 8.6-17.2 m circumference and 52-77 m length, two arms of 200 m each, filled with sand in situ. Crest depth, fill height and volume: not found. *Outcome:* the manufacturer states that erosion is "well-controlled"; no independent monitoring found. *Cost:* not found. *Citations:* ACE Geosynthetics (n.d.), read; Chien et al. (2013) was not accessible (abstract level only); Chien et al. (2014) as reported by Bar and Drimer (2023), full text not accessed. *Level:* VENDOR.

D.4 Sigandu Beach, Batang Regency, Central Java, Indonesia

Type: low-crested geotextile-tube breakwaters (LCB) for shore protection, not a surf reef. *Year:* installation from September 2014 (October 2014 in the performance section); paper 2015 (Sulaiman et al., 2015, pp. 510, 518). *Purpose:* an eroding tourist beach, attributed to the jetty of the Batang commercial port that blocks the westward longshore transport (Sulaiman et al., 2015, pp. 511, 518). *Material and size:* geotextile tubes of about 2 m width, 1.5 m height and 20 m length as a generic type; 14 units installed (the text lists 5 + 5 + 5 tubes in three groups, which is 15), three gaps of 20 m, about 75 m from the coast, set at about mean sea level (no other crest depth is given) (Sulaiman et al., 2015, pp. 514-516). *Outcome:* about one month after installation, salients and tombolos had formed behind the tubes, with on average 17,627 m^3 of sediment accreted over 280 m of beach 75 m wide (about 84 cm thick); the authors attribute the speed of the accretion largely to the port jetty and to dredged sediment, so the amount cannot be credited to the structure alone. Some tubes had settled and tilted at the gap ends from toe scour by gap currents, but the structures were in good condition overall (Sulaiman et al., 2015, pp. 517-518). A later paper by the same group (Sulaiman et al., 2017) was not read. *Cost:* not stated. *Citations:* Sulaiman et al. (2015) (read in full; journal pages); the vendor page Solmax (2023) gives tube dimensions of 9.5 m circumference, 1.5 m filled height and 25 m length. *Level:* READ.

D.5 Living Breakwaters, Tottenville, Staten Island, New York

Type: partially submerged breakwaters of armour stone and ecologically enhanced concrete units (ECOncrete armour units and tide pools), with "reef ridges", "reef streets" and crenulated crests for habitat, plus a shoreline restoration; coastal risk reduction and ecology, no surfing, in Raritan Bay at Tottenville (Marrone et al., 2019, pp. 1044-1045). *Year:* design-modelling paper 2019; construction from 2021, completion announced in September 2024 (Hochul, 2024). *Design and modelling:* the paper describes the modelling programme only (wave transformation, GENESIS shoreline change, FUNWAVE wave attenuation, Delft sediment transport, FLOW-3D, and 1:20 and 1:80 physical models); it gives no breakwater dimensions, crest levels or costs. The design target was wave heights below 1 m (3 ft) in the lee under 100-year conditions with 0.8 m of sea-level rise, which the final design achieved in the model; the shoreline model showed erosion mitigated, with accretion near the most exposed neighbourhood, and the sediment model showed lower bed shear stress at the shoreline and negligible effects on the navigation channel (Marrone et al., 2019, pp. 1048, 1053–1054). These are model results, not measurements. *Size and cost:* 2,400 linear feet of breakwaters, USD 111 million (Hochul, 2024). *Outcome:* too recent for performance data. *Other design paper:* Baker et al. (2018), abstract only. *Relevance:* a rock-plus-eco-concrete submerged breakwater, comparable to the ECOncrete applications in Israel; it shows how a balance between longshore transport and storm protection is modelled. *Level:* READ (paper); size and cost come from a press release.

D.6 Portuguese multifunctional artificial reef (design study)

Type: preliminary design of a multifunctional artificial reef (MFAR; coastal protection and surfing) for the northwest coast of Portugal, a delta-shaped submerged breakwater of geotextile sand containers on a platform; a design study with numerical and physical tests, and no evidence that it was built (ten Voorde et al., 2009, pp. 69, 78). *Year:* paper 2009; a follow-up on the parameters in the vertical plane is Antunes do Carmo et al. (2011) (not read). *Size and crest depth:* delta angle 66 degrees, side slope 1:10 with a side length of at least one-fourth of the local wavelength, valid only for a sea bed flatter than 1:50; crest submergence 1.5 m for a design breaking wave of 2.0 m, with a minimum reef height of 2.0 to 2.5 m and a platform depth of 4.0 m; in the worked example, 175 m from the delta base and 250 m from the apex to the shore, the ride is about 82 m (ten Voorde et al., 2009, pp. 69, 75–76, 78). *Position:* the distance from the apex to the undisturbed shoreline should exceed 1.5 times the natural surf-zone width, after Ranasinghe and Turner (2006) (Figure 8a) (ten Voorde et al., 2009, pp. 77-78). *Outcome and cost:* none (not built); no cost is stated. *Relevance:* the main design guideline in the literature for a submerged surfing reef that must also protect the coast. *Level:* READ.

D.7 Babilonia beach, Guardamar del Segura, Alicante, Spain (design study)

Type: design method for a multifunctional artificial reef, applied as a design study to the eroding Babilonia beach, Guardamar del Segura (Alicante), on the Spanish Mediterranean coast; not built as far as the paper states (López et al., 2016, p. 910). *Year:* paper 2016. *Design:* a V-shaped reef of geotextile containers, 60 m in the cross-shore direction, 63.4 m wide at the top, internal angle 24 degrees, offshore slope 1:10, in 4 m of water with the crest at 0.7 m depth, breaking zone about 100 m from the coast and a maximum ride of about 65 m; it is combined with a 240 m dike at the -5 m contour, 329 m from the shoreline, to form a hemitombolo at the marina mouth (López et al., 2016, pp. 916-919). *Predicted outcome:* the authors' model predicts a maximum beach accretion of 33.2 m over 270 m of beach, and a wave for beginner to intermediate surfers; they state that the reef "could be used 60 % of the time" by surfers of low to intermediate skill, the 60 % being the occurrence of the surfing wave conditions in the study area (López et al., 2016, pp. 910, 921–922). These are modelled predictions, not measurements. *Cost:* volume about 1,920 m^3; the cost is estimated from the unit costs of other reefs (43 to 332

EUR/m^3) at 82,560 to 637,440 EUR, most probably about 403,200 EUR (López et al., 2016, p. 921). *Relevance:* the closest published analogue to a microtidal Mediterranean surf reef; the authors note that a microtidal site keeps the reef operable and rarely hidden (López et al., 2016, p. 921). *Level:* READ.

D.8 Ashkelon North geotubes, Israel

Type: submerged geotextile-tube structure of 48 tubes in five clusters at the foot of the eroding Ashkelon cliff; urgent, temporary marine protection until permanent breakwaters are approved. It was the first geotube project in Israel (Bar and Drimer, 2023, p. 2, citing a 2016 design report by Gouaud that could not be located online; not verified). *Dates:* the State Comptroller of Israel (2020, PDF pp. 71, 83) gives July 2017 as the contractor's start and September 2019 as completion; the Government Company for the Protection of Mediterranean Sea Cliffs (Government Company for the Protection of Mediterranean Sea Cliffs, 2026) lists April 2018 to April 2019; Rinat (2019) says the tubes were laid in summer 2018. *Size:* 48 tubes (State Comptroller of Israel, 2020); 50 according to Rinat (2019); tube dimensions are not stated in any source read. *Cost:* about NIS 11 million (State Comptroller of Israel, 2020); NIS 10 million (Rinat, 2019). *Outcome:* "several of the geotubes were damaged and torn shortly after being laid in the sea, after the winter storms, and had to be repaired" (State Comptroller of Israel, 2020, p. 71). In early February 2019 the contractor reported severe damage of unknown cause and halted laying; the last storm "was not so severe" (Rinat, 2019). The number of destroyed tubes is not stated in any source read; no source gives separate counts for damage by storms and by boats. During maintenance in late 2021, 12 geotubes were replaced (State Comptroller of Israel, 2024, p. 48). The 2024 report states that the geotube marine protection in Ashkelon North "collapsed" (p. 29) and that geotubes were considered "less effective" there, with five rock breakwaters planned (p. 53). The same report lists general geotube threats: poor design assumptions, poor filling, and damage by propeller blades, anchors and divers' knives (p. 9). *Surfers and ecology:* the local surfers' organisation opposed the tubes (Rinat, 2019). No monitoring results were found. *Netanya:* no source shows a geotube built or contracted at Netanya, whose cliff protection is rock breakwaters (State Comptroller of Israel, 2020, p. 71; 2024, p. 48; Government Company for the Protection of Mediterranean Sea Cliffs, 2026). Ashkelon North is therefore the site used in this review: it is the site where the sources document geotubes torn in the first winter after laying (State Comptroller, 2020, p. 71). *Level:* READ (two official reports and one press article).

D.9 Ashkelon South (area 39), Israel

Geotubes were not built. A December 2017 committee decision limited stage 1 to sand feeding; from September 2023 the company has piloted an "armoured geotube" (fabric with metal conductors for low-voltage mineral precipitation), designed in response to "the low durability and survivability of the fabric sleeves used at Ashkelon North" (State Comptroller of Israel, 2024, pp. 54, 55, 57). *Level:* READ.

D.10 Hof HaTzuk ("Cliff Beach"), Tel Aviv

A submerged geotube breakwater was planned (tender 2019), opposed by surfers and an environmental group, and stopped at tender stage in 2022 on the Ministry's recommendation; it was not built (State Comptroller of Israel, 2024, p. 11; the 2019 press reports were not re-opened). *Level:* READ for the 2024 report, secondary for the rest.

D.11 Ashkelon geotube trial, 2005-2006

An earlier trial by the Ashkelon Economic Company (about NIS 1.7 million, tubes of about 20 m) suffered storm damage and vandalism according to a Haaretz report of 6 April 2006 as recorded in an earlier internal note; the report was not re-opened and the entry is not verified. *Level:* secondary.

D.12 Types named without a site

Bar and Drimer (2023, p. 1) mention floating breakwaters, Bragg breakwaters, submerged reef balls and an artificial mangrove-root system as alternatives that “may present potential advantages in cost, hydrodynamics and environmental features”. No project data are given and none of the underlying papers was read.

Appendix E. Profile cuts and peel-angle geometry per reef

This appendix shows, on the plan of each reef, the lines that the quantitative sections rely on. Figure E1 draws them in the same plan frame and style as Figure 1 (shoreline at the bottom, seaward up, view looking seaward, plan shape of the default outline). Three kinds of line are drawn.

The main profile cut (black dashed line) is the line along which the two-dimensional cross-section of the three-dimensional model is read; it is the profile of Figure 1 and the basis of the ramp length, depth gap and slopes of Figures 4, 5 and 7 to 9. Where the reef has two arms, the cut runs along one of them (Narrowneck: the south arm; Mount Maunganui: the south, long arm), because the cut through the other arm clips the tip of the structure and does not cross the part where the wave is ridden. For Albany, Cables and Borth the cut was placed along the surfed section of the reef (the reasons are given in the next paragraph); for the other reefs the cut is the line through the shallowest crest cell (or the middle of the footprint) that comes closest to the crest depth of the master table (Appendix A). Every cut was sampled every 1 m on the model grid and runs perpendicular to the local shoreline, except where stated.

Placement of the cuts at Albany, Cables and Borth. Albany: the automatic line (x = 48 m of the model frame) crosses the short, shallow take-off plateau at the offshore end of the reef, not the ridge that is surfed, so the main cut was moved to the middle of the long -2.0 m AHD ridge (“crest 110 m at -2 m AHD”, City of Albany, n.d.), at x = 17 m. Cables: the main cut passes through the centroid of the footprint and is perpendicular to the fitted 0 m AHD shoreline (offshore bearing 277.8°T), so it is 17.5° off the model’s y axis. Borth: the cut at x = -43.5 m is 10 m south of the line used for Figure 1 in an earlier version of this review; no source says where the Borth wave breaks, so the position is a judgement (Appendix A, section A.9).

The second cut. Figure 4 originally drew a second, hollow symbol for several reefs. A second cut is kept only where it has a clear physical meaning. Albany is the only such case: the main cut is the ride section along the long ridge, and the second cut (x = 48 m, red dotted line) passes through the take-off peak, the short and shallow part of the reef where the surfer catches the wave and which probably acts as a refraction and focusing element for the wave that then runs along the ridge (a reading of the design plan, not a measured fact; Royal HaskoningDHV, 2015). The second cuts of Narrowneck (a cut through the north arm that clips the tip, ramp length 16 m) and Pratte’s (a cut through the tip of the apex, ramp length 1 m) are artefacts of where the shallowest model cell happens to lie, and were dropped.

The arm and the incident crest. The peel angle of Figure 5 is built from the angle $\varphi0$ between the incident wave crest and the reef arm that the wave peels along (refracted to the break depth by Snell’s law, Appendix A and Section 4). The arm is drawn as a solid black line at the true bearing of the arm axis given in the shape files of the reef (Appendix A); for Pratte’s, where no bearing is available, the arm is the principal axis of each half of the V outline. The incident crest (blue dashed line, through each arm and through the arrow offshore of the reef) is perpendicular to the arrow, which gives the direction in which the waves are assumed to travel. $\varphi0$ is obtained in one of three ways, written beside each reef name in Figure E1 and in the tables below: *sourced* (Cables: the design peel angle of 45° between each arm and the wave crest; Pattiaratchi, 1999; Lyons, 1992, as cited by Mead and Borrero, 2011); *derived from a design drawing* (Pratte’s: the arms are drawn at 44.7° and 44.4° to the axis that is the assumed wave direction; Skelly Engineering, 2000, as reproduced by Raised Water Research, n.d.-f; Boscombe: the reef axis against the crest of the 191° design swell, geometry only; Mead et al., 2010); or *assumed* in red (Narrowneck, Mount Maunganui, Borth, Palm Beach, Albany: no incident direction is documented, so the swell is assumed to arrive parallel to the shoreline, that is the crest is parallel to the beach and $\varphi 0$ is the angle between the arm and the shoreline). The assumption is the usual first guess (refraction turns a swell towards parallel to the shore), but it ignores any oblique approach and therefore probably overstates $\varphi 0$ where the real swell is oblique (for example Narrowneck, where the dominant swell is from the south-east, Nettle, 2017; with an assumed 49° obliquity $\varphi 0$ would be 41°, the dashed variant of Figure 5).

Cut positions are given in the horizontal frame of the model of each reef (x alongshore, y offshore; Appendix A) and as the alongshore offset of the cut from the middle of the plan drawn in Figure E1 (positive: to the right when looking seaward). Bearings are true. The offshore (shore-normal) direction of the cut is the seaward bearing of the local shoreline normal.

E.1 Narrowneck, Gold Coast

Item	Value
Main cut	south arm, x = -40 m, reef run y = 232 to 332 m, offshore bearing 86.2°T; 39 m right of the plan middle
Second cut	none (the north-arm cut, x = 46 m, clips the arm tip and was dropped)
Arm axes	north arm 104.0°T (72.2° to the shoreline), south arm 86.2°T (90.0°); both from the traced outline (Appendix A, A.3)
Incident direction	assumed shore-normal, from 86°T (crest parallel to the shoreline, 356.2°T)
$\varphi 0$	81.1° for both arms (mean of the two arm angles, converging seaward by 17.8°); alternative 41° for a south-easterly swell
Status	assumed; the south-east swell direction is qualitative (Nettle, 2017)

E.2 Cables Reef, Perth

Item	Value
Main cut	oblique cut through the footprint centroid (x = -5.0 m, y = 274.8 m), perpendicular to the fitted 0 m AHD shoreline (7.8°T), offshore bearing 277.8°T; at the plan middle (+1 m)
Second cut	none
Arm axes	north arm 35°T (27° to the shoreline), south arm 128°T (60°); georeferenced design outline (Appendix A, A.2); apex angle 87°
Incident direction	the crest that meets both arms at the same angle: swell from about 262°T, 16° from shore-normal (derived from the design angle, not a measured direction)
$\varphi 0$	45° for both arms (43.5° as drawn)
Status	sourced: design peel angle of about 45° (Pattiaratchi, 1999; Lyons, 1992, as cited by Mead and Borrero, 2011); the real swell direction is not confirmed

E.3 Pratte's Reef, El Segundo

Item	Value
Main cut	centreline through the apex, x = 0 m, reef run y = 93 to 103 m; offshore direction taken as shore-normal (about 65°T); at the plan middle
Second cut	none (the crest-line cut at the apex tip, ramp length 1 m, was dropped)

Item	Value
Arm axes	two arms at 44.7° (north) and 44.4° (south) to the V axis; the V axis is taken as shore-normal, so the arms lie at about 45° to the shoreline (Skelly Engineering, 2000)
Incident direction	along the V axis, the direction that the drawing designates as the wave direction (assumed shore-normal)
$\varphi 0$	44.7° and 44.4°
Status	derived from the design drawing; the real swell direction is not confirmed

E.4 Mount Maunganui Reef

Item	Value
Main cut	south (long) arm, x = -22.5 m, reef run y = 287 to 313 m, offshore bearing 46.2°T; 22 m right of the plan middle
Second cut	none
Arm axes	west (northern) arm 86°T (50.1° to the shoreline of 136.2°T), south arm 11°T (54.9°); apex angle 75° (Appendix A, A.5)
Incident direction	assumed shore-normal, from 46°T
$\varphi 0$	50.1° and 54.9°
Status	assumed; the swell direction is not confirmed in the sources read

E.5 Boscombe Surf Reef

Item	Value
Main cut	x = 2.0 m through the shallowest crest cell, reef run y = 187 to 252 m, offshore bearing 173.4°T; 3 m right of the plan middle
Second cut	none
Arm axes	one axis: the long axis of the outline, 28.1°T (55.3° to the shoreline of 83.4°T)
Incident direction	design swell from 191°T (travelling towards 11°T), 17.6° from shore-normal, no refraction
$\varphi 0$	72.9° (reef axis against the crest line, bearing 101°T)
Status	derived: geometry of the traced outline against the design swell of Mead et al. (2010); refraction is ignored

E.6 Borth coastal-defence reef

Item	Value
Main cut	x = -43.5 m, reef run y = 264 to 382 m, offshore bearing 269.6°T, through the north (hook) mound; 68 m right of the plan middle (the plan includes the south oval mound)
Second cut	none
Arm axes	north mound long axis 63.8°T (64.2° to the shoreline of 359.6°T); the south oval is parallel to the shore and carries no arm
Incident direction	assumed shore-normal, from 270°T
$\varphi 0$	64.2°

Item	Value
Status	assumed; no wave direction is available for Borth, and no source locates the break

E.7 Palm Beach reef

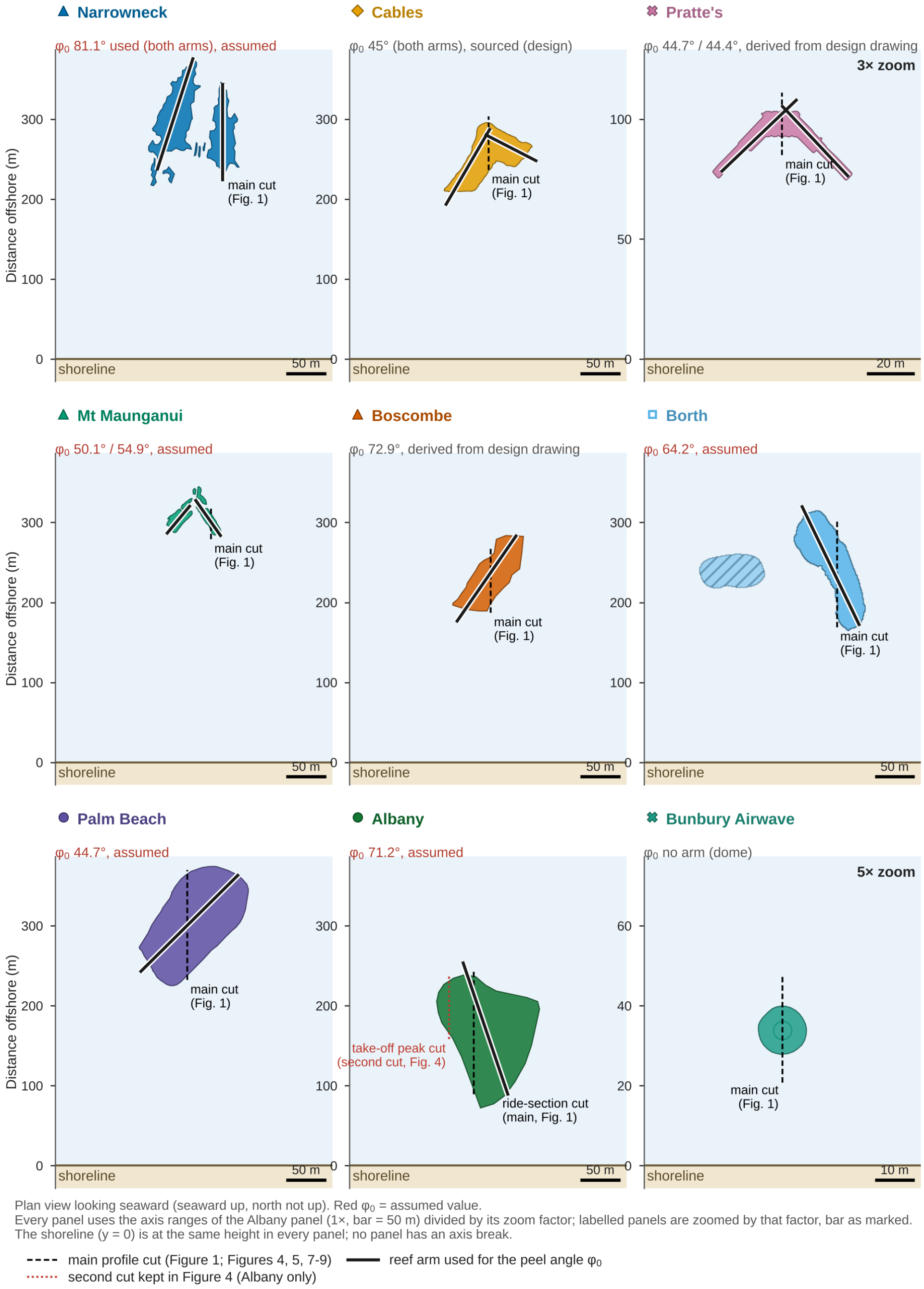
Item	Value
Main cut	x = 9.0 m through the shallowest crest cell, reef run y = 240 to 362 m, offshore bearing 64.3°T; 8 m left of the plan middle
Second cut	none
Arm axes	one axis: the long axis of the toe outline, 109.6°T (44.7° to the shoreline of 334.3°T); the concept design gives a crest orientation of 105° (Mortensen et al., 2015)
Incident direction	assumed shore-normal, from 64°T (crest parallel to the shoreline)
$\varphi0$	44.7°
Status	assumed; the south-east swell is only qualitative

E.8 Southern Ocean Surf Reef, Albany

Item	Value
Main cut	ride-section cut, x = 17.0 m through (17, 118), reef run y = 92 to 231 m, offshore bearing 120.3°T, along the long -2.0 m AHD ridge; 18 m left of the plan middle
Second cut	take-off peak cut, x = 48.0 m, reef run y = 161 to 224 m, 49 m left of the plan middle, through the shallow offshore plateau (-1.0 m AHD); the only second cut kept in Figure 4
Arm axes	crest (shoreward) edge of the traced wedge 281.5°T (71.2° to the shoreline of 30.3°T); the design crest line is 278.1°T (City of Albany, n.d.; Royal HaskoningDHV, 2015)
Incident direction	assumed shore-normal, from 120°T; the real swell is probably oblique (refraction in King George Sound), which would lower $\varphi0$
$\varphi0$	71.2°
Status	assumed; the 3D surface of Albany is a fitted design rule (low confidence, Appendix A, A.12)

E.9 Bunbury Airwave

Item	Value
Main cut	through the centre of the 12 m circular dome, x = 0 m, reef run y = 32 to 43 m; orientation not known
Second cut	none
Arm axes	none (dome)
Incident direction	not defined
$\varphi0$	not computed
Status	no peel angle; Bunbury is omitted from Figure 5



Plan view looking seaward (seaward up, north not up). Red φ_0 = assumed value.
Every panel uses the axis ranges of the Albany panel (1×, bar = 50 m) divided by its zoom factor; labelled panels are zoomed by that factor, bar as marked.
The shoreline (y = 0) is at the same height in every panel; no panel has an axis break.

Figure E1. Plan-view outlines of the nine reefs with a profile cut, each with the main profile cut (dashed black; Figure 1), the reef arm used for the peel angle $\varphi0$ (solid black), and, for Albany, the second cut kept in Figure 4 (dotted red). Seaward is up; north is not up. All panels share one baseline scale, that of the Albany panel (1×; 50 m scale bar): the horizontal window is 347 m wide and the vertical window runs from 28 m behind the shoreline to 347 m offshore, and every panel shows these ranges divided by its zoom factor. Panels at a different zoom are labelled in the top right (Pratte's 3×, Mount Maunganui 3×, Bunbury 5×; scale bar as marked); all other panels are 1×. The shoreline ($y = 0$) is at the same height in every panel and zoom is about the shoreline. Where a reef lies beyond the window it is shown after a y-axis break (marked //, omitted range stated): Narrowneck ($y = 103$ to 185 m omitted), Mount Maunganui (21 to 268 m) and Palm Beach (111 to 193 m); the two strips share one scale. Panels are centred on the reef cut. Red $\varphi0$ = assumed value; reef colours as in the main figures.

E.10 Lines for the breaking length after the break

Table 3 reads the breaking length after the break on five lines per reef (Figure E2; rules in Appendix A, item 4). The Figure 1 cut is the main cut above. The three shore-normal lines divide the alongshore extent of the reef footprint at 25, 50 and 75 % from its left end, looking seaward; they show how much the result depends on where a single cross-section is taken (Palm Beach 19 to 28 m, Albany 29 to 98 m, Narrowneck 6 to 38 m). The longest crest line follows the shallow ridge from its most seaward point, the take-off, to its far end. It measures a different quantity: on a peeling reef the wave breaks progressively along the arm, so the crest line gives the length of ride over which the arm keeps finding breaking depth, not how long a broken wave keeps breaking as it travels shoreward. Its rule (ridge = reef within 1.0 m of the shallowest point) is a geometric judgement; with 0.5 or 1.5 m the crest-line medians change by up to a factor of about two at Cables, Mount Maunganui and Pratte's (more at Borth, whose crest dries), and are within 10 % at Palm Beach, Narrowneck and Boscombe; the rank correlation with the outcome is 0.69 for both (0.62 for 1.0 m). At Palm Beach the crest line runs obliquely across the reef for 92 m, which is the 100 m or so of reef seen along the ride in Figure 1; at Albany it runs 155 m from the take-off plateau along the −2 m ridge.

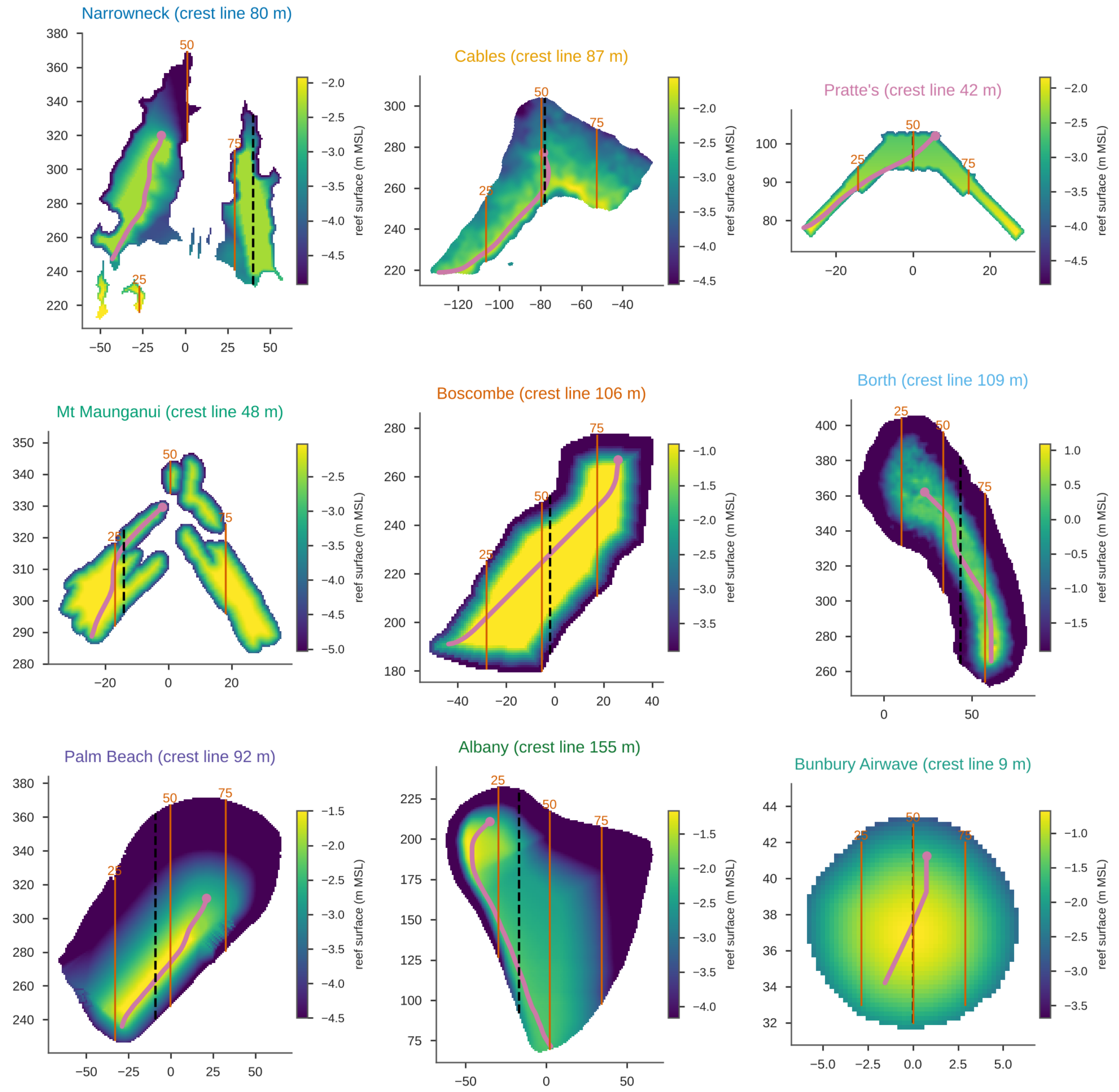


Figure E2. Lines used for the breaking length after the break (Section 4.3, Table 3), drawn on the plan of each reef model (colour: reef surface, m MSL; footprint cells at least 0.10 m thick; axes in metres of the model frame, plan looking seaward, seaward up, the left of each panel is the left looking seaward). Dashed black: the Figure 1 cut (hidden under the 50 % line at Pratte's and Bunbury). Orange: shore-normal lines at 25, 50 and 75 % of the alongshore extent of the footprint from its left end. Pink: the longest crest (ride) line, from the most seaward ridge cell (dot) along the cells within 1 m of the shallowest reef cell to the ridge cell farthest from it along the ridge; its length is in the panel title. Borth: northern structure only. DERIVED in this review (method in Appendix A).

Appendix F. Working hypotheses and their status

Before the analysis, eight working hypotheses were listed from field experience of built reefs and of the Haifa shore and from the literature, labelled (a) to (h) in Section 1.3. They were used only as questions to investigate; the Discussion (Section 8) rests on the computations and the evidence, not on the hypotheses. This appendix keeps track of each hypothesis: what the evidence of Sections 3 to 7 says about it, and whether it is consistent with published work or new to this review (verdicts by this review). Hypothesis (h) is split into two parts, the small tide (h-A) and mobile sand at Haifa (h-B). Six hypotheses are supported, at least in direction; two are partly supported; and (h-B) is split, with the burial risk supported and the sand trap not supported as a sand source. None is contradicted as a whole.

Hypothesis	Status	Main evidence	Literature
(a) Reefs need volume; small reefs such as Pratte's do not work	supported (descriptive)	volume ranks with outcome, $\rho = 0.72$ ($p = 0.021$, $n = 10$) stated and 0.93 ($p = 0.005$, $n = 8$) modelled; all reefs that do not work are 4,900 m^3 or less, both built reefs that work 25,000 m^3 or more; Burkitts (a reshaped natural reef) is the exception (Section 4.5)	consistent with Shand (2011, as cited in Mead and Borrero, 2011) and Borrero and Nelsen (2003); the threshold is new
(b) Little measurable effect on sand; Gold Coast claims confounded by nourishment	supported where testable; Gold Coast cannot be decided	observed lee response 0.02 to 0.06 of the Black and Andrews (2001) prediction (Boscombe +3.4 m against 84 m); Palm Beach "retained" 550,000 m^3 against about 470,000 m^3 placed (Sections 5.3-5.4)	consistent with Ranasinghe and Turner (2006) and Blacka et al. (2013)
(c) Soft structures do not last in the breaking zone; geotubes are cheapest and support marine life	durability supported; cost partly; ecology untested	6 of 7 fabric reefs failed or were lost against 0 of 4 rock reefs (Fisher $p = 0.015$; confounded by date and budget); A\$56 against A\$457 per m^3 to build, but renewal and removal add cost (Section 6)	durability consistent with Mead (as quoted in Geosynthetics Magazine, 2023); cost and ecology claims new

Hypothesis	Status	Main evidence	Literature
(d) Reef waves work on few days; Boscombe too steep, mostly for bodyboarders	supported	surfable 0 to 0.70 of the time even when a suitable swell is present; Pratte's 1 to 5 breaking events a year; the Boscombe 1:3 face collapses in every case; stand-up surfers called it "a bodyboarders wave", while advanced stand-up surfers also rode it, so "mostly" bodyboarders holds (Sections 4.3, 4.6)	consistent with Mead and Borrero (2011) and Rendle (2016)
(e) Wave power at the break lifts and drops the tube, so most tubes tear	partly	critical state just before breaking with simultaneous uplift (Bar & Drimer, 2023); sealed-base tubes 1.0 to 1.65 m high lift at 2.3 to 3.9 m depth; tearing not modelled, and reported causes are point damage, undersizing and construction faults (Section 6.3)	uplift consistent with Bar and Drimer (2023); tearing mechanism new and untested
(f) The depth gap alone does not decide; a long ramp gives a good wave, a sudden rise a too-hollow one	supported	same breaking depth (1.78 m) at Palm Beach and Boscombe, but ξ_b 0.56 against 3.00; on the surf cuts, ramp length $\rho = 0.93$ ($p = 0.005$) and mean slope 0.77 ($p = 0.039$) rank with outcome, crest depth does not (0.39, $p = 0.36$); Albany, read on its ridge (99 m, 1:19.8), plots with Palm Beach (Section 4.2)	ramp is a known component (Mead and Black, 2001a, as summarised by Blacka et al., 2013); the quantification is new
(g) Beginners need spilling waves that reefs do not make; difficulty set by peel angle; about 30 degrees for good surfers	supported	spilling breakers 0 to 29 % of on-reef cases; only one case (Borth, $H_0 = 3$ m, 61.5 degrees, above the height range of its rating) reaches the 60 degrees of a learner; the 30 degrees matches the rating-7 limit of 29 degrees (Section 4.4)	consistent with Hutt et al. (2001); Walker's (1975) 30 degrees is a hydrodynamic minimum, not a skill class

Hypothesis	Status	Main evidence	Literature
(h-A) Israel: the small tide gives consistency and smaller loads	partly	consistency supported (0.87 to 0.98 of the no-tide surfable time kept) but the average does not rise (mean change +0.01); storms, not the tide, set the loads, and the Ashkelon North tubes tore (Sections 4.6, 7.2-7.3)	Netanya tubes not confirmed (Ashkelon North instead; State Comptroller of Israel, 2020); consistency effect new
(h-B) Israel: sand at Haifa is mobile (about 1 m seasonal bed change), so it can bury a reef, but could also be trapped	burial supported as a risk; sand trap not supported	the 1 m comes from field observations by the author at the Haifa shore (Appendix G), backed by Google Earth satellite imagery that shows more sand accumulating north of the breakwaters; it is not a survey and has not been published, but the published seasonal shoreline swing implies 0.3 to 1.3 m; a 1 m rise removes 19 to 40 % of a reef's relief; the sand trap stores less than 4 % of one year's net drift (Section 7.4)	seasonal shoreline swing from Zviely et al. (2009); bed change and sand trap new

Table F1. Working hypotheses (a) to (h) and their status (verdicts by this review).

Two cases deserve comment. The Pratte's testimony separates the two failures that (a) and (c) run together: the witnesses describe a wave that almost never broke, and one says the bags were intact when removed (Dan Harmon, 2026; grade D), whereas the removal report found fabric that "could be torn by hand" (Blacka et al., 2013, pp. 116-117). Either way the reef failed first as a wave, through its size and its site in the swell shadow of Catalina Island, which supports (a) more directly than (c). Albany, the best wave in the sample, first appeared to contradict (f): on the automatic profile through its −1 m take-off plateau it rose at about 1:4 and fell in the collapsing band. On the cut through the long −2 m ridge where the wave is ridden (Figure E1, Appendix E) it has a 99 m ramp at 1:19.8 and plunges like Palm Beach, so it supports (f); it is still "intermediate-to-advanced on most days" (Mercer, 2025), consistent with its median vortex ratio of 1.75 ("extreme").

Appendix G. Field observations of surfable conditions and take-off depth at Haifa

This appendix documents the author's own field observations at Haifa, which Section 7 uses for the wave conditions and the water depth of a planned reef. The observations are notes from regular surf sessions over about ten weeks, from late July to early October 2025, and are not a survey. They were recorded as a slide deck: one slide gives water depths measured in the surf for the minimum surfable waves, and six slides mark, on charts of the Shikmona wave-buoy record, the stretches during which waves were surfable, borderline or not surfable. The numeric values read from the charts and the depths are kept in the data set of this review. Depths are treated separately in Section G.5, because no depth was measured with an instrument.

G.1 What was recorded

1. **Colour-coded sea states (chart slides 3 to 8, charts A to F).** For each session period the author took the time series of the Shikmona wave buoy off Haifa, published by the Israel Marine Data Center (ISRAMAR, 2025), and drew a coloured box over the stretch of the record in which a given surfability was observed. The chart gives the significant wave height H_s, the maximal wave height H_{max} and the period of the maximal waves, which is called T_p below. The boxes therefore mark *when* a condition held; H_s, H_{max} and T_p in Tables G1 and G2 were read from the chart inside each box. They are wave conditions at the buoy, not water depths.
2. **Depths at the peak for the minimum surfable waves (slide 2, Section G.5).** Water depths measured by the author standing in the surf on days with H_s 0.45 to 0.60 m, for two bounds of wave height and three positions of the break, with the distance from the measured depth to the break.
3. **Take-off depth (author's field estimate, Section G.5).** An estimate of the water depth at the take-off in borderline conditions, made by the author in the water on some days.

G.2 Colour classification

The classification is the author's own judgement from regular surf sessions at Haifa; no instrument was involved and no second observer checked it. The colours mean:

- **green**: surfable; the minimum conditions for surfing;
- **yellow**: borderline; surfable only in the bigger sets that arrive about once in 15 minutes;
- **orange**: the upper limit of red; not surfable, but a little more, or a perfectly shaped wave in a set that arrives about once in 15 minutes, would have been surfable;
- **red**: not surfable.

The wording for yellow (chart slides 4 to 6) and orange (chart slides 7 and 8) is written on the slides in Hebrew; the meanings of green and red were confirmed by the author on 20 October 2026. The colours were assigned after the sessions, by judgement. They are not defined by a wave height alone: surfability depends on height *and* period (and on wind and seabed shaping), so the minimum conditions of a colour are given below as height-period pairs.

G.3 Method and sources

The wave data are screenshots of the ISRAMAR Shikmona buoy page (ISRAMAR, 2025) ("Shikmona Buoy Significant Wave Height and Period of maximal waves"; H_s and H_{max} in metres on the right axis, the period of the maximal waves in seconds on the left axis, local time). The charts cover six periods, 27 July to 9 October 2025, within a sample of about ten weeks:

Slide (chart)	Period shown	"Last observed" stamp on the screenshot
3 (A)	27 Jul to 1 Aug 2025	none
4 (B)	2 to 7 Aug 2025	6 Aug 2025, 19:00 UTC
5 (C)	11 to 15 Aug 2025	15 Aug 2025, 09:00 UTC
6 (D)	3 to 9 Sep 2025	none (a splice in the screenshot omits part of 7 to 8 Sep)
7 (E)	15 to 19 Sep 2025	19 Sep 2025, 11:00 UTC
8 (F)	5 to 9 Oct 2025	9 Oct 2025, 05:00 UTC

The values in Tables G1 and G2 were read by this review from the chart images, not by the author. The slides were rendered without the coloured boxes, the plot frame was located by its axis lines and calibrated on the axis ranges, and the two curves and the period dots were then extracted by colour inside the area of each box (pixel column by column). The reading error is about 0.05 m in H_s and H_{max}, about 0.3 s in T_p (the period dots are discrete and 2 to 14 fall inside a box), and about 1 to 2 hours in time, because the boxes were drawn by hand and their edges include some of the neighbouring record. For each box the tables give the lowest H_s inside the box, the H_{max} at the same time, and T_p as the median of the period dots within about 1.25 h of that time; the range of all period dots inside the box is also given. Because box edges can include a lower neighbouring value, the lowest H_s of a box is a lower bound on the minimum condition, and the 20th percentile of H_s (in the CSV) is less sensitive to the edges.

The author's own readings of the orange boxes, for comparison, were H_s 0.6 m, H_{max} 0.8 m, T_p 5.5 s (chart 8, F); H_s 0.45 m, H_{max} 0.5 m, T_p 7.5 s (chart 7, E); and H_s 0.5 m, H_{max} 0.55 m, T_p 7 s (chart 7, E). The pairs of this review agree within the reading error for chart 7 (H_s 0.42 to 0.53 m, T_p 6.9 to 7.3 s; the H_{max} of 0.50 m is within 0.05 m of the 0.54 m read here). For chart 8 the box starts at H_s about 0.57 m and H_{max} about 0.73 to 0.80 m with period dots at 3.8 to 4.5 s, and the dots near 5.5 s occur later in the box (around 12:00 to 16:00 on 5 October), where H_s and H_{max} are higher. The author's pair mixes the two ends of the box; both readings are given in Tables G1 and G2.

Swell direction, noted by the author beside each chart, was 284 degrees on charts A and E (E: 300 degrees from 14:30 on 19 September), 284 changing to above 292 degrees on chart B, and 292 to 294 degrees on charts C and D; none is noted for chart F.

G.4 Results: wave conditions by colour

Slide (chart)	Chart period	Box (2025)	Colour	Hs (m)	Hmax (m)	Tp (s)	Tp range in box (s)
3 (A)	27 Jul-1 Aug	28 Jul, 00:30-06:30	red	0.38	0.49	6.5	6.2-7.3
3 (A)	27 Jul-1 Aug	28 Jul, 06:30-09:00	green	0.44	0.57	6.3	6.2-6.5
3 (A)	27 Jul-1 Aug	29 Jul, 10:00-16:00	red	0.42	0.54	5.1	4.6-5.7

Slide (chart)	Chart period	Box (2025)	Colour	Hs (m)	Hmax (m)	Tp (s)	Tp range in box (s)
3 (A)	27 Jul-1 Aug	30 Jul 09:00 to 31 Jul 18:30	green	0.63	0.81	7.8	5.4-8.2
4 (B)	2-7 Aug	3 Aug 18:30 to 4 Aug 22:00	green	0.53	0.69	7.1	6.2-7.7
4 (B)	2-7 Aug	4 Aug 22:00 to 5 Aug 11:30	yellow	0.46	0.59	6.5	6.2-7.3
4 (B)	2-7 Aug	5 Aug 11:30 to 6 Aug 21:00	red	0.40	0.51	5.9	5.2-6.5
5 (C)	11-15 Aug	11 Aug, 00:30-12:30	yellow	0.41	0.52	6.9	6.5-7.7
5 (C)	11-15 Aug	11 Aug 12:00 to 12 Aug 22:00	green	0.42	0.54	6.9	3.9-8.2
5 (C)	11-15 Aug	13 Aug 04:30 to 14 Aug 00:00	yellow	0.42	0.54	5.3	4.1-6.5
5 (C)	11-15 Aug	14 Aug 00:00 to 15 Aug 10:30	red	0.17	0.22	3.7	3.3-6.2
6 (D)	3-9 Sep	3 Sep 11:00 to 4 Sep 00:30	green	0.80	1.03	6.9	6.2-7.7
6 (D)	3-9 Sep	4 Sep, 05:30-12:30	green	0.73	0.93	7.3	6.2-7.7
6 (D)	3-9 Sep	4 Sep, 15:30-20:00	yellow	0.67	0.86	7.3	6.9-7.7
6 (D)	3-9 Sep	5 Sep 03:30 to 7 Sep 17:30	red	0.34	0.44	6.5	4.5-7.3
6 (D)	3-9 Sep	8 Sep, 14:30-22:00	yellow	0.41	0.53	6.5	5.2-7.7
7 (E)	15-19 Sep	16 Sep, 07:30-12:30	orange	0.42	0.54	6.9	6.2-7.3
7 (E)	15-19 Sep	16 Sep, 17:00-22:00	orange	0.44	0.56	7.1	5.9-7.3
7 (E)	15-19 Sep	17 Sep, 14:00-21:30	orange	0.53	0.68	7.3	6.5-7.7
7 (E)	15-19 Sep	18 Sep, 06:30-11:00	orange	0.42	0.54	7.3	6.9-7.7
7 (E)	15-19 Sep	18 Sep, 15:30-20:00	orange	0.44	0.56	7.1	6.5-7.3
7 (E)	15-19 Sep	19 Sep, 01:30-11:00	orange	0.48	0.62	6.9	4.5-7.3
8 (F)	5-9 Oct	5 Oct, 00:30-16:30	orange	0.57	0.73	3.9	3.8-5.7
8 (F)	5-9 Oct	7 Oct 23:00 to 8 Oct 17:30	red	0.13	0.16	4.8	3.9-5.7

Table G1. Coloured stretches of the Shikmona record (chart slides 3 to 8; read from the charts by this review: H_s and H_{max} $±$0.05 m, T_p $±$0.3 s). H_s is the lowest value in the box, H_{max} the value at that time, T_p the period of the maximal waves near that time.

Colour	H_s (m)	H_{max} (m)	T_p (s)	Box (chart)
green	0.42	0.54	6.9	11 to 12 Aug (C)
green	0.44	0.57	6.3	28 Jul, 06:30-09:00 (A)
green	0.53	0.69	7.1	3 to 4 Aug (B)
yellow	0.42	0.54	5.3	13 to 14 Aug (C)
yellow	0.41	0.53	6.5	8 Sep (D)
yellow	0.41	0.52	6.9	11 Aug (C)
yellow	0.46	0.59	6.5	4 to 5 Aug (B)
orange	0.42	0.54	6.9	16 Sep, morning (E)
orange	0.42	0.54	7.3	18 Sep, morning (E)
orange	0.44	0.56	7.1	16 and 18 Sep, evening (E)

Colour	H_s (m)	H_{max} (m)	T_p (s)	Box (chart)
orange	0.57	0.73	3.9 (3.8 to 4.5 for the first hours; 5.4 to 5.7 later in the box)	5 Oct (F)
red (not surfable), highest of the six boxes	0.42	0.54	5.1	29 Jul (A); the other red boxes have H_s 0.40 m (H_{max} 0.51 m, T_p 5.9 s, 5 to 6 Aug, B), 0.38 m (0.49 m, 6.5 s, 28 Jul, A), 0.34 m (0.44 m, 6.5 s, 5 to 7 Sep, D), 0.17 m and 0.13 m

Table G2. Minimum conditions in which each colour was observed (H_s, H_{max}, T_p pairs; the lowest-energy boxes of each colour, ordered by $H_s^2 T_p$ as a rough energy measure).

Three points follow from the tables. First, the minimum height at which waves were surfable at Haifa in this sample is about H_s 0.4 to 0.45 m with H_{max} about 0.5 to 0.55 m, at periods of 6.5 to 7.5 s; green, yellow and orange boxes reach the same low heights at similar periods, so height alone does not separate the colours. Second, the colours do separate when height and period are read together with the other factors named in G.2 (set size, wind, seabed shaping): the one orange box at a short period (chart F, 3.8 to 4.5 s) has H_s 0.57 m, higher than the long-period orange boxes (0.42 to 0.53 m), which indicates that a short period needs more height for the same classification. Third, the yellow box of 13 to 14 August has the shortest period near its minimum (5.3 s, dots 4.1 to 6.5 s) at H_s 0.42 m, so a single height-period boundary per colour cannot be drawn from six charts; the pairs above are minimum conditions that were observed, not thresholds. The red box of 29 July (chart A) has H_s 0.42 m at T_p 5.1 s and was not surfable, whereas green boxes at the same height had T_p 6.3 to 6.9 s, which is again consistent with period mattering. The sample has no surfable record below H_s 0.4 m.

G.5 Water depths at the peak and at the take-off (author's field measurements)

No depth was measured with an instrument. Two sets of depths were taken by standing in the water and reading where the water reached on the body, whose heights were measured afterwards with a tape.

Depth at the peak for the minimum surfable waves (slide 2, "Sea level measures - For minimum 'Surfable' waves"). On days when the Shikmona buoy gave H_s 0.45 to 0.60 m and H_{max} 0.6 to 0.8 m from 285 to 294 degrees, the author stood at the peak once the wave had reached shallow water (about one second after it broke) and read the water depth there. The wave heights of those days were taken from the forecasts, Surfline (the former Magicseaweed forecast) with 0.6 to 0.9 m and 4Surfers with 0.65 to 0.8 m ("knee high") at 6 to 8 s (4Surfers, 2025; Surfline, 2025), and checked against the buoy (ISRAMAR, 2025). Three depths were recorded for a low and a high bound of the wave height (Table G3).

Wave-height bound (buoy)	Min depth (m)	Breaking depth (m)	No breaking (m)	H_s/h at breaking	H_{max}/h at breaking
low: H_s 0.45 m, H_{max} 0.6 m	1.15	1.30	1.45	0.35	0.46
high: H_s 0.60 m, H_{max} 0.8 m	1.35	1.50	1.70	0.40	0.53

Table G3. Water depth at the peak for the minimum surfable waves at Haifa (author's field measurements, slide 2; the ratios are derived in this review).

The terms are those of the slide. *Min depth* is the depth at which the smallest part of the wave that can still be surfed (the smallest shoulder) breaks; *breaking depth* is the depth at the peak where the wave breaks; *no breaking* is the depth seaward of which the wave did not break, and its upper value is the depth at which the largest sets begin to break. The side view of the slide adds "2 m until break" with an accuracy of $±$0.5 m: the wave travels about 2 m from the depth where it was measured to the point where it breaks, measured in the same conditions. Over the two bounds the depths span 1.15 to 1.70 m, and the wave breaks at 1.30 to 1.50 m. The ratios in Table G3 use the buoy heights, not the breaker heights, which are larger after shoaling; they are therefore not breaker indices and are given only to compare with the model of Section 7.5.

Take-off depth in borderline conditions. On some days the author estimated the water depth at the wave peak by standing in the water at the take-off position and noting where the water reached on the body; that body height was measured afterwards with a tape. On this basis the author estimates that the **minimum water depth at the take-off for yellow and orange conditions is about 1 m (about chest height)**, **decreasing to about 0.5 m towards the end of the ride**.

Both sets are field measurements by a single observer. The uncertainty of the take-off estimate is of the order of $±$0.2 m; that of the slide-2 depths is not stated, apart from the $±$0.5 m of the length measure. The two sets agree in order: the borderline (yellow and orange) seas, with heights at the low end of Table G2, take off in about 1 m, slightly less than the 1.15 m minimum depth of the smallest surfable shoulder at H_s 0.45 m. No datum or tide correction was applied; the tidal range at Haifa is small (about 0.4 m, Section 7.1), so the tide is of the same order as the estimate's error, and the water-level anomaly of the day (weather, season) may add to it. The days of the measurements are not recorded and they are not tied to a particular box in Table G1; the slide-2 depths are stated for the two height bounds and the take-off estimate for the yellow and orange conditions in general. Section 7 uses them as order-of-magnitude figures for the water depth over the crest of a reef that is meant to be surfable at the lower end of the wave-height range: the wave breaks at 1.30 to 1.50 m (1.15 to 1.70 m from the smallest shoulder to the largest sets), takes off in about 1 m in borderline seas and needs about 0.5 m near the end of the ride.

G.6 Annotated buoy records

Figures G1 to G6 reproduce the author's annotated chart slides (slides 3 to 8 of the author's presentation, charts A to F), rendered with the coloured boxes drawn by the author over the ISRAMAR chart. Each box marks a stretch of the record in which the author classified the surf at Haifa by the colour scheme of Section G.2. A single legend (Figure G0, below) applies to all six charts; it was added by this review. The chart, the boxes and the author's handwritten swell-direction notes are unchanged. Figure G7 shows the depth-measure slide (slide 2) used in Section G.5.

🟩 green: surfable in all conditions (minimum)

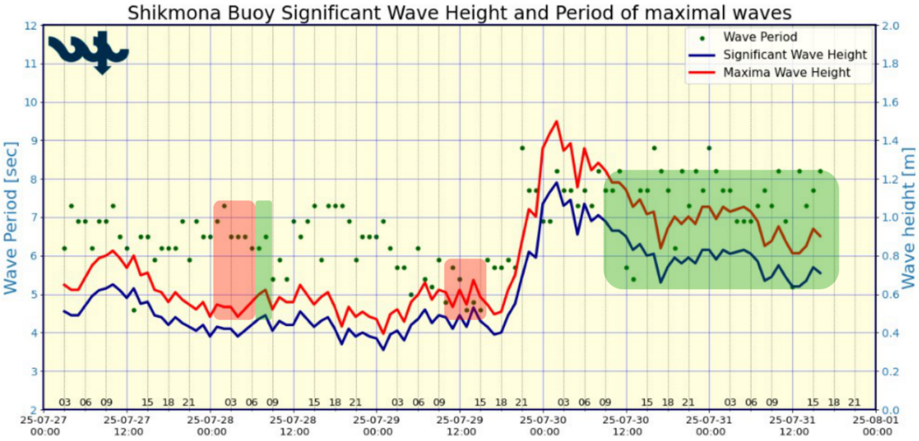
🟨 yellow: surfable in great conditions, or in normal conditions only in the big sets (about once per 15 min)

🟧 orange: surfable only in great conditions (seabed shaping, wind) and only in the big sets

🟥 red: not surfable

— Hs: significant wave height (right axis, m) — Hmax: maximal wave height (right axis, m) ● Tp: period of the maximal waves (dots, left axis, s)

Figure G0. Legend for Figures G1 to G6. Box colours (the author's field classification of the surf): green, surfable in all conditions (minimum); yellow, surfable in great conditions or, in normal conditions, only in the big sets that arrive about once in 15 minutes; orange, surfable only in great conditions (seabed shaping, wind) and only in the big sets; red, not surfable. Curves: blue, H_s (right axis, m); red, H_{max} (right axis, m); green dots, T_p (left axis, s).



degree 284

degree 284

Figure G1. Chart A (slide 3): ISRAMAR Shikmona buoy record, 27 July to 1 August 2025, local time (ISRAMAR, 2025). Blue line: significant wave height H_s; red line: maximal wave height H_{max} (both on the right axis, m); green dots: period of the maximal waves T_p (left axis, s). The coloured boxes are the author's field classification of the surf, not an ISRAMAR product; legend as in Figure G0 above.

G.7 Limitations

- **Single observer, no instrument.** The colours are a personal judgement of surfability by one surfer, which depends on skill, board and expectation. There is no inter-observer check. The depth figures in G.5 are likewise single-observer estimates.
- **Buoy, not breaker.** Heights and periods are from the Shikmona buoy, offshore of the break, and were read from images. Shoaling, refraction and the sheltered position of Haifa Bay (Section 7.1) mean the height at the breaker differs from the buoy value. The period dots show the period of the maximal waves, not of the whole spectrum.
- **Conditions, not depths.** Tables G1 and G2 are wave conditions. The depths of Table G3 and the take-off estimate are not tied to any colour box or date, and the wave heights of the slide-2 days come from forecasts and the buoy, not from the break.
- **Box edges.** Boxes were drawn by hand and their edges include neighbouring record, so the lowest values of a box can come from its edge; the pairs in Table G2 are lower bounds of the minimum conditions.
- **Datum.** The depths have no datum or tide level, and the error from this is of the order of the tidal range (about 0.4 m).

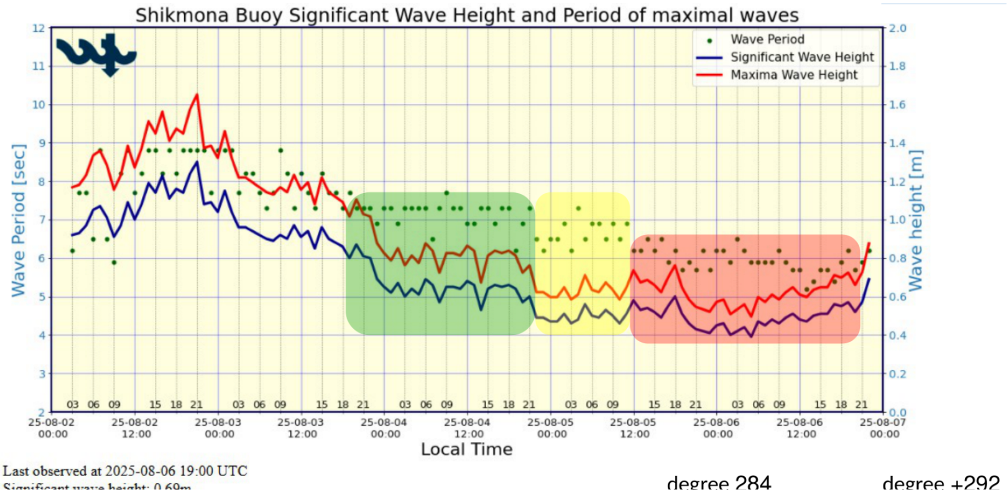


Figure G2. Chart B (slide 4): ISRAMAR Shikmona buoy record, 2 to 7 August 2025, local time (ISRAMAR, 2025). Blue line: significant wave height H_s; red line: maximal wave height H_{max} (both on the right axis, m); green dots: period of the maximal waves T_p (left axis, s). The coloured boxes are the author's field classification of the surf, not an ISRAMAR product; legend as in Figure G0 above.

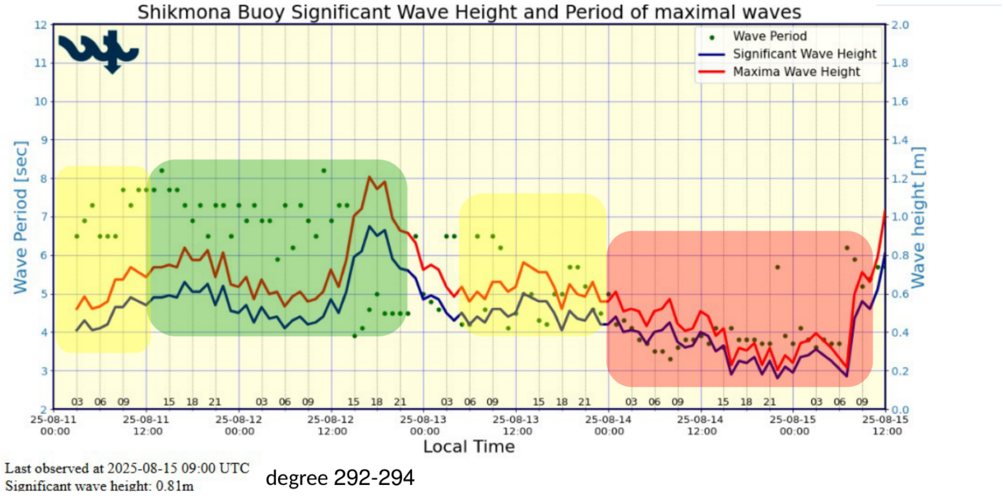


Figure G3. Chart C (slide 5): ISRAMAR Shikmona buoy record, 11 to 15 August 2025, local time (ISRAMAR, 2025). Blue line: significant wave height H_s; red line: maximal wave height H_{max} (both on the right axis, m); green dots: period of the maximal waves T_p (left axis, s). The coloured boxes are the author's field classification of the surf, not an ISRAMAR product; legend as in Figure G0 above.

- **Short and uneven sample.** About ten weeks, 27 July to 9 October 2025 (summer to early autumn, no winter swell), six chart periods and 24 coloured boxes. Chart D has a gap between 7 and 8 September where the screenshot is spliced; charts A and D carry no capture stamp; the swell direction is missing for chart F.

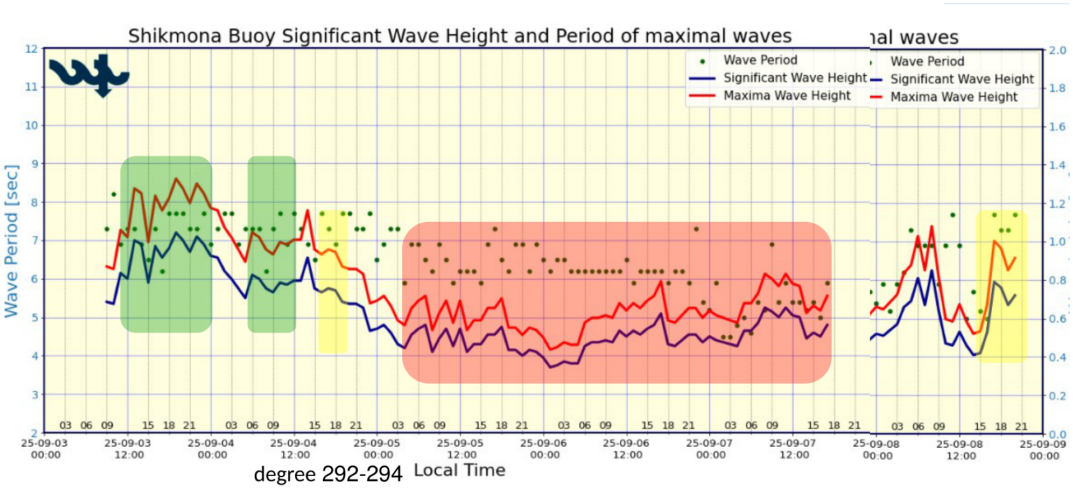


Figure G4. Chart D (slide 6): ISRAMAR Shikmona buoy record, 3 to 9 September 2025 (the screenshot has a splice that omits part of 7 to 8 September), local time (ISRAMAR, 2025). Blue line: significant wave height H_s; red line: maximal wave height H_{max} (both on the right axis, m); green dots: period of the maximal waves T_p (left axis, s). The coloured boxes are the author's field classification of the surf, not an ISRAMAR product; legend as in Figure G0 above.

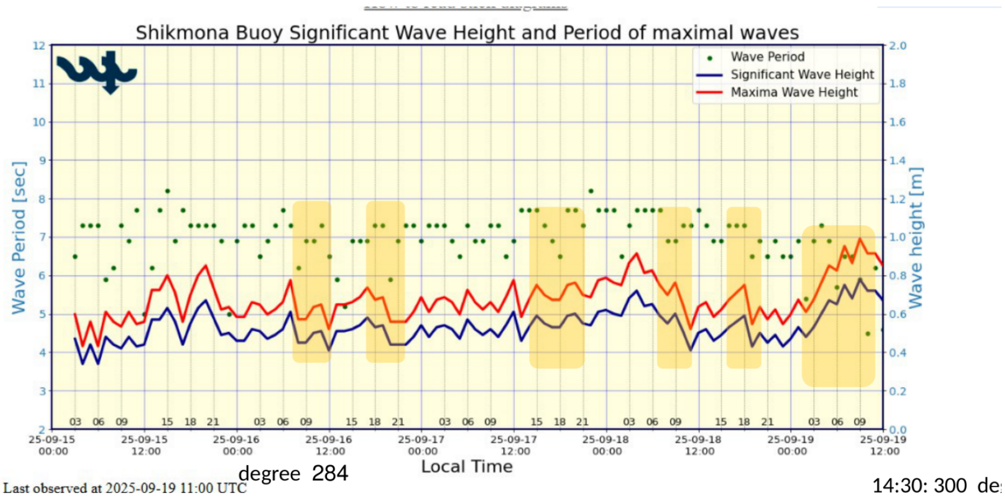


Figure G5. Chart E (slide 7): ISRAMAR Shikmona buoy record, 15 to 19 September 2025, local time (ISRAMAR, 2025). Blue line: significant wave height H_s; red line: maximal wave height H_{max} (both on the right axis, m); green dots: period of the maximal waves T_p (left axis, s). The coloured boxes are the author's field classification of the surf, not an ISRAMAR product; legend as in Figure G0 above.

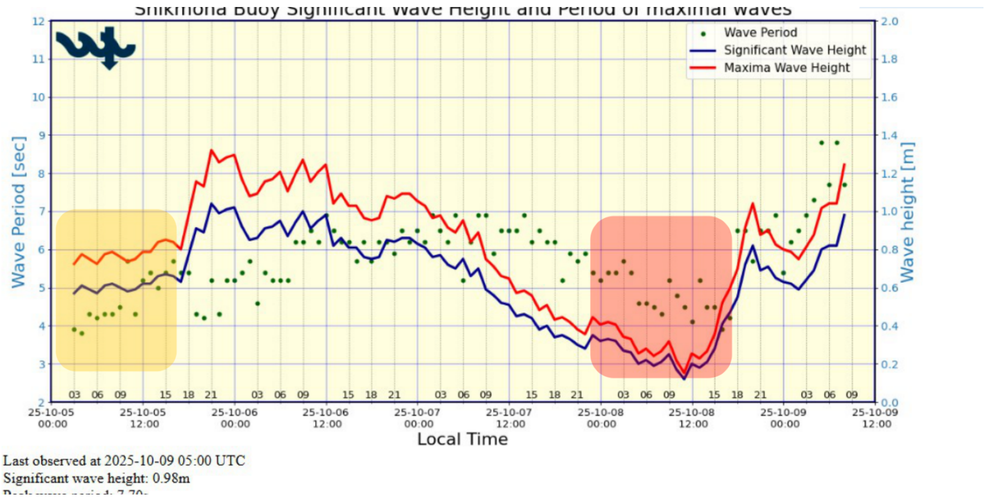
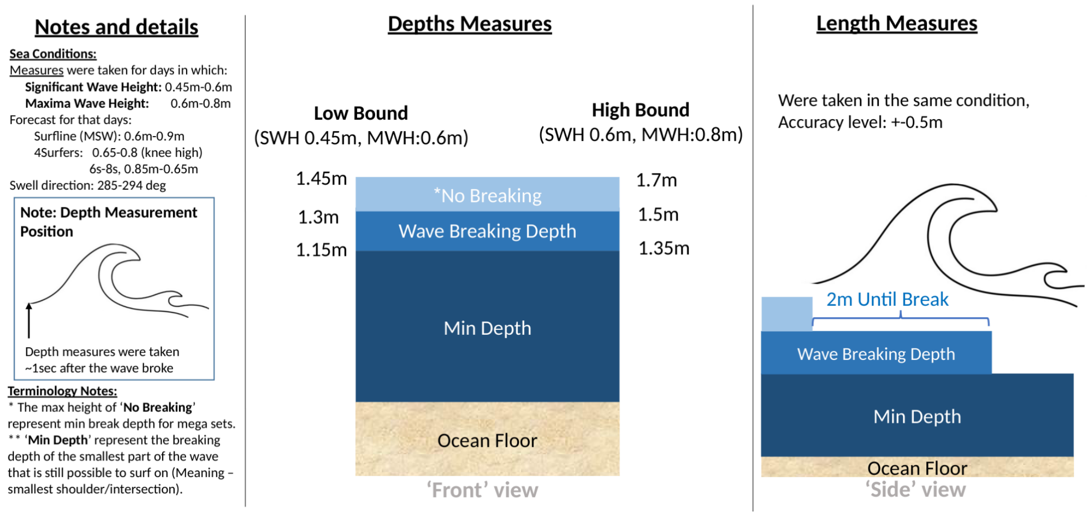


Figure G6. Chart F (slide 8): ISRAMAR Shikmona buoy record, 5 to 9 October 2025, local time (ISRAMAR, 2025). Blue line: significant wave height H_s; red line: maximal wave height H_{max} (both on the right axis, m); green dots: period of the maximal waves T_p (left axis, s). The coloured boxes are the author's field classification of the surf, not an ISRAMAR product; legend as in Figure G0 above.



Notes and details

Sea Conditions:
Measures were taken for days in which:
Significant Wave Height: 0.45m-0.6m
Maxima Wave Height: 0.6m-0.8m
Forecast for that days:
Surfline (MSW): 0.6m-0.9m
4Surfers: 0.65-0.8 (knee high)
6s-8s, 0.85m-0.65m
Swell direction: 285-294 deg

Note: Depth Measurement Position

Depth measures were taken ~1sec after the wave broke

Terminology Notes:
* The max height of '**No Breaking**' represent min break depth for mega sets.
** '**Min Depth**' represent the breaking depth of the smallest part of the wave that is still possible to surf on (Meaning – smallest shoulder/intersection).

Depths Measures

Low Bound (SWH 0.45m, MWH:0.6m)		High Bound (SWH 0.6m, MWH:0.8m)
1.45m	*No Breaking	1.7m
1.3m	Wave Breaking Depth	1.5m
1.15m	Min Depth	1.35m
	Ocean Floor	

'Front' view

Length Measures

Were taken in the same condition,
Accuracy level: +-0.5m

'Side' view

Figure G7. Depth measures (slide 2 of the author's presentation): water depths measured by the author on days with H_s 0.45 to 0.60 m for the low and high bound of wave height, front and side view, with the distance to the break (accuracy $±$0.5 m). The values are the author's field measurements; see Section G.5.